HITACHI
Inspire the Next



Train Protection
VF. CONCEPT
SFRS

Responsible Division:	Responsible Unit:	Document Type:	Distribution Status:	Document State:

	Name	Signature	Date
Prepared:	LUCA GALGANO Prepared	29/11/2021	
Checked:	DANIELE FERRARI	29/11/2021	
Checked:			
Approved:	GIUSEPPE CARRINO Head of Vehicle Engineering	29/11/2021	
Released:	GIUSEPPE CARRINO Deputy Project Engineer	29/11/2021	

Copyright © Hitachi Rail - All Rights Reserved

Identity Number:		
FA02P021958B		
Effective Date:	Revision:	Language:
29/11/2021	02.00	EN

Revisionlog

Revision	Date	Changes	Signed
00.02	15/05/2019		G. Carrino
00.03	31/05/2019	Updating after review of the 29/05/2019	G. Carrino
00.04	19/07/2019	Updating by HR STS review from document “FA02P021958B_SFRS_ALA-Bistandard_Rev00.02_review_RBE JIS” Updating by BT observations	G. Carrino
00.05	23/09/2019	-Updating based to BT request by e-mail 31.07.2019 15:17 -Added the check on the Z-(BG)IS rotating selector misalignment -Added the requiremente about emergency speed indication -Added the recorded of the analog signals by ATESS JRU (line voltage, max line current) -Added the check of coherence for all France signalling selectors -Added the events requirement aligned to diagnostic Event file (yellow marker)	G. Carrino
00.06	29/09/2019	-Modify L_DEGAGEMENT_US from 202m to 180 m -Added all sequences cancel procedures changing phase, catenary and system -Added requirements VY(CO)URG and VY(CO)Z lamps power supply and parking mode functionality	G. Carrino
00.07	12/10/2019	Vrescue change selector from Z(CO)DJE to Z-ETCS Added requirement that the lowering of pantographs and the main switch opening are also commanded requested by ATP (in addition at driver commands). Added tables of max current selectable from AC and DC lines and the digital status of the line volttage absence/not absence to recorder in ATESS (JRU). Updated TRS.294 / TRS.313 / TRS.295 / TRS.296 / TRS.297 / TRS.298 Updated Cancellation phase separation sections and separation section sequences in manual mode Updated Cancellation phase separation sections and separation section sequences in automatic mode. Updated Voltage separation sections and separation section sequences shall be activated by external sensors Updated Cancellation voltage separation sections and separation section sequences in manual mode Updated Cancellation voltage separation sections and separation section sequences in automatic mode Added reference RD5 - ICD Added reference RD7 - EventReport	G. Carrino
00.08	24/1172019	Updated requirement 657 Added Table 7: Parameters Phase and Catenary Changing parameters phase and voltage catenary changing (table 7) Added Table 4: Parameter for the function S7A TVM – ETR1000 application Updated IDs: TRS.264, TRS303, TRS265, TRS304, TRS301, TRS305, TRS306, TRS307, TRS308, TRS309, TRS266, TRS267, TRS268, TRS269 TRS370, TRS254, TRS111, TRS273, TRS274, TRS275, TRS276, TRS277, TRS279, TRS280, TRS282, TRS284, TRS285, TRS286, TRS288, TRS289, TRS310, TRSxxx, TRS311, TRS312, TRS286, TRS293, TRS294, TRS313, TRS295, TRS314, TRS296, TRS315, TRS397, TRS316, TRS298, TRS317, TRS321, TRS318, TRS319, TRS299, TRS320, TRS300. Updated Chapters: 6 / 4.1.4.1.2 / 4.1.6.2.1 / 4.1.6.3.1	G. Carrino
00.09	21/12/2019	Concept mumbers assiged: 972; 825; 975; 976; 977; 978; 979; 980; 981; 982; 983; 984; 985; 986; 987; 988; 989; 990; 991; 992; 993; 994; 995; 996; 997; 998; 999; 1000; 1001; 1002; 1003; 1004; 1005; 1006; 1007; 1008; 1009; 1010; 1011; 1012; 1013; 1014; 1015; 1016; 1017; 1018; 1019; 1020; 1021; 1022; 1023; 1024; 1025; 1026; 1027; 1028; 1029; 1030; 1031; 1032; 1033; 1034; 1035; 1036; 1037; 1038; 1039; 1040; 1041; 1042; 1043; 1049; 1050; 1061; 1062; 1063; 1064; 1065; 1066; 1067; 1068; 1069; 1070; 1071; 1072; 1073; 1074; 1075; 1077 Concepts deleted: 181; 266; 298; 357; 396; 418; 434; 575; 598; 600; 602; 604; 606; 608; 610; 611; 614; 615; 651; 652; 1004 Concepts modified: 348; 349; 392; 444; 672; 727; 780; 821; 945; 946; 947; 959; 960; 961 Updated chapter 6, erased double diagram	G. Carrino
1.0	25/02/2020	4.1.3.1 Modified table Forward/Reverse	G. Carrino
2.0	29/11/2021	Added concepts for multiple configuration: - Modified §4.1.3.2 introducing Multiple information input to TVM and modified SIL for concepts 370, 371, 372, 373, 374 and 1079 - §4.1.5.1.2 added requirement inherent KVB phase transition for mutiple configuration - §4.1.6.1.2 added requirement inherent KVB voltage transition for mutiple configuration	G. Carrino

Content

1 Introduction 6

1.1 Train Configuration 6

1.2 Definitions and Abbreviations 7

2 System Overview 9

2.1 System Configuration 9
2.1.1 Sub-System Categories 9
2.1.2 Command Rack and Panel 10

2.2 Signalling Sub-systems 11
2.2.1 The KVB equipment 11
2.2.2 The EUROCAB Bi-Standard equipment 11
2.2.2.1 EUROCAB Sub-system 11

2.3 Additional Sub-systems 12
2.3.1 The Cabin equipment 12
2.3.2 The Cabinet equipment 12
2.3.3 The MIOs modules equipment 12
2.3.4 The Fans equipment 12
2.3.5 The ATESS equipment 12
2.3.6 The MVB network 12
2.3.7 The Profibus network 12
2.3.8 The GSM-R data equipment 12
2.3.9 The KARM GPS receiver equipment 12

3 System Interfaces Synthesys 13

3.1 Presentation 13

3.2 Interface Tables 13
3.2.1 External Interfaces Synthesis 13
3.2.2 Internal Interfaces Synthesis 14

3.3 EUROCAB-level interfaces in ATP 15
3.3.1 KVB equipment 15
3.3.2 Cabin equipment 16
3.3.3 EUROCAB equipment 17
3.3.4 MIOs equipment 18
3.3.5 Fans equipment 19
3.3.6 ATESS equipment 20
3.3.7 MVB network equipment 21
3.3.8 Profibus network equipment 22
3.3.9 GSM-R DATA equipment 23
3.3.10 KARM-GPS receiver equipment 23
3.3.11 Train equipment 24

3.4 EVC-level interfaces in EUROCAB 25
3.4.1 EVC equipment 25
3.4.2 DMI equipment 27
3.4.3 BTM equipment (predisposition only) 27
3.4.4 Radars equipment 28

3.4.5 Odo sensor equipment 28
3.4.6 Euroantenna equipment 29
3.4.7 TVM sensors equipment 29
3.4.8 Profibus network equipment 30

4 Functional requirements 31

4.1 Train Protection 31
4.1.1 Drivers Isolation Device 31
4.1.1.1 Function Design 31
4.1.1.2 Normal condition 31
4.1.1.3 Failure cases 31
4.1.1.4 Failure detection 31
4.1.1.5 Failure handling 31
4.1.1.6 Requirements 32
4.1.1.7 Isolation Selectors 40
4.1.1.8 Start up Mission Selectors 43
4.1.1.9 Toggle Switch and Pushbuttons 44
4.1.2 Emergency Brake Control 46
4.1.2.1 Emergency Brake Function Test 46
4.1.2.2 VACMA Emergency Brake Function 48
4.1.2.3 RPS Emergency Brake Function 53
4.1.2.4 KVB Emergency Brake Function 56
4.1.2.5 TVM Emergency Brake Function 59
4.1.3 Train Input to ATP 63
4.1.3.1 Function design 63
4.1.3.2 Requirements 64
4.1.4 ATP output Train Technical Functions 65
4.1.4.1 TVM supervision 65
4.1.4.2 KVB supervision 73
4.1.5 Phase Transition 78
4.1.5.1 Phase Transition with KVB Sequence and signalling 78
4.1.5.2 Phase Transition with TVM Manual Sequence and Signalling 82
4.1.5.3 Phase Transition with TVM Automatic Sequence and signalling 84
4.1.6 Voltage Transition 88
4.1.6.1 Voltage Transition with KVB Sequence and signalling 88
4.1.6.2 Voltage Transition with TVM Manual Sequence and signalling 94
4.1.6.3 Voltage Transition with TVM Automatic Sequence and signalling 97

4.2 Juridical Recorder Unit (ATESS) 102
4.2.1 Function design 102
4.2.1.1 Normal condition 102
4.2.1.2 Failure Cases 102
4.2.1.3 Failure Detection 102
4.2.1.4 Failure Handling 102
4.2.1.5 Requirements 102

5 Signal Interface 107

5.1 I/O Interface 107

5.2 Bus Interface 111
5.2.1 TCMS SW to ATESS JRU (MVB) 111

6 References 112

Table List

Table 1: External interfaces synthesis 13
Table 2: Internal interfaces synthesis 14
Table 3: Parameter for the function S7A TVM (brake performance degradation) 33
Table 4: Parameter for the function S7A TVM – ETR1000 application 33
Table 5 :Cabine activation information to ATP 63
Table 6: Direction Control information to ATP 63
Table 7: Parameters Phase and Catenary Changing 66

Figure List

Figure 1:V300ZEFIRO-ETR1000IF 6
Figure 2: ERT1000 Onboard French ATP system and EUROCAB system composition 9
Figure 3:Command Rack and Panel presence in a subsystem 10
Figure 4: KVB system internal composition and KVB system connections 15
Figure 5: Cabin system internal composition and cabin system connections 16
Figure 6: EUROCAB system internal composition and EUROCAB system connections 17
Figure 7: MIO system internal composition and MIO system connections 18
Figure 8: Fans system internal composition and Fans system connections 19
Figure 9: ATESS system internal composition and ATESS system connections 20
Figure 10:ATP MVB network system internal composition and MVB network system connections 21
Figure 11: *ATP MVB layers actors* 21
Figure 12:ATP Profibus network internal composition and Profibus system connections 22
Figure 13:ATP Profibus network subscribers interconnection order 22
Figure 14: ATP Profibus network layers actors 22
Figure 15: GSM-R data system internal composition and GSM-R data system connections 23
Figure 16: KARM-GPS receiver system composition and KARM-GPS receiver system connections 23
Figure 17: Train system composition and train system connections 24
Figure 18: EVC system internal composition and EVC system connections 25
Figure 19:Odometry speed sensors connections to EVC ODMV boards 26
Figure 20: DMI system internal composition and DMI system connections 27
Figure 21: BTM system internal composition and BTM system connections 27
Figure 22: Radars system internal composition and radars system connections 28
Figure 23: Wheel sensor system internal composition and wheel sensors system connections 28
Figure 24: Euroantenna system internal composition and Euroantenna system connections 29
Figure 25:TVM sensors system internal composition and TVM sensors system connections 29
Figure 26:EUROCAB Profibus network internal composition and Profibus system connections 30
Figure 27:EUROCAB Profibus network subscribers interconnection order 30
Figure 28: EUROCAB Profibus network layers actors 30
Figure 29:Phase Transition section signal sequence for switching off the main switch via the KVB. 79
Figure 30:Phase Transition TVM Manual main switch off (TVM 1/p) 82
Figure 31:Phase Transition TVM Automatic main switch off, Announcement TVM (n/p) 84
Figure 32:Phase Transition TVM Automatic main switch off, Execution TVM (n/p) 85
Figure 33:Voltage Transition section signal sequence for lowering of the pantograph via the KVB. 89
Figure 34:Voltage Transition TVM “Manual” with TVM300 94
Figure 35:Voltage Transition TVM Announcement phase Automatic sequence 97
Figure 36:Voltage Transition TVM Execution phase Automatic sequence 98

1 Introduction

This document is a functional specification for the control of the *French ATP Bi-Standard ERTMS/TVM Signalling System* for V300Zefiro Italy- France for Customer Trenitalia.

1.1 Train Configuration

Each V300Zefiro train consists of eight cars in a fixed formation including a driver's cab at each end:

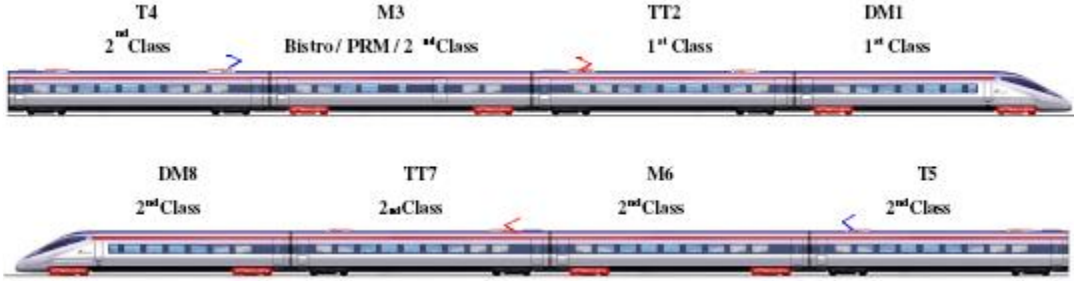


Figure 1:V300ZEFIRO-ETR1000IF

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and Abbreviations

3G	3rd Generation
ACT	Autonomous Compact Terminal
ACK	Acknowledgment
aka	also known as
ASIG	Amplification des Signaux (French words for signal amplification)
STS	Hitachi Rail STS
ATESS	Acquisition Temporelle des Evénements de conduite Statique de Sécurité (French words for temporal acquisition of static driving safety events)
ATP	Automatic Train Protection
AWG	American Wire Gauge
BRS	Brush function (= BR)
BTM	Balise Transmission Module
CB	Circuit Breaker
CCTE	Carte Compacte de Traitement ERTMS (French words for ERTMS processing compact board)
CD	Circuit Diagram
CSSP	Carte de Sortie Série Parallèle (French words for KVB extra logical output board)
CTODL	Current Time Odometric Data Line
CTV	Compatibilité avec la TéléVision embarquée (French words for embedded television compatibility)
DC	Direct Curent
DIS	Driver Information system
DMI	Driver Machine Interface
DRU	Data Recorder Unit
EB	Emergency Braking
EB	Emergency Brake
ENSTA	Enregistreur Statique (French words for static recorder)
ERTMS	European Rail Traffic Management System
ETCS	European Train Control System
EUROCAB	European Cabin
EVC	European Vital Computer
EVENT	Diagnostics that will be stored in the Event database within TCMS SW
FFFIS	Form Fit Function Interface Specifications
FU	Freinage d'Urgence (French words for emergency braking)
GPS	Global Positioning System
GPS – KARM	KARM (TVM arming control) module with a GPS receiver (N.A.)
GSM-R	Global System for Mobile communication – Railways
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps
I/O	Input/Output
IDU	Interface Display Unit
JRU	Juridical Recording Unit
KARM	Contrôle (K) d'Armement de la TVM (French words for TVM arming control)
MIO	I/O Interfaces
KVB	Contrôle (K) de Vitesse par Balise (French words, signalling system in France)
LCB	Line circuit breaker
LTM	Loop Transmission Module
MTORE	Module Tout Ou Rien ERTMS (French words for digital output module)
MVB	Multifunction Vehicle Bus
NC	Normally Closed
NO	Normally Opened
ODMV	Odometrie Mesure de Vitesse (French words for Odometric speed mesurement board)
Profibus	Process Field Bus
PST	Panier Supervision Train (French words for train management rack)
Radar	Radio Detection And Ranging

RGxxx	Radio Guide type xxx (coaxial cable type)
RPS	Répétition Ponctuelle des Signaux (French words for intermittent signal repeater, France = BRS)
RS	Recommanded Standard
RSC	Ripetizione Segnali Continua
SAB	System Architecture Breakdown
SAL	Systeme d'Alerte Lumineux (French words for warning light system)
SB	Service brake
SCMT	Sistema Controllo Marcia Treno
SFRS	Software Functional Requirements Specification
SIS	System Interfaces Specifications
STM	Specific Transmission Module
TBD	To Be Defined
TBL	Transmission Balise Locomotive (French words for track to train transmission, signalling system in Belgium)
TCMS	Train Control and Management System
TVM	Transmission Voie Machine (French words for track to train transmission, signalling system in France and Belgium for high speed lines)
UEVAL	Unité d'Evaluation (French words for KVB processing unit)
UM	Unité Multiple (French words for multiple unit)
US	Unité Simple (French words for single unit)

2 System Overview

2.1 System Configuration

2.1.1 Sub-System Categories

The Onboard ATP system is composed of two subsystem categories:
the signalling subsystems,
the additional subsystems.

The following figure shows for the ETR1000 (or V300ZEFIRO) trains, the different equipment composing the EUROCAB, the onboard ATP and the ATP.

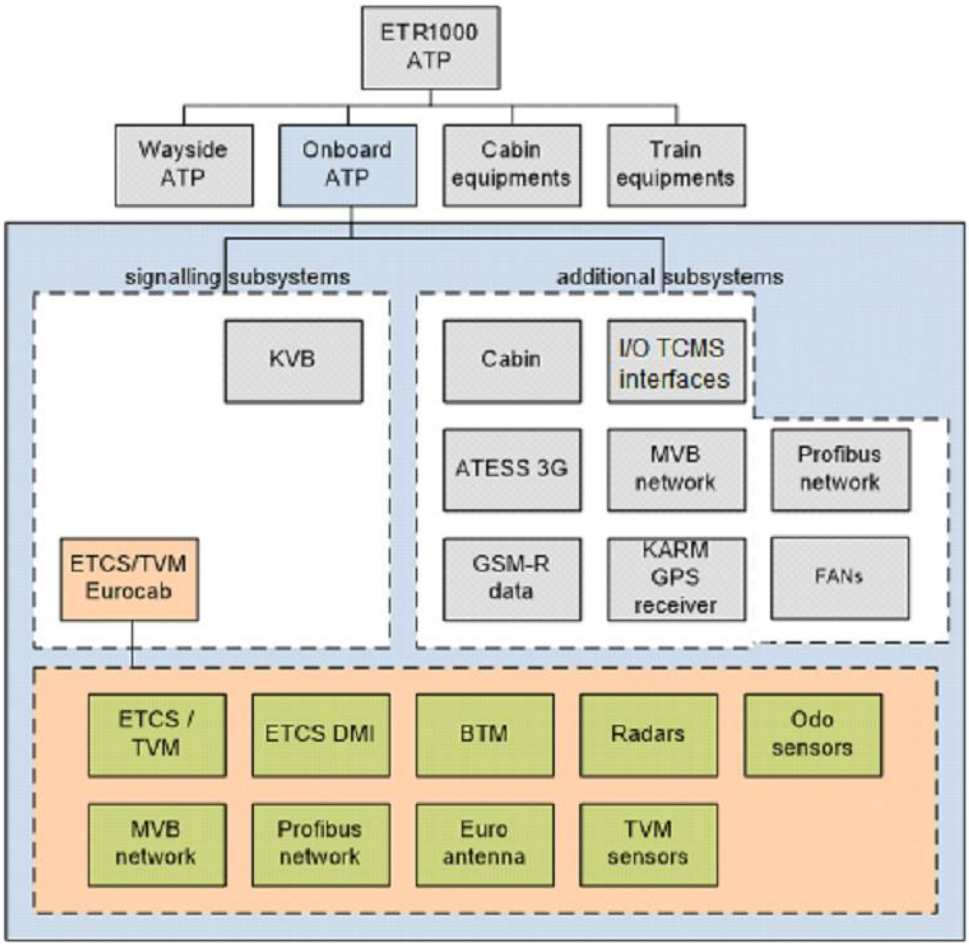


Figure 2: ERT1000 Onboard French ATP system and EUROCAB system composition

2.1.2 Command Rack and Panel

All the useful parts of the different command rack are associated to the considered subsystem.
Considering a sub system, the connections related to the use of the fault switch and the circuit breaker are the following:

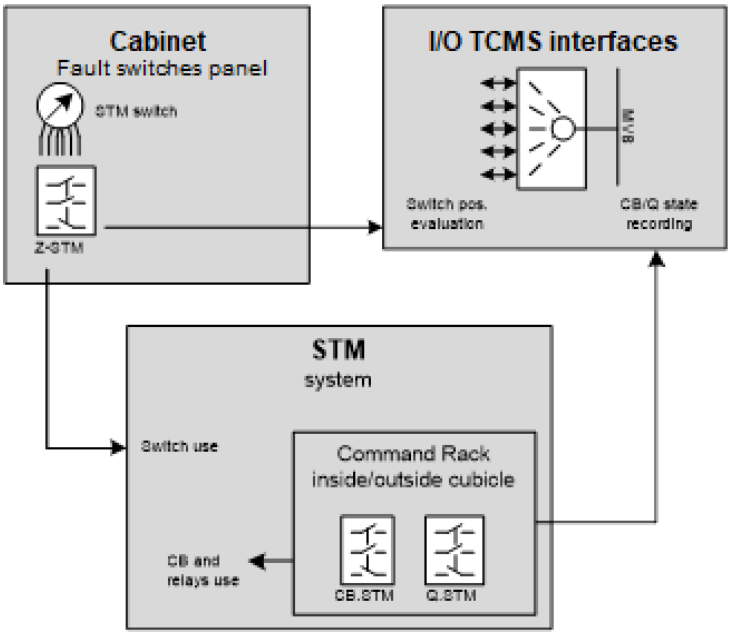


Figure 3:Command Rack and Panel presence in a subsystem

All the selectors are entirely located in the Fault Selector Panel.
There are as many wires exiting from the fault switch panel as needed/required by the different onboard subsystems.

2.2 Signalling Sub-systems

The signalling subsystems category is composed of six equipment.

2.2.1 The KVB equipment

This system is provided by ALSTOM France. It performs KVB signalling functions in France.

2.2.2 The EUROCAB Bi-Standard equipment

This system is provided by ASTS.
It performs TVM signalling functions in France, and ERTMS level 0, 1 and 2 signalling functions wherever a country has lines equipped with operational ERTMS tracks equipment.
It also manages all the transitions from (and to) the national STM equipment.
The subsystems composing the EUROCAB are described in §2.2.2.1.

2.2.2.1 EUROCAB Sub-system

The EUROCAB subsystem is composed of seven equipment.
All the following equipment composing the EUROCAB system are provided by ASTS France.

2.2.2.1.1 The ETCS/TVM (aka EVC) equipment

This equipment is the main part of the EUROCAB subsystem. It oversees managing all the signalling equipment.

2.2.2.1.2 The DMI equipment

This equipment oversees displaying in the cabin all the ETCS signalling information and most of the national systems signalling information to the driver.

2.2.2.1.3 The BTM equipment

This equipment oversees providing the Eurobalise information from the Euroantenna to the EVC.
Note: BTM and Euroantenna are not used in configuration "Reduit National".

2.2.2.1.4 The Radars equipment

This equipment oversees providing odometric information to the EVC.

2.2.2.1.5 The Wheel sensors equipment

This equipment oversees providing odometric information to the EVC.

2.2.2.1.6 The Euroantenna equipment

This equipment oversees receiving the Eurobalises and euroloops data.

2.2.2.1.7 The TVM sensors equipment

This equipment oversees providing TVM information to the EVC.

2.3 Additional Sub-systems

The additional subsystems category is composed of nine equipment.
These equipments are described hereafter:

2.3.1 The Cabin equipment

This equipment provides switches, push buttons and indicators acting on the system (except DMIs).

2.3.2 The Cabinet equipment

Fault switch panel comprises isolation switches and MCBs which are in DM1/8 cab vestibule area.

2.3.3 The MIOs modules equipment

This equipment provides discrete-to-MVB and MVB-to-discrete signal conversion facilities.

2.3.4 The Fans equipment

This equipment represents all the fans used to cool the cubicle 601 and 606 internal environment.

2.3.5 The ATESS equipment

This equipment performs the storage function (ENSTA/JRU).
It also provides:

- The tachymetry function providing the speed to the KVB and to the recorder.
- The RPS or BRS function (French National ATP system).
- The VACMA vigilance function
- The GPS data (time and position) determination function.

2.3.6 The MVB network

The MVB network is considered as pseudo equipment. It is defined here to link more than two equipment together.
The MVB network is composed of the ATP/Train gateways and the terminators

2.3.7 The Profibus network

The Profibus network is considered as pseudo equipment. It is defined here to link more than two equipment together.
The Profibus network is composed of the terminators.

2.3.8 The GSM-R data equipment

This equipment performs all the functions related to the GSM-R DATA transmission between the train and the trackside equipment.
Note: GSM-R Data equipment is not used in configuration "Reduit National".

2.3.9 The KARM GPS receiver equipment

Using an embedded TVM equipped trackside information map and a GPS signal from the ATESS, this equipment provides the KAR/DKAR command used by the EUROCAB for its TVM arming control function.

3 System Interfaces Synthesys

3.1 Presentation

The system interfaces are classified in two different groups corresponding to two different levels of descriptions.
The first group deals with the interfaces between the EUROCAB equipment and the external equipment and between all the external equipment.
The second group deals with the interfaces between the sub-systems composing the EUROCAB system.
Each interface consists of one the following kind of link:

- Discrete I/O, for simple Boolean information.
- Analogue I/O, for sensor information.
- PROFIBUS link, which is a standard bus.
- MVB link, which is a standard bus.
- Serial links (RS232 / RS422 / RS485).
- Other specific interfaces such as coaxial links.

3.2 Interface Tables

The two following tables (external interfaces and internal interfaces tables) are “oriented”.
It means that the cross stating that it exists information transiting from system A to system B is different from the cross stating that it exists information transiting from system B to system A.
The right turning arrow at the left top corner is present to clarify the orientation.
The source systems, from which data is coming, are listed vertically.
The destination systems, to which data is going, are listed horizontally.

3.2.1 External Interfaces Synthesis

The following table shows for which couple of equipment, an external interface exists:

- between the EUROCAB equipment and the external equipment (green cells)
- between the external equipment (grey cells)

		Onboard ATP (signalling)		Onboard ATP additional								Train Equ.	
		1	2	7	8	9	A	B	C	D	E	F	G
		ERTMS/TVM Eurocab	KVB	Cabin	ATESS	GSM-R data	MIO	MVB network	Profibus network	KARM GPS receiver	Fans	Radio (phonie)	Train
1	ERTMS/TVM Eurocab		x		x	x	x	x	x			x	x
2	KVB	x			x		x					x	x
7	Cabin	x	x		x		x			x			
8	ATESS	x	x				x	x	x	x			x
9	GSM-R data	x			x					x			
A	MIO				x			x					
B	MVB network	x			x		x						
C	Profibus network	x			x								
D	KARM GPS receiver	x					x						
E	Fans						x						
F	Radio (phonie)	x	x										
G	Train	x											

Table 1: External interfaces synthesis

3.2.2 Internal Interfaces Synthesis

The following table shows the interfaces between the sub-systems composing the EUROCAB system.

		1	2	3	4	5	6	7	8	9
		ETCS/TVM EVC	ETCS DMI	BTM	Radars	Wheel sensors	Euro antenna	TVM sensors	MVB network	Profibus network
1	ETCS/TVM EVC		x	x					x	x
2	ETCS DMI	x							x	
3	BTM						x			x
4	Radars	x								
5	Wheel sensors	x								
6	Euro antenna			x						
7	TVM sensors	x								
8	MVB network	x	x							
9	Profibus network	x		x						

Table 2*:* Internal interfaces synthesis

3.3 EUROCAB-level interfaces in ATP

This section is listing the external interfaces, for each of the subsystems composing the ATP system.
When the interface description is already defined in another section, the “(link)” suffix is added to the interface title.
In this case, follow the link to read the interface description.

3.3.1 KVB equipment

The KVB system is composed of several equipment located in different parts of the train.
From this document point of view, the modules composing the KVB system are shown below:

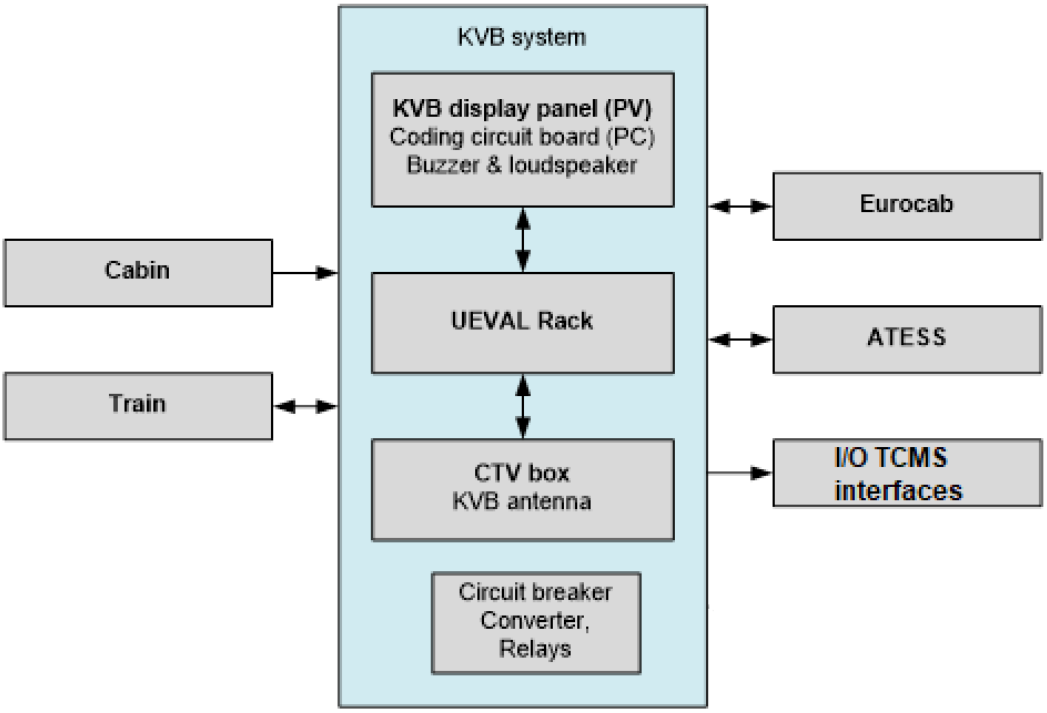


Figure 4: KVB system internal composition and KVB system connections

The Z-KVB selector is not considered as part of KVB system. It is considered as part of the Cabinet equipment.
The KVB system is connected to the following systems:

the Cabinet, for the fault switch position use,
the Train, for the braking system and the US/UM (predisposition only) info use
the EUROCAB for the TVM info use,
the ATESS for recording of some push buttons and indicators,
the MIOs modules for the MVB diffusion of KVB data and circuit breaker states.

It includes some relays located in the Command Rack (for CSSP, SO, SF, VAL, FRAB, INHRPS).
It also includes some circuit breaker located in the Command Rack or in the train whose states are read by MIOs and forwarded to end receivers via MVB.

3.3.2 Cabin equipment

The Cabin system is composed of several equipment located in the train cabin.
From this document point of view, the two modules composing the cabin system are shown below:

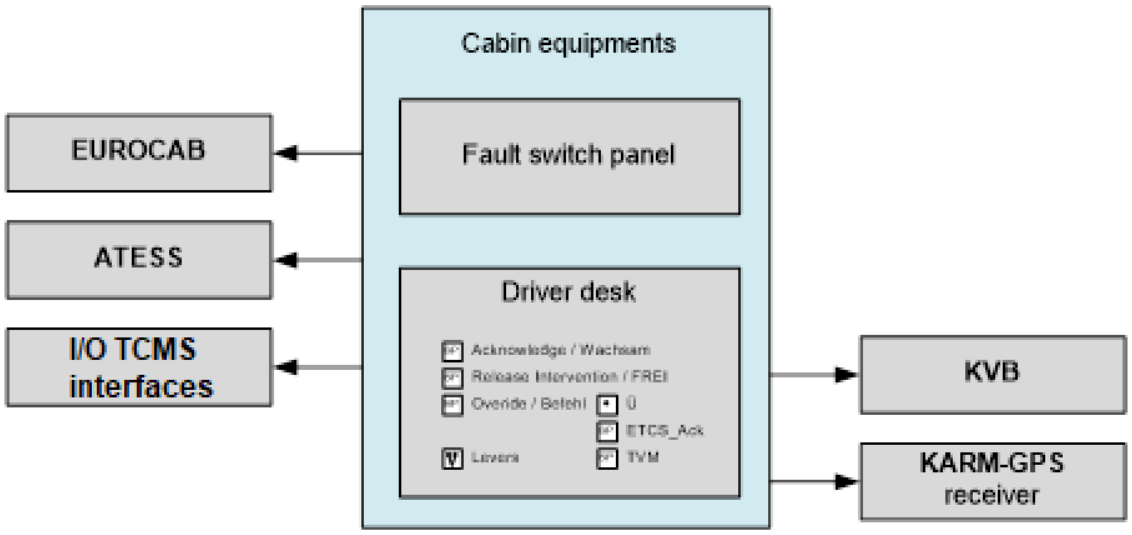


Figure 5: Cabin system internal composition and cabin system connections

The DMI screen present in the driver desk is not considered as part of the Cabin system. It is considered as part of the EUROCAB system.
The KVB display panel present in the driver desk is not considered as part of the Cabin system. It is considered as part of the KVB system.
The Cabin system is connected to the following equipment:

the EUROCAB system, for fault switch position use, the TVM DM/VOIE1/VOIE2 and ETCS_Acknowledge push buttons,

the ATESS system, for fault switch position use, the “release intervention” button used for RPS to reenergize the self-holding of the Q-RPS relay and the Acknowledge/VACMA push button, also the direction controller for recording.

the KVB system, for the fault switch position use and the “Release Intervention” pushbutton,

the MIOs system, for the fault switches position recording and the push button position and direction controller evaluation,

3.3.3 EUROCAB equipment

The EUROCAB system is composed of several equipment's located in different parts of the train.
From this document point of view, the modules composing the EUROCAB system are shown below:

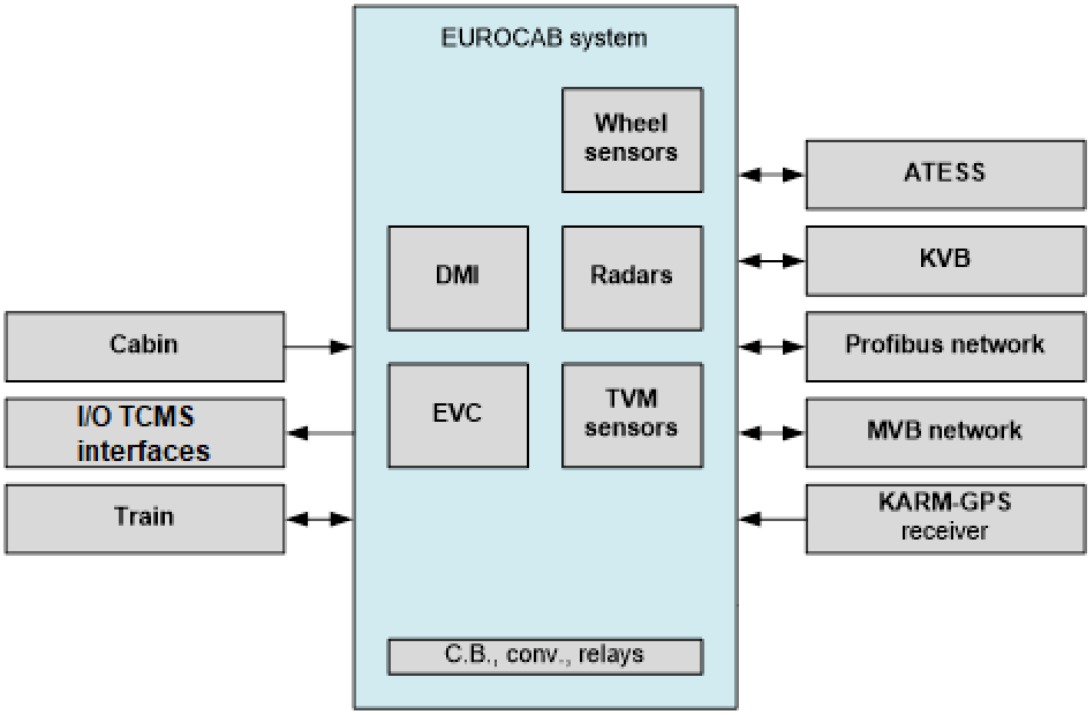


Figure 6: EUROCAB system internal composition and EUROCAB system connections

The Z(FU)ETCS, Z(SEL)DMI, Z-EXPL, Z-Bistandard, Z-KARM, Z-ETCS, Z(ES)TEST, and Z(CO)DJE switches are not considered as part of EUROCAB system. They are considered as part of the Cabin equipment's.
The EUROCAB system is connected to the following systems:

- the Cabinet, for the fault switch position use,
- the MIOs modules for the MVB diffusion of EUROCAB data and circuit breakers states.
- the Train, for the braking system use,
- the ATESS, for the brush info, the JRU and the odometry data,
- the KVB, for the TVM info use, the TVM arming control command and the KVB VI command,
- the Profibus network for the Profibus communications,
- the KARM-GPS receiver for the TVM arming control command.

It includes some relays located in the command rack (for CSSP, SO, SF, VAL, FRAB, INHRPS, Q1-ETCS, Q2-ETCS, Q_LN). It also includes some circuit breaker located in the command rack or in the train whose states are read by MIOs and forwarded to end receivers via MVB.

3.3.4 MIOs equipment

The MIOs system is composed of several equipment's located in two Cubicles named 601 and 606 in the electronic room of DM1/8 cars.
From this document point of view, the four modules composing the MIOs system are shown below:

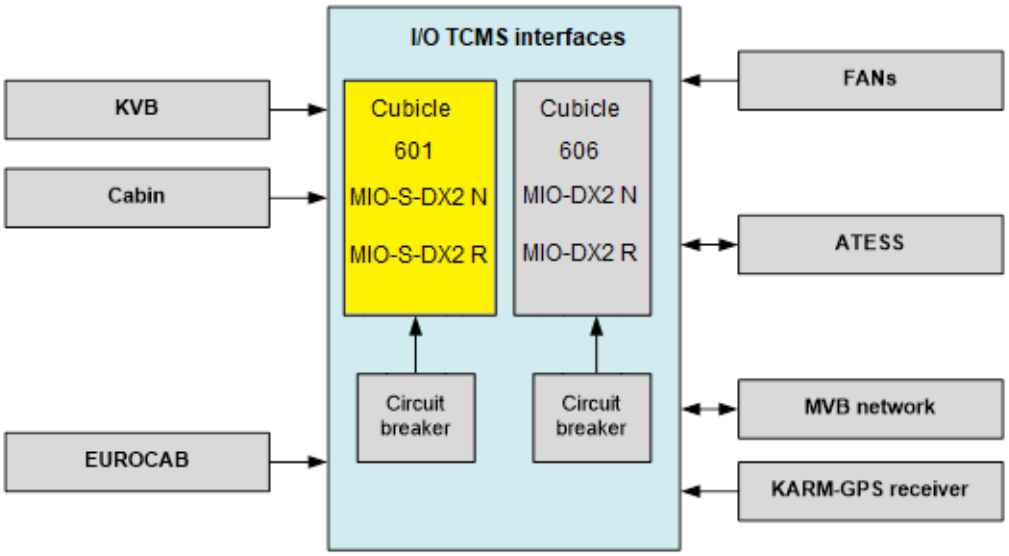


Figure 7: MIO system internal composition and MIO system connections

The MIO system is connected to the following systems:

the KVB, for the MVB diffusion of KVB data and circuit breakers states,
the Cabinet and Cubicles, for the MVB diffusion of the all the fault switches states, the direction controller, some push buttons (Release Intervention, Acknowledge, Override EoA),
the EUROCAB, for the MVB diffusion of EUROCAB data and circuit breakers states,
the Fans, for the MVB diffusion of Fans circuit breakers states,
the ATESS, for the MVB diffusion of ATESS data and circuit breakers states,
the MVB network, for MVB communications.
the KARM-GPS receiver, for the circuit breakers states.

This section of the document only contains links to other sections where the distant equipment defines the interface, except for MIO to MIO and general interfaces.
2 x MIO-S-DX2 (safety module) stands for the MIOs module located in cubicle 601.
2 x MIO-DX2 stands for the MIOs module located in cubicle 606.

The represented circuit breakers are dedicated to MIO usage.

3.3.5 Fans equipment

The Fans system is composed of several equipment located in different cubicle 601 and 606.
From this document point of view, the modules composing the Fans system are shown below:

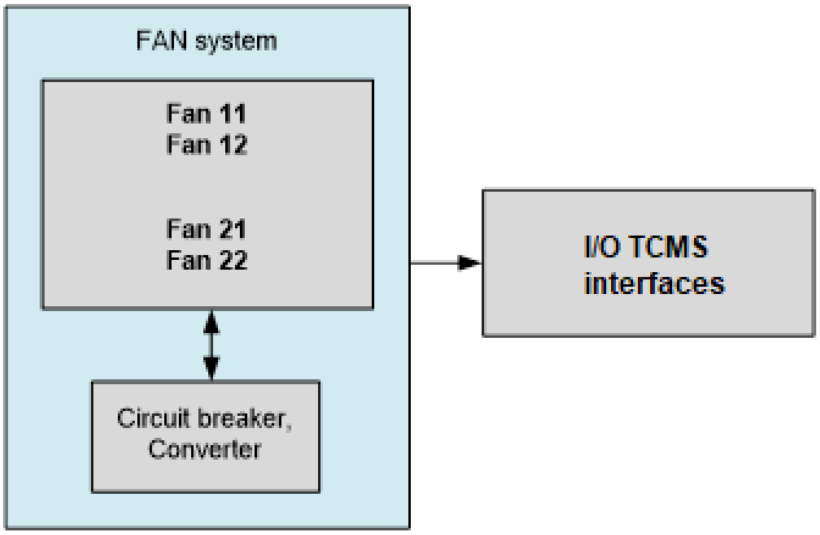


Figure 8: Fans system internal composition and Fans system connections

The Fans system is connected to the MIOs modules for the MVB diffusion of fans circuit breakers states.
The Fans system uses its own internal DC/DC converter from 110VDC to produce 24VDC.
It also includes some circuit breakers located in the command rack whose states are read by MIOs and forwarded to end receivers via MVB.

3.3.6 ATESS equipment

The ATESS system is composed of several equipment's located in different parts of the train.
From this document point of view, the 4 modules composing the ATESS system are shown below:

VACMA
Recorder
BRS
Tachymetry

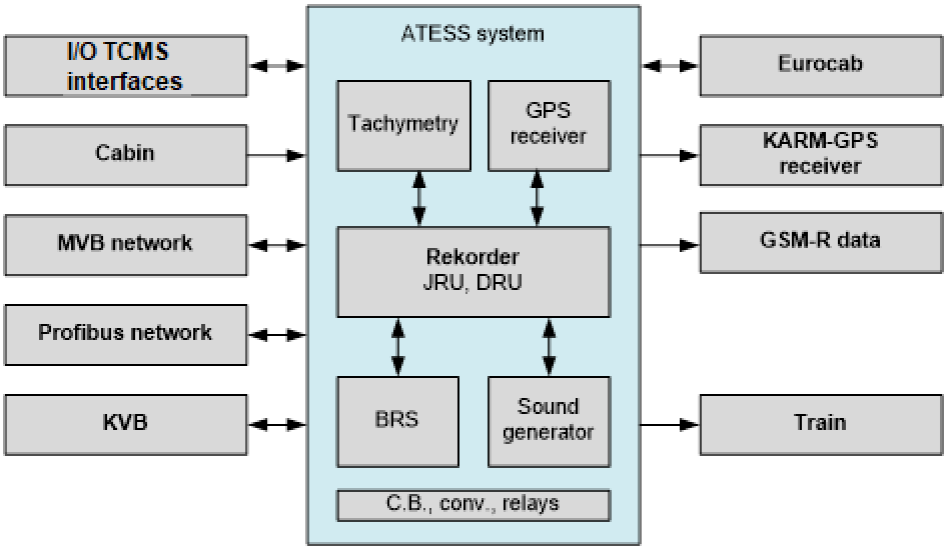


Figure 9: ATESS system internal composition and ATESS system connections

The Z-RS, Z-ENR and Z-VIT switches are not considered as part of ATESS system. They are considered as part of the Cabin equipment's.
The ATESS system is connected to the following systems:

the MIOs modules for the MVB diffusion and reception of ATESS data (discrete inputs for recording) and circuit breakers states.
the Cabin, for the fault switch position use, direction controller and some push button directly connected to ATESS discrete input for data recording, "/override EoA" push button for RPS, "Release information" for reenergizing self-holding or RPS EB relay,
the MVB network, for the MVB communications,
the Profibus network, for the Profibus communications,
the KVB for some push buttons and indicators use,
the EUROCAB, for the brush info use,
the KARM-GPS receiver, for the 3km/h info,
the GSM-R data, for radio communication use,
the Train, for the braking system use,

The ATESS system integrates one 110V DC to 72 DC converter (CV1) used by the RPS sound generator.
It also includes some circuit breakers and relays (Q_RPS) located in the command rack or in the train whose states feedback are read by MIOs and forwarded to end receivers via MVB.
The arbitrary choice has been made to locate BRUSH inside the ATESS subsystem.

3.3.7 MVB network equipment

This section describes the ATP MVB network equipment's of the ATP environment.
The MVB network is considered as pseudo equipment's. It is composed of the ATP/Train gateways and the terminators.

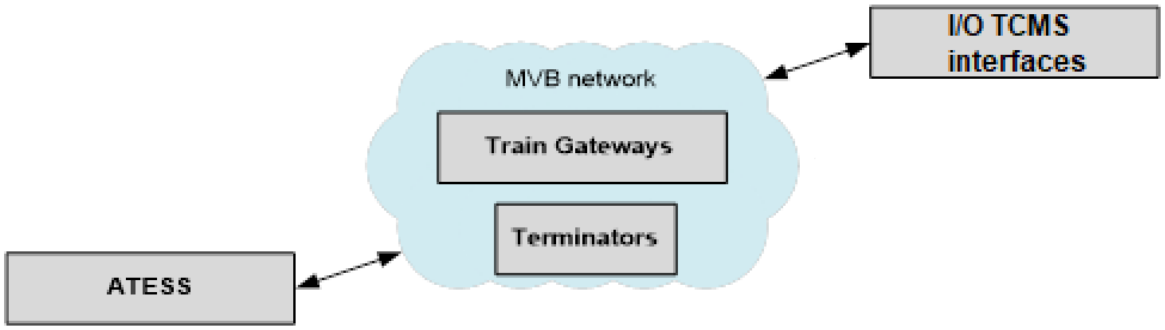


Figure 10:ATP MVB network system internal composition and MVB network system connections

The following figure shows in red color which part of the MVB actors are concerned at this considered level of description.

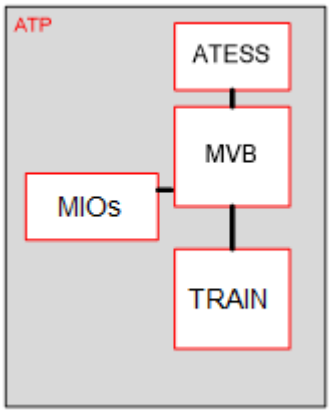


Figure 11*: ATP MVB layers actors*

3.3.8 Profibus network equipment

This section describes the Profibus network equipment's of the ATP environment.
The Profibus network is considered as pseudo equipment. It is composed of the terminators.

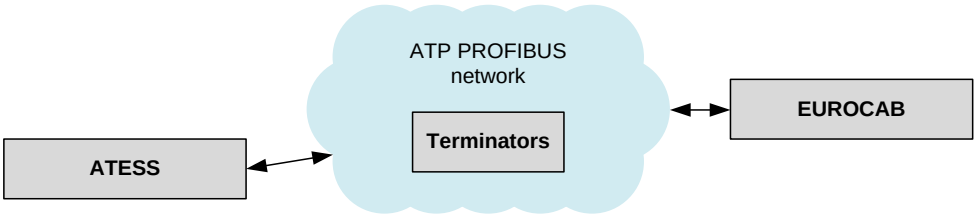


Figure 12:ATP Profibus network internal composition and Profibus system connections

The following figure shows the order in which the ATP profibus subscribers are interconnected in the Profibus chain.

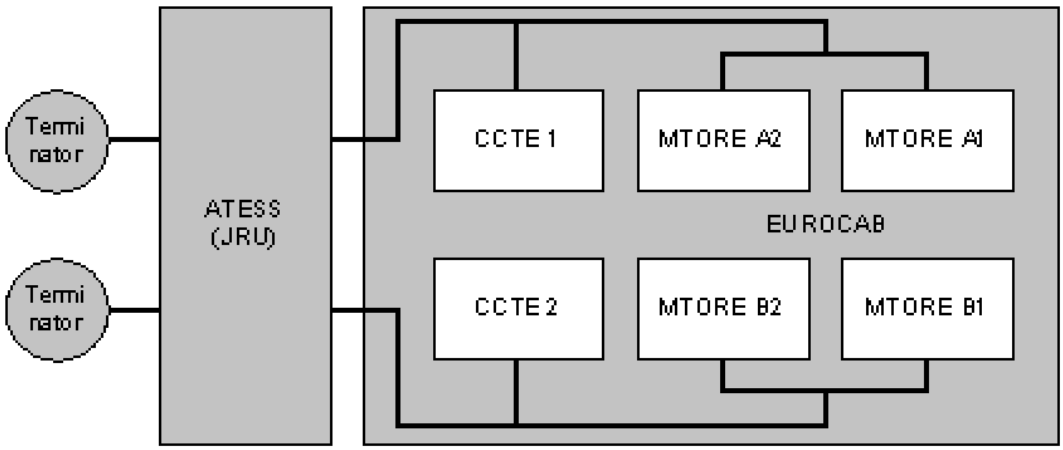


Figure 13:ATP Profibus network subscribers interconnection order

The following figure shows in red color which part of the profibus actors are concerned at this considered level of description.

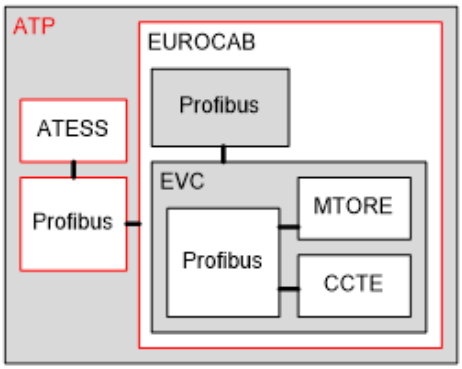


Figure 14: ATP Profibus network layers actors

3.3.9 GSM-R DATA equipment

The GSM-R data system is composed of several equipment located in different parts of the train.
From this document point of view, the modules composing the GSM-R data system are shown below:

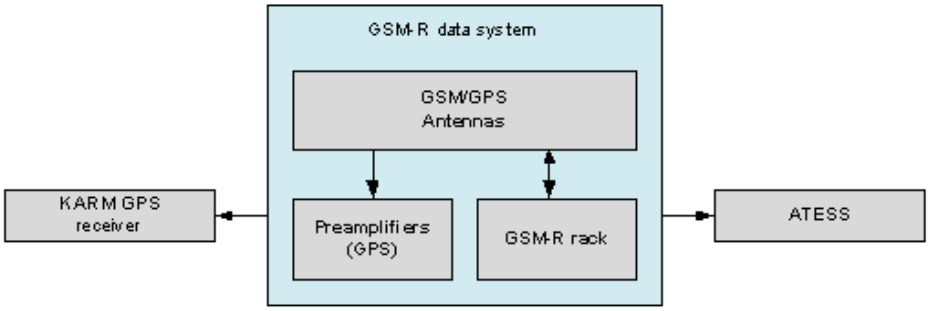


Figure 15: GSM-R data system internal composition and GSM-R data system connections

The GSM-R data system is connected to the following systems:

the KARM-GPS receiver, for GPS data use,
the ATESS, for GPS data use.

The KARM GPS receiver and the ATESS equipment do only use the GPS part of the combined GSM-R(data)/GPS antennas. They are not part of the GSM-R(data) system for ETCS level 2.

3.3.10 KARM-GPS receiver equipment

The KARM-GPS receiver system is composed of one equipment.
From this document point of view, the module composing the KARM-GPS receiver system is shown below:

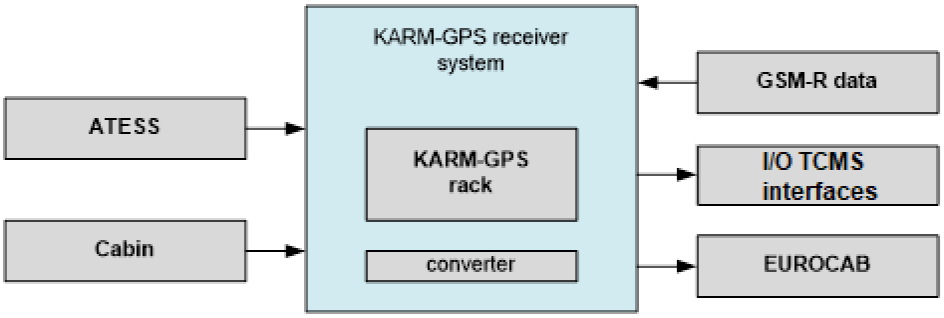


Figure 16: KARM-GPS receiver system composition and KARM-GPS receiver system connections

The KARM-GPS receiver system is connected to the following systems:

the ATESS, for the 3km/h info use,
the GSM-R data, for the GPS data use.
the MIOs modules for the MVB diffusion of KARM-GPS receiver circuit breaker states.

The KARM system integrates one 110V DC to 72 DC converter (CVK).

3.3.11 Train equipment

The Train system is composed of several equipment located outside the cubicle 601 and 606.
From this document point of view, the module composing the Train system is shown below:

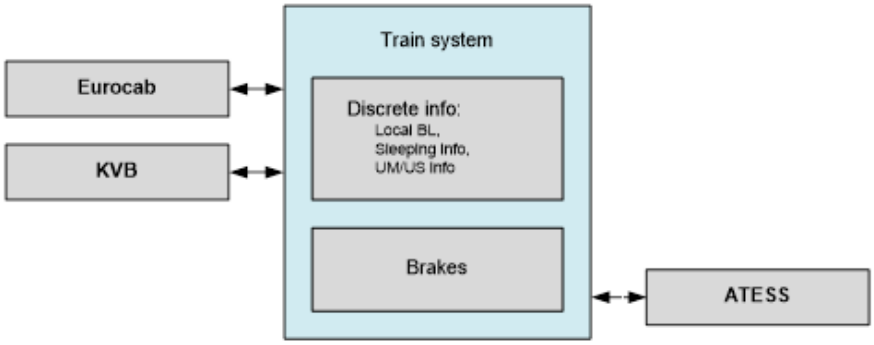


Figure 17: Train system composition and train system connections

3.4 EVC-level interfaces in EUROCAB

This section is listing the external interfaces, for each of the subsystems composing the EUROCAB system.
When the interface description is already defined in another section, the “(link)” suffix is added to the interface title. In this case, follow the link to read the interface description.
The states of the circuit breakers used by the EUROCAB subsystems and recorded by MIOs are not mentioned in this section.
The multiple uses of the switches located in the fault switch panel are also not mentioned in this section.

3.4.1 EVC equipment

From this document point of view, the modules composing the EVC system are shown below:

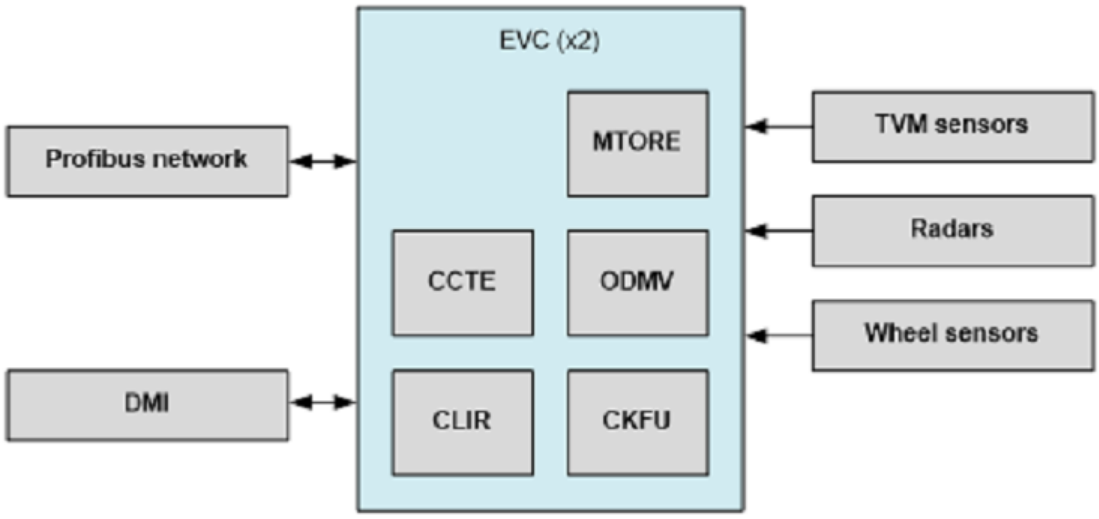


Figure 18: EVC system internal composition and EVC system connections

The EVC system is connected to the following systems:
- the Profibus network, for Profibus communications,
- the DMI, for the communication with the driver,
- the TVM sensor, to get information for TVM,
- the Radars, as source for speed info,
- the Wheel sensors, to get speed info,

The outputs of the odometry speed sensor number 2 are connected directly to the EVC ODMV boards.
The outputs of the odometry speed sensor number 1 are connected to an ASIG box, then connected to the EVC ODMV boards. The ASIG box is a signal amplifier.

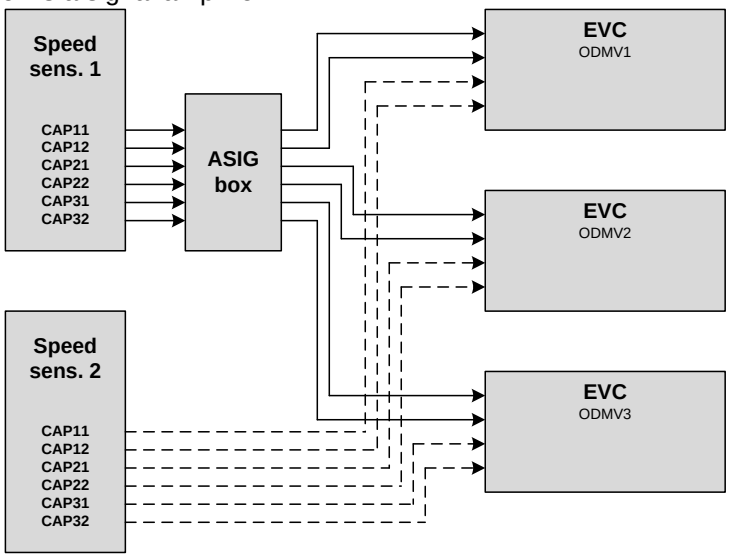


Figure 19:Odometry speed sensors connections to EVC ODMV boards

3.4.2 DMI equipment

From this document point of view, the modules composing the DMI system are shown below:

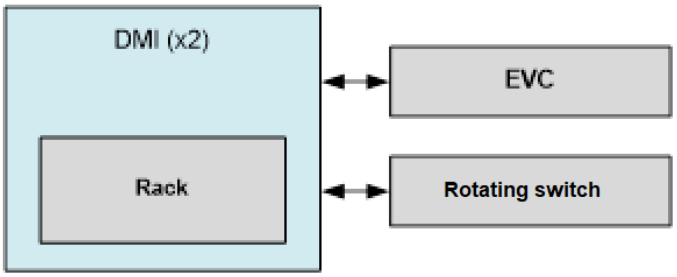


Figure 20: DMI system internal composition and DMI system connections

The DMI system is connected to the following systems:

the EVC, for the handling of EVC communication request with the driver,

3.4.3 BTM equipment (predisposition only)

The BTM system is composed of two equipment located outside the cubicle.
From this document point of view, the module composing the BTM system is shown below:

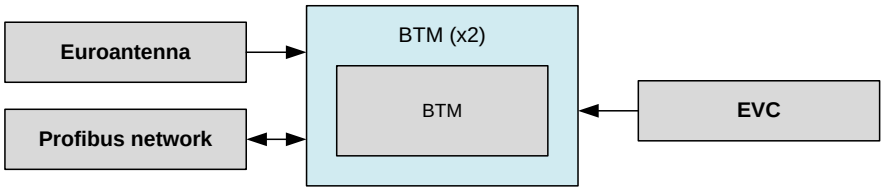


Figure 21: BTM system internal composition and BTM system connections

The BTM system is connected to the following systems:

the Profibus network, for Profibus communications with EVC,
the EVC, to receive synchronization data from CTODL link,
the Euroantenna, to receive Eurobalise signal.

3.4.4 Radars equipment

The radars system is composed of two equipment located outside the cubicle.
From this document point of view, the three modules composing the radars system is shown below:

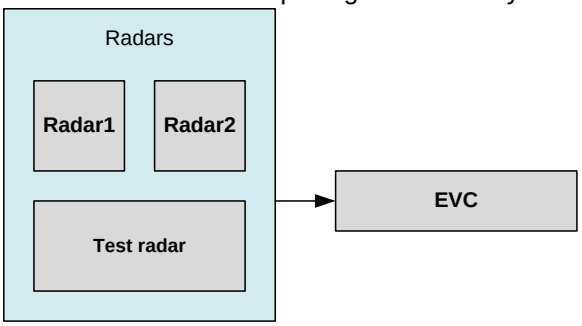


Figure 22: Radars system internal composition and radars system connections

The Radars system is connected to the following systems:

the EVC, to provide ground speed info.

3.4.5 Odo sensor equipment

The wheel sensors system is composed of two equipment located outside the cubicle.
From this document point of view, the three modules composing the radars system is shown below:

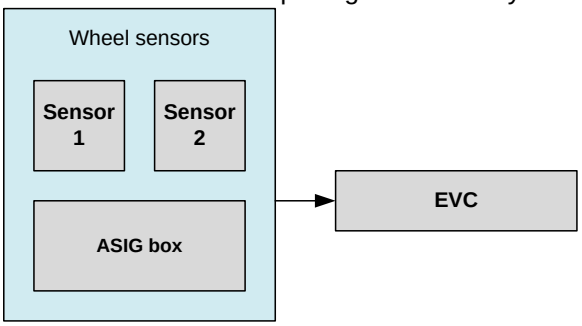


Figure 23: Wheel sensor system internal composition and wheel sensors system connections

The Wheel sensors system is connected to the following systems:

the EVC, to provide speed info.

3.4.6 Euroantenna equipment

The Euroantenna system is composed of one equipment located outside the cubicle.
From this document point of view, the module composing the Euroantenna system is shown below:

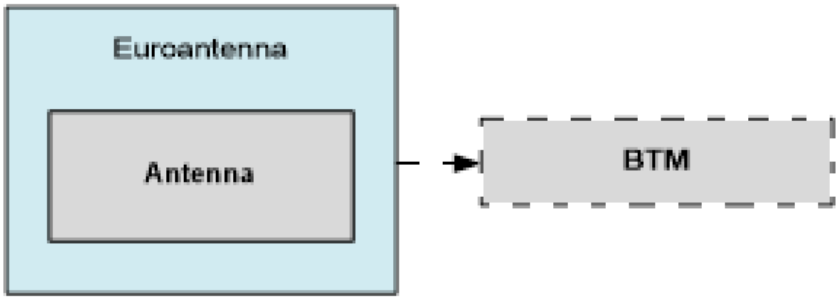


Figure 24: Euroantenna system internal composition and Euroantenna system connections

The Euroantenna system is connected to the following systems:
the BTM, to provide Eurobalise signal to the BTM.

3.4.7 TVM sensors equipment

The TVM sensors system is composed of twelve sensors located outside the cubicle.
From this document point of view, the modules composing the TVM sensors system are shown below:

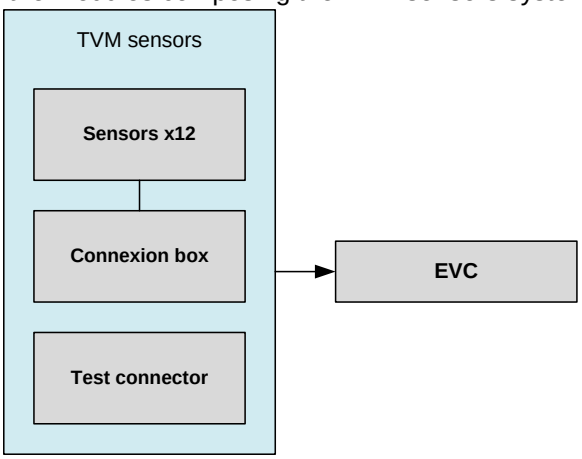


Figure 25:TVM sensors system internal composition and TVM sensors system connections

The TVM sensors system is connected to the following systems:
the EVC, to provide TVM info and TVM stimuli.

3.4.8 Profibus network equipment

This section describes the Profibus network equipment of the EUROCAB environment.
The Profibus network is considered as pseudo equipment. It is composed of the terminators.

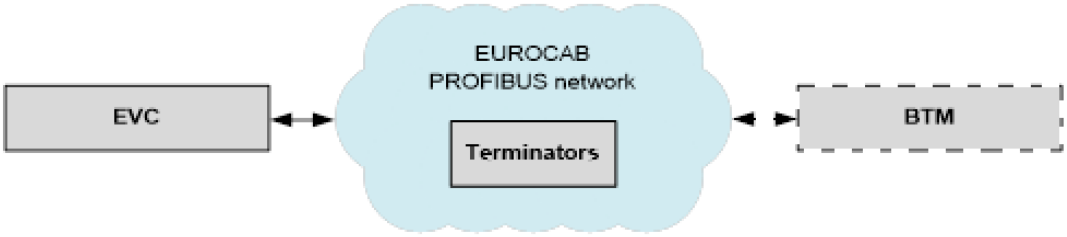


Figure 26:EUROCAB Profibus network internal composition and Profibus system connections

The following figure shows the order in which the EUROCAB Profibus subscribers are interconnected (connections between EUROCAB and BTM are predisposition only).

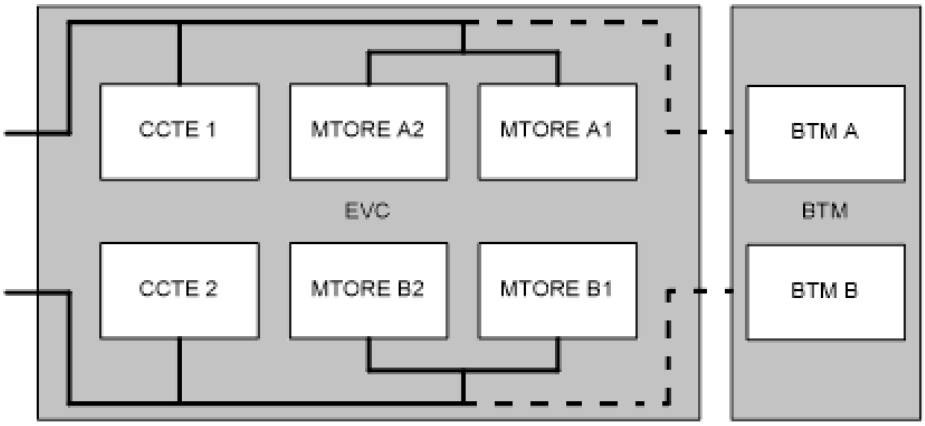


Figure 27:EUROCAB Profibus network subscribers interconnection order

The following figure shows in red color which part of the PROFIBUS actors are concerned at this level of description.

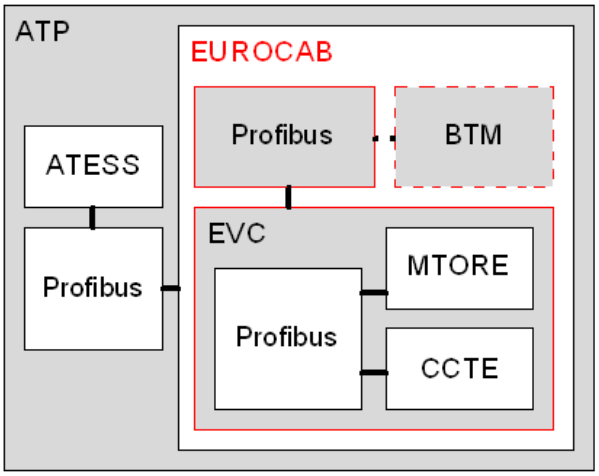


Figure 28: EUROCAB Profibus network layers actors

4 Functional requirements

4.1 Train Protection

4.1.1 Drivers Isolation Device

4.1.1.1 Function Design

ATP France System combine different signalling sub-system, for each one of this in case of faults are foreseen isolation switches by passing the fault.
There are switches that use different degraded modes of this sub-systems of signalling and switches to test and select operating mode of at start of the mission.

4.1.1.2 Normal condition

The driver at the start of the mission perform all test and selection of the switches in the start mission phase.

4.1.1.3 Failure cases

A. Sub-system failure => The Driver operates Isolation switches

4.1.1.4 Failure detection

A. Failure self-evident

4.1.1.5 Failure handling

A. The sub-system shall be isolated and shall be repaired during maintenance

4.1.1.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.264
Description	A consistency check is performed at TCMS level to define if the information acquired by JD-S025 [Z-(BG)IS] selector is consistent with the number of bogie actually isolated
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.171	TCMS shall read the status of Bogie isolation selector Z-(BG)IS **Pos. 0** (BSTD_DATAENTRYN_1=1, BSTD_DATAENTRYN_2=1, BSTD_DATAENTRYN_3=1) & (BSTD_DATAENTRYR_1=1, BSTD_DATAENTRYR_2=1, BSTD_DATAENTRYR_3=1) **Pos. 1** (BSTD_DATAENTRYN_1=0, BSTD_DATAENTRYN_2=1, BSTD_DATAENTRYN_3=1) & (BSTD_DATAENTRYR_1=0, BSTD_DATAENTRYR_2=1, BSTD_DATAENTRYR_3=1) **Pos. 2** (BSTD_DATAENTRYN_1=1, BSTD_DATAENTRYN_2=0, BSTD_DATAENTRYN_3=1) & (BSTD_DATAENTRYR_1=1, BSTD_DATAENTRYR_2=0, BSTD_DATAENTRYR_3=1) **Pos. 3** (BSTD_DATAENTRYN_1=0, BSTD_DATAENTRYN_2=0, BSTD_DATAENTRYN_3=1) & (BSTD_DATAENTRYR_1=0, BSTD_DATAENTRYR_2=0, BSTD_DATAENTRYR_3=1) **Pos. 4** (BSTD_DATAENTRYN_1=1, BSTD_DATAENTRYN_2=1, BSTD_DATAENTRYN_3=0) & (BSTD_DATAENTRYR_1=1, BSTD_DATAENTRYR_2=1, BSTD_DATAENTRYR_3=0) **Pos. 5** (BSTD_DATAENTRYN_1=0, BSTD_DATAENTRYN_2=1, BSTD_DATAENTRYN_3=0) & (BSTD_DATAENTRYR_1=0, BSTD_DATAENTRYR_2=1, BSTD_DATAENTRYR_3=0) **Pos. 6** (BSTD_DATAENTRYN_1=1, BSTD_DATAENTRYN_2=0, BSTD_DATAENTRYN_3=0) & (BSTD_DATAENTRYR_1=1, BSTD_DATAENTRYR_2=0, BSTD_DATAENTRYR_3=0) **Pos. 7** (BSTD_DATAENTRYN_1=0, BSTD_DATAENTRYN_2=0, BSTD_DATAENTRYN_3=0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

	& (BSTD_DATAENTRYR_1=0, BSTD_DATAENTRYR_2=0, BSTD_DATAENTRYR_3=0)				
2F_09.02.Zefiro-IF.CONCEPT.172	TCMS shall check if selector position is consistent with the number of bogie actually isolated according to the table 4 (row related to the position versus number of isolated bogies).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.972	In case of inconsistency from COMPCEPT.172, TCMS shall send a diagnose message to DMI.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

N° Application TVM	Z-(BG) IS (JD-S025)							
	Position 1 Number 0 on the slelector	Position 2 Number 1 on the slelector	Position 3 Number 2 on the slelector	Position 4 Number 3 on the slelector	Position 5 Number 4 on the slelector	Position 6 Number 5 on the slelector	Position 7 Number 6 on the slelector	Position 8 Number 7 on the slelector
1	Vmax	160 km/h	160 km/h	Vrescue	Vrescue	Vrescue	Vrescue	Vrescue
2	Vmax	160 km/h	160 km/h	Vrescue	Vrescue	Vrescue	Vrescue	Vrescue
3	Vmax	160 km/h	160 km/h	Vrescue	Vrescue	Vrescue	Vrescue	Vrescue
4	Vmax	160 km/h	160 km/h	Vrescue	Vrescue	Vrescue	Vrescue	Vrescue
5								
6								
7								

Table 3: Parameter for the function S7A TVM (brake performance degradation)

N° Appli TVM	Position 1 - NO BG (*) isolated - N° 1 BG isolated - N° 2 BG isolated	Position 2 N° 3 BG isolated	Position 3 N° 4 BG isolated	Position 4 > N° 4 BG isolated	Position 5 > N° 4 BG isolated	Position 6 > N° 4 BG isolated	Position 7 > N° 4 BG isolated	Position 8 > N° 4 BG isolated
2	*300 km/h*	*160 km/h*	*160 km/h*	*20 km/h*	*20 km/h*	*20 km/h*	*20 km/h*	*20 km/h*

Table 4: Parameter for the function S7A TVM – ETR1000 application

Note: in case of multiple configuration, speed limit defined in the table 4 will be defined later.

ID	2F_09.02.Zefiro-Europe.TRS.303
Description	A check is performed at TCMS level to define the JD-S025 [Z-(BG)IS] rotating selector misalignment between the redundancy contacts by digital inputs control
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.664	TCMS shall read the status of Z-(BG)IS selector BSTD_DATAENTRYN_1 BSTD_DATAENTRYN_2 BSTD_DATAENTRYN_3 and BSTD_DATAENTRYR_1 BSTD_DATAENTRYR_2 BSTD_DATAENTRYR_3	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.665	Compare the digital inputs N with the digital inputs R	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.666	Verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.265
Description	If JD-S013 [Z(CO)DJE] selector is on the position 'A + B Off' then TCMS shall inhibit the KVB-CSSP function (Train Technical Function). See 4.1.1.7.1
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.174	TCMS shall read the status of Z(CO)DJE selector	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.638	TCMS shall read the position 'ON'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.639	TCMS shall read the position 'A Off'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.640	In case TCMS reads Z(CO)DJE selector in the position 'A Off', TCMS shall consider not valid the following PSTDA outputs: Main Switch Off Lower Pantograph Catenary Indication	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.641	TCMS shall read the position 'B Off'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.642	In case TCMS reads Z(CO)DJE selector in the position 'B Off', TCMS shall consider not valid the following PSTDB outputs: Main Switch Off Lower Pantograph Catenary Indication	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.175	TCMS shall read the position 'A + B Off'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.643	In case TCMS reads Z(CO)DJE selector in the position 'A+B Off', TCMS shall consider not valid the following outputs (for PSTDA and PSTDB): Main Switch Off Lower Pantograph Catenary Indication	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.644	In case TCMS reads Z(CO)DJE selector in the position 'A+B Off', TCMS shall inhibit display of the KVB pictograms	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.176	TCMS shall inhibit command through digital output VY-SECT-AUTO	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.177	TCMS shall inhibit command through digital output VY-BPT-KVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.178	TCMS shall inhibit Tunnel Mode command through IP network	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.304
Description	If JD-S020 [Z-ETCS] selector is on the position "A + B Off" or JD-S019 [Z-BISTANDARD] selector is on the position "OFF" then TCMS shall indicated the Emergency Speed Indication
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.668	TCMS shall read the selector Z-BISTANDARD in position 'Off'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.669	TCMS shall read the selector Z-ETCS in position 'A+B Off'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.670	In case TCMS reads Z-BISTANDARD selector in the position 'OFF', TCMS shall indicated on DMI the emergency speed indication (Vrescue)	Derived Requirement	4S_09.03.05.-.TCMS Software	-	SIL0
2F_09.02.Zefiro-IF.CONCEPT.671	In case TCMS reads Z-ETCS selector in the position 'A+B Off', TCMS shall indicated on DMI the emergency speed indication (Vrescue)	Derived Requirement	4S_09.03.05.-.TCMS Software	-	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.301
Description	TCMS shall acquire isolation selector status and verify the their coherence
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.180	TCMS shall read the selector Z-KARM position BSTD_KARM_SWON BSTD_KARM_SWOFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF .CONCEPT.181	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.975	If the signals are in the below state BSTD_KARM_SWON=1 BSTD_KARM_SWOFF=0 Then the selector is set in 'On'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.976	If the signals are in the below state BSTD_KARM_SWON=0 BSTD_KARM_SWOFF=1 Then the selector is set in 'Off''	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.977	In case of incoherence between Z-KARM signal position (both signals at zero or at one) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.182	TCMS shall read the selector Z-BISTANDARD position BSTD_ETCSGEN_SWON BSTD_ETCSGEN_SWOFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.183	If the signals are in the below state BSTD_ETCSGEN_SWON=1 BSTD_ETCSGEN_SWOFF=0 Then the selector is set in 'On'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.978	If the signals are in the below state BSTD_ETCSGEN_SWON=0 BSTD_ETCSGEN_SWOFF=1 Then the selector is set in 'Off''	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.979	In case of incoherence between Z-BISTANDARD signals position (both signals at zero or at one) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.672	TCMS shall read both input signals of the rotating selector Z(SEL)DMI signals BSTD_SELDMI_1 BSTD_SELDMI_2	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.980	If the signals are in the below state BSTD_SELDMI_1=1 BSTD_SELDMI_2=0 Then the selector is set in 'Normal''(left TVM DMI is On)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.981	If the signals are in the below state BSTD_SELDMI_1=0 BSTD_SELDMI_2=1 Then the selector is set in 'Off'' Then the selector is set in 'Right''(right TVM DMI is On)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.982	In case of incoherence between Z(SEL)DMI signals position (both signals at zero or at one) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.983	TCMS shall read through digital input the status of ALA BISTANDARD selector BSTD_SEL_I-F_1 BSTD_SEL_I-F_2 BSTD_SEL_I-F_3 selector postions	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

2F_09.02.Zefiro-IF.CONCEPT.984	TCMS shall read through digital input the status of Z(CO)DJE selector BSTD_CODJE_SWAOFF BSTD_CODJE_SWN BSTD_CODJE_SWBOFF BSTD_CODJE_SWABOFF	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.985	If the signals are in the below state BSTD_CODJE_SWAOFF=1 BSTD_CODJE_SWN=0 BSTD_CODJE_SWBOFF=0 BSTD_CODJE_SWABOFF=0 Then the selector is set in 'A-Off'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.986	If the signals are in the below state BSTD_CODJE_SWAOFF=0 BSTD_CODJE_SWN=1 BSTD_CODJE_SWBOFF=0 BSTD_CODJE_SWABOFF=0 Then the selector is set in 'Normal'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.987	If the signals are in the below state BSTD_CODJE_SWAOFF=0 BSTD_CODJE_SWN=0 BSTD_CODJE_SWBOFF=1 BSTD_CODJE_SWABOFF=0 Then the selector is set in 'B-Off'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.988	If the signals are in the below state BSTD_CODJE_SWAOFF=0 BSTD_CODJE_SWN=0 BSTD_CODJE_SWBOFF=0 BSTD_CODJE_SWABOFF=1 Then the selector is set in 'A+B-Off'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.989	For all different sequences of Z(CO)DJE other than the concepts 985, 986, 987, 988 (see above), TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.990	TCMS shall read through digital input the signal status of selector Z-VIT BSTD_ATESS_SWON BSTD_ATESS_SWOFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.991	If the signals are in the below state BSTD_ATESS_SWON =1 BSTD_ATESS_SWOFF =0 Then the selector is set in 'ON''(ATESS is On)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.992	If the signals are in the below state BSTD_ATESS_SWON =0 BSTD_ATESS_SWOFF =1 Then the selector is set in 'OFF''(ATESS is OFF)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.993	In case of incoherence between Z-VIT signals position (both signals at zero or at one) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.994	TCMS shall read through digital input signals the status of selector Z(FU)ETCS BSTD_EBL_SW1OFF BSTD_EBL_SWON BSTD_EBL_SW2OFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.995	If the signals are in the below state BSTD_EBL_SW1OFF=1 BSTD_EBL_SWON=0 BSTD_EBL_SW2OFF=0 Then the selector is set in ''1-OFF'' ETCS1 is excluded	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.996	If the signals are in the below state BSTD_EBL_SW1OFF=0 BSTD_EBL_SWON=1 BSTD_EBL_SW2OFF=0 Then the selector is set in ''ON'' ETCS1-2 are inserted	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.997	If the signals are in the below state BSTD_EBL_SW1OFF=0 BSTD_EBL_SWON=0 BSTD_EBL_SW2OFF=1 Then the selector is set in ''2-OFF'' ETCS2 is excluded	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

2F_09.02.Zefiro-IF.CONCEPT.998	For all different sequences of Z(FU)ETCS other than the concepts 995, 996, 997(see above), TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.999	TCMS shall read through digital input signals of the selector Z-EXPL (ETCS MODE) BSTD_ETCSMO_SWOFF BSTD_ETCSMO_SWON	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1000	If the signals are in the below state BSTD_ETCSMO_SWON=1 BSTD_ETCSMO_SWOFF =0 Then the selector is set in ‘‘Inserted’’, ERTMS/TVM are inserted	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1001	If the signals are in the below state BSTD_ETCSMO_SWON=0 BSTD_ETCSMO_SWOFF =1 Then the selector is set in ‘‘Excluded’’, ERTMS/TVM are exluded and TVM is inserted	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1002	In case of incoherence between Z-EXPL signals position (both signals at zero or at one) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1003	TCMS shall read through digital input the status of Z(ES)ETCS selector postions signals BSTD_ETCSTVMTEST_ET BSTD_ETCSTVMTEST_N BSTD_ETCSTVMTEST_TVM BSTD_ETCSTVMTEST_ODM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT. 1004	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.1005	If the signals are in the below state BSTD_ETCSTVMTEST_ET=1 BSTD_ETCSTVMTEST_N=0 BSTD_ETCSTVMTEST_TVM=0 BSTD_ETCSTVMTEST_ODM=0 Then the selector is set in ‘‘ET’’, ETCS test is inserted	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1006	If the signals are in the below state BSTD_ETCSTVMTEST_ET=0 BSTD_ETCSTVMTEST_N=1 BSTD_ETCSTVMTEST_TVM=0 BSTD_ETCSTVMTEST_ODM=0 Then the selector is set in ‘‘NORMAL’’, normal mode (no test).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1007	If the signals are in the below state BSTD_ETCSTVMTEST_ET=0 BSTD_ETCSTVMTEST_N=0 BSTD_ETCSTVMTEST_TVM=1 BSTD_ETCSTVMTEST_ODM=0 Then the selector is set in ‘‘TVM’’, TVM test is inserted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1008	If the signals are in the below state BSTD_ETCSTVMTEST_ET=0 BSTD_ETCSTVMTEST_N=0 BSTD_ETCSTVMTEST_TVM=0 BSTD_ETCSTVMTEST_ODM=1 the selector is set in ‘‘Odometrie’’, Odometrie test is inserted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1009	For all different sequences of Z(CO)DJE other than the concepts 1005, 1006, 1007, 1008 (see above), TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.305
Description	TCMS shall indicated the status of which track arming
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.674	TCMS shall read the input signal BSTD_TVM_VOIE1 and show on the TCMS DMI, the status on which TVM system has been manual arming on track 1	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.675	TCMS shall read the input signal BSTD_TVM_VOIE2 and show on the TCMS DMI, the status on which TVM system has been manual arming on track 2	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.676	TCMS shall read the input signal BSTD_TVM_VOIEDM and show on the TCMS DMI, the status on which TVM system hasn't been manual arming	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.677	TCMS shall read the input signal BSTD_QLN-2_RELAY and show on the TCMS DMI, the status of TVM arming	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.678	TCMS shall read the input signal BSTD_QDEFKARM_DIA and show on the TCMS DMI, the status of TVM arming control default	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.306
Description	TCMS shall indicated the status of ETCS_Acknowledge
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.680	TCMS shall read the input signals BSTD_ETCS_ACK_N BSTD_ETCS_ACK_R	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1010	If the signals are in the below state BSTD_ETCS_ACK_N=1 BSTD_ETCS_ACK_R=1 the ETCS_Acknowledge is requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1011	If the signals are in the below state BSTD_ETCS_ACK_N=0 BSTD_ETCS_ACK_R=0 the ETCS_Acknowledge is not requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1012	In case of incoherence between acknowledge signals BSTD_ETCS_ACK_N=0 BSTD_ETCS_ACK_R=1 or BSTD_ETCS_ACK_N=1 BSTD_ETCS_ACK_R=0 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.307
Description	TCMS shall indicated if ones or more circuit breakers of the signalling circuit are tripped in the cabinet and in the command racks
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.682	TCMS shall read the input signal BSTD_609A_FCBAUX of the daisy chain and verify that no circuit breaker of the cabinet 609a tripped	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.683	TCMS shall read the input signal BSTD_COM1_FCBAUX of the daisy chain and verify that no circuit breaker of the command rack 1 tripped	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.684	TCMS shall read the input signal BSTD_COM2_FCBAUX of the daisy chain and verify that no circuit breaker of the command rack 2 tripped	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.685	TCMS shall read the input signal BSTD_COM3_FCBAUX of the daisy chain and verify that no circuit breaker of the command rack 3 tripped	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.308
Description	TCMS shall indicate if the cubicles ventilation is in failure
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.687	TCMS shall read the input signal BSTD_FAN601 and shall show on TCMS DMI the message of the ventilation failure of the cubicle 601	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.688	TCMS shall read the input signal BSTD_FAN606 and shall show on TCMS DMI the message of the ventilation failure of the cubicle 606	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.309
Description	TCMS shall indicated the status of the converter 110/72 VDC
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.690	TCMS shall read the input signal BSTD_CVRACK1_FEED and show on the TCMS DMI the status of the converter 110/72 number 1	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.691	TCMS shall read the input signal BSTD_CVRACK2_FEED and show on the TCMS DMI the status of the converter 110/72 number 2	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

4.1.1.7 Isolation Selectors

4.1.1.7.1 Selector JD-S013 - Z(CO)DJE

Sealed on ‘Normal’ position.

4.1.1.7.1.1 Function

Isolation selector for train technical function (main switch off and lower pantograph functions) executed by ERTMS/TVM equipment

4.1.1.7.1.2 Description

Selector on the position ‘On’ shall indicates that train technical functions ‘Main Switch Off’, ‘Lower Pantograph’ and ‘Catenary Indication’ are activated on both channel.
Selector on the position ‘A Off’ shall isolate train technical functions ‘Main Switch Off’, ‘Lower Pantograph’ and ‘Catenary Indication’ on channel A.
Selector on the position ‘B Off’ shall isolate train technical functions ‘Main Switch Off’, ‘Lower Pantograph’ and ‘Catenary Indication’ on channel B.
Selector on the position ‘A + B Off’ shall isolate train technical functions ‘Main Switch Off’, ‘Lower Pantograph’ and ‘Catenary Indication’ on both channel.
Note: This inhibition is executed by the train control system. If there is no distinction at train level between 2 chains or 2 channels, the 2 positions ‘A Off’ and ‘B Off’ are equivalent to the position ‘On’.
If Z(CO)DJE selector on the position ‘A + B Off’ then DMI shall inhibit the display of pictograms related to train technical functions for electrical traction (MSO and LP) in “STM France – TVM” level.
If Z(CO)DJE selector on the position ‘A + B Off’ then TCMS shall inhibit the KVB-CSSP function (Train Technical Function).

4.1.1.7.2 Selector JD-S020 - Z-ETCS/TVM

Sealed on ‘Normal’ position.

4.1.1.7.2.1 Function

-Isolation of ERTMS/TVM equipment

4.1.1.7.2.2 Description

Selector on the position ‘Normal’ shall power on both channel of the Bi-Standard equipment.
Selector on the position ‘Isolation of chain A’ shall isolate only the channel A of the Bi-Standard equipment.
Selector on the position ‘Isolation of chain B’ shall isolate only the channel B of the Bi-Standard equipment.
Selector on the position ‘Isolation of chains A & B’ shall isolate both channels (A and B) of the Bi-Standard equipment.
Note:When operating Z-ETCS/TVM selector, the user shall wait for 10 seconds in Isolation position before re-energization of Bi-standard equipment.

4.1.1.7.3 Selector JD-S015 - Z(FU)ETCS

Sealed on ‘Normal’ position.

4.1.1.7.3.1 Function

Isolation selector of EB channels triggered by ERTMS/TVM equipment

4.1.1.7.3.2 Description

Selector on the position ‘Normal’ shall connect both EB loops (1 and 2) to the Bi-Standard equipment.
Selector on the position ‘Isolation of EB channel 1’ shall isolate EB loop 1 to the Bi-Standard equipment.
Selector on the position ‘Isolation of EB channel 2’ shall isolate EB loop 2 to the Bi-Standard equipment.

4.1.1.7.4 Selector JD-S019 - Z-BISTANDARD

4.1.1.7.4.1 Function

Power off/on of both ETCS/TVM equipment (all components)

4.1.1.7.4.2 Description

Selector on the position ‘Normal’ shall power on all components of the Bi-Standard ERTMS/TVM (i.e. EVC, DMIs and Radars).
Selector on the position ‘Isolation’ shall power off all components of the Bi-Standard ERTMS/TVM.
Note: The Z-BISTANDARD selector is used to restart properly ERTMS/TVM equipment in degraded situation (e.g. ERTMS/TVM in System Failure mode) without stopping other ATP or train equipment and avoids the generation of defects inside the internal diagnosis memory of ERTMS/TVM equipment.

4.1.1.7.5 Selector JD-S027 - Z-VA

Sealed on ‘Normal’ position.

4.1.1.7.5.1 Function

Isolation of VACMA function.
This selector overrides VACMA emergency brake intervention too.

4.1.1.7.5.2 Description

Selector on the position ‘Normal’ shall power on VACMA function.
Selector on the position ‘Isolation’ shall power off VACMA function.By pass VACMA intervention on EB loop

4.1.1.7.6 Selector JD-S025 - Z-(BG)IS

4.1.1.7.6.1 Function

Isolation of brakes bogies

4.1.1.7.6.2 Description

Selector represent the status of the braking performance needed for performing function S7A TVM.
S7A TVM function is used in case of degraded braking perfomance to limit authorized speed.
A consistency check is performed at TCMS level to define if the information acquired by JD-S025 [Z-(BG)IS] selector is consistent with the number of bogie actually isolated.

Bogie isolated	0	1	2	3	4	5	6	7
Data Entry 1	1	0	1	0	1	0	1	0
Data Entry 2	1	1	0	0	1	1	0	0
Data Entry 3	1	1	1	1	0	0	0	0

4.1.1.7.7 Selector JD-S018 - Z-KARM

Sealed on ‘Normal’ position.

4.1.1.7.7.1 Function

Isolation selector for TVM arming control

4.1.1.7.7.2 Description

Selector on the position ‘On’ shall allow KARM function.
Selector on the position ‘Off’ shall inhibit KARM function.

4.1.1.7.8 Selector JD-S017 - Z-KVB

Sealed on ‘Normal’ position.

4.1.1.7.8.1 Function

Isolation of KVB.
This selector overrides KVB ermergency brake intervention too.

4.1.1.7.8.2 Description

Selector on the position ‘Normal’ shall power on KVB equipment.
Selector on the position ‘Isolation’ shall power off KVB equipment.

4.1.1.7.9 Selector JD-S022 - Z-RPS

Sealed on ‘Normal’ position.

4.1.1.7.9.1 Function

Isolation of RPS function.
This selector overrides RPS emergency brake intervention too.

4.1.1.7.9.2 Description

Selector on the position ‘Normal’ shall power on RPS function.
Selector on the position ‘Isolation’ shall power off RPS function.By pass RPS intervention on EB loop

4.1.1.7.10 Selector JD-S016 - Z-VIT

4.1.1.7.10.1 Function

Isolation of ATESS tachymetry function

4.1.1.7.10.2 Description

Selector on the position ‘Normal’ shall power on ATESS tachymetry.
Selector on the position ‘Isolation’ shall power off ATESS tachymetry.

4.1.1.7.11 Selector JD-S021 - Z-ENR

Sealed on ‘Normal’ position.

4.1.1.7.11.1 Function

Isolation of recorder (ATESS and JRU)

4.1.1.7.11.2 Description

Selector on the position ‘Normal’ shall power on ATESS recorder.
Selector on the position ‘Isolation’ shall power off ATESS recorder.

4.1.1.7.12 Selector JD-S014 - Z(EXPL)

Sealed on ‘TVM’ position.

4.1.1.7.12.1 Function

Running mode selector for ERTMS/TVM equipment
(ERTMS levels 0, 1 and 2 enabled or disabled for Start of Mission and dynamic level transitions)

4.1.1.7.12.2 Description

Selector on the position « TVM » by default (sealed).
Selector on the position « ERTMS/TVM» not applicable.

4.1.1.8 Start up Mission Selectors

4.1.1.8.1 Selector JD-S012 - Selector I-F

4.1.1.8.1.1 Function

Selection ALA- BISTANDARD

4.1.1.8.1.2 Description

Selector on the position ‘I’ shall activate the Italian ATP system
Selector on the position ‘F’ shall activate the French ATP system.

4.1.1.8.2 Selector JD-S024 - Z(SEL)DMI

4.1.1.8.2.1 Function

Selection of DMI displaying signalling information

4.1.1.8.2.2 Description

Selector on the position ‘Left’ shall activate the DMI located at the left in the driver’s desk. Right DMI shall be in standby.
Selector on the position ‘Right’ shall activate the DMI located at the right in the driver’s desk. Left DMI shall be in standby.
Selector position is used only if both DMIs are available. If failure is detected by the Bi-Standard, failover is automatic and selector position is ignored.

4.1.1.8.3 Selector JD-S023 - Z(ES)ETCS

4.1.1.8.3.1 Function

Test selector for ERTMS/TVM equipment

4.1.1.8.3.2 Description

Selector on the position ‘Normal’ shall stop any TVM or ETRMS tests.
Selector on the position ‘TVM’ shall launch tests of the TVM.
Selector on the position ‘Odometric’ shall launch tests of the TVM with external odometry.
Position ‘ETCS’ is a reserve.

4.1.1.8.4 Selector JD-S030 - Z(ES)VA

4.1.1.8.4.1 Function

Test of VACMA function

4.1.1.8.4.2 Description

Selector on the position ‘Normal’ shall stop the test of VACMA function.
Selector on the position ‘Test’ shall start the test of VACMA function.

4.1.1.9 Toggle Switch and Pushbuttons

4.1.1.9.1 Toggle switch JD-A068 BP-DM

4.1.1.9.1.1 Function

TVM manual disarming

4.1.1.9.1.2 Description

Action on BPDM switch shall disarm TVM.

4.1.1.9.2 Toggle switch JD-A067 BP-AMV2

4.1.1.9.2.1 Function

TVM manual arming on track 2

4.1.1.9.2.2 Description

Action on BPAMV2 switch shall arm TVM on track 2.

4.1.1.9.3 Toggle switch JD-A066 BP-AMV1

4.1.1.9.3.1 Function

TVM manual arming on track 1

4.1.1.9.3.2 Description

Action on BPAMV1 switch shall arm TVM on track 1.

4.1.1.9.4 Pushbutton JD-A074 BP-AC-SF

4.1.1.9.4.1 Function

Acknowledgement button VACMA, ATESS, TVM

4.1.1.9.4.2 Description

Acknowledgement button SF of BRS

4.1.1.9.5 Pushbutton JD-A073 BP-DJ

4.1.1.9.5.1 Function

Release Intervention KVB, RPS

4.1.1.9.5.2 Description

Action on Release intervention push button shall allow the driver to release KVB/RPS brake intervention.

4.1.1.9.6 Pushbutton JD-A069 BP-ACQ

4.1.1.9.6.1 Function

ETCS_Acknowledgement button TVM, DM1-2

4.1.1.9.6.2 Description

Action on BP-ACQ button shall allow the driver to acknowledge message on DMI (e.g. Standstill supervision during Start of Mission phase).

4.1.1.9.7 Pushbutton JD-A086 BP-R-URG

4.1.1.9.7.1 Function

VACMA armament button

4.1.1.9.7.2 Description

Pressed after VACMA intervention of EB to rearm the system

4.1.1.9.8 Pushbutton JD-A084 BP(A-AL)VA

4.1.1.9.8.1 Function

VACMA alarm cancellation button

4.1.1.9.8.2 Description

Action on Release intervention push button shall allow the driver to release VACMA brake intervention.

4.1.2 Emergency Brake Control

4.1.2.1 Emergency Brake Function Test

Each following equipment / component of the ATP system may command the emergency brake:
The TVM/ERTMS equipment (through 2 independent channels),
The KVB equipment (through 1 single channel),
The RPS function in ATESS 3G (through 1 single channel),
The VACMA function in ATESS 3G (through 1 single channel).

4.1.2.1.1 Function Design

Before activating the cab all this sub-system is tested accordingly to the French standard [SAM-F-004].

4.1.2.1.2 Normal condition

At the power on of the train, before to active the driver desk,
a warning light, lighting on, shall inform the driver that the contacts relevant the emergency brake request for each safety system are correctly set.
a warning light, lighting on, shall inform the driver that no one safety system is isolated.
At the activation of the driver desk, the relevant lamps shall be switched off by the train.
VY(CO)URG
VY(CO)Z
In case of one of the ATP system installed in the train (VACMA, RPS, KVB and TVM) is not triggering an Emergency Brake request the train shall switch off the visual indication called VY(CO)URG on the desk.
In case of at least one of the ATP system installed in the train (VACMA, RPS, KVB and TVM) is isolated the VY(CO)Z shall be lighted off by the train.

4.1.2.1.3 Failure cases

A. VY(CO)URG test fails
B. VY(CO)Z test fails

4.1.2.1.4 Failure detection

A. Failure is self-evident (lamp switched off).
B. Failure is self-evident (lamp switched off).

4.1.2.1.5 Failure handling

A. The failure shall be repaired during maintenance
B. The failure shall be repaired during maintenance

4.1.2.1.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.266
Description	The ATP system shall notice to the driver before cab activation through visual signal lighting off on the driver desk that one or more ATP sub systems are not triggering EB.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.692	The lamp VY(CO)Z shall be supply when the the train battery inserted	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	$\leq 10^{-6}$
2F_09.02.Zefiro-IF.CONCEPT.693	The lamp VY(CO)Z shall be supply also and parking mode functionality	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	$\leq 10^{-6}$
2F_09.02.Zefiro-IF.CONCEPT.225	Before cab activation the status of the ATP sub system isolation switch (isolated) or contact NC relay EB of each ATP sub-system shall be light on visual signallisation on the desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	$\leq 10^{-6}$

ID	2F_09.02.Zefiro-Europe.TRS.267
Description	The ATP system shall notice to the driver before cab activation through visual signalization lighting off on the driver desk that one or more ATP sub systems are isolated from EB intervention.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.694	The lamp VY(CO)URG shall be supply when the the train battery inserted	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	$\leq 10^{-6}$
2F_09.02.Zefiro-IF.CONCEPT.695	The lamp VY(CO)URG shall be supply also and parking mode functionality	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	$\leq 10^{-6}$
2F_09.02.Zefiro-IF.CONCEPT.227	Before cab activation the status of the ATP sub system isolation switch (non isolated) shall be light on visual signallisation on the desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	$\leq 10^{-6}$

4.1.2.2 VACMA Emergency Brake Function

The (DVD) for the “France corridor” VACMA is implemented inside the control unit (ATESS 3G) within the card VAC.

4.1.2.2.1 Function Design

The Driver Vigilance Device (DVD) monitors the presence and vigilance of the driver through the activation of the controls in the cabin. VACMA shall require the emergency brake in case of Dead Man Reaction. In this case the train shall apply the emergency braking.

4.1.2.2.2 Normal condition

In case the driver vigilance or presence is not confirmed the DVD switch on the acoustical warning by the ATESS Sound generator. The Driver must confirm the presence acting on pedal switch or acknowledge pushbutton, master controller, horn switch. If none of the command is not performed in the end it applies the emergency brake and light the lamp on the desk. The event is then recorded by the train recorder JRU and after 30s a message is sent to the ground by the cab radio.

4.1.2.2.3 Failure cases

A. ATESS VACMA applies the brake when not needed => The train is stopped
B. ATESS VACMA fails to apply the brake when commanded=>The train is not stopped when needed

4.1.2.2.4 Failure detection

A. Failure self-evident
B. Failure can be detected during the periodic test initiated by the driver

4.1.2.2.5 Failure handling

A. The ATESS VACMA shall be isolated and the system shall be repaired during maintenance
B. If ATESS VACMA failure is detected during the periodic test by the driver, the mission is cancelled

4.1.2.2.6 Requirements

ID	2F_09.02.Zefiro_Europe.TRS.268
Description	The driving personnel shall confirm presence and vigilance by activating the safety pedal, in case of non-activation it can be confirmed by activating any of the following Horn Selector, Master controller, Acknowledgement Pushbutton on the driver desk
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.242	The status of the pedal (operated/not operated) shall be provided to the DVD (ATESS 3G).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.243	The status of the control on the traction brake lever (operated/not operated) shall be provided to the DVD (ATESS 3G).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.244	The status of the control on the horn toegle swicth (operated/not operated) shall be provided to the DVD (ATESS 3G).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.245	The status of the control on the acknowledgement push buttons (operated/not operated) shall be provided to the DVD (ATESS 3G) BSTD_ATESS_ACK_N BSTD_ATESS_ACK_R	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	$\leq 10^{-6}$
2F_09.02.Zefiro-IF.CONCEPT.1013	TCMS shall read the status of ATESS acknowledgement push buttons BSTD_ATESS_ACK_N BSTD_ATESS_ACK_R and verify the coherence, show the diagnose message on DMI in case of incoherence.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.269
Description	An acoustic warning shall be applied if the driver does not confirm the presence.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.247	The DVD shall command a dedicated sound in the cabin through ATESS Sound generator when the DVD controls are not operated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1014	The status of the control on the cancellation VACMA allarm shall be provided to ATESS BSTD_VACMA_CANC_OFF BSTD_VACMA_CANC_ON	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	$\leq 10^{-6}$
2F_09.02.Zefiro-IF.CONCEPT.248	TCMS shall read the status of VACMA alarm cancellation push button BSTD_VACMA_CANC_OFF BSTD_VACMA_CANC_ON	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1015	If the signals are in the below state BSTD_VACMA_CANC_OFF=1 BSTD_VACMA_CANC_ON =0 then VACMA alarm cancellation is not requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1016	If the signals are in the below state BSTD_VACMA_CANC_OFF=0 BSTD_VACMA_CANC_ON =1 then VACMA alarm cancellation is requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1017	In case of incoherence between VACMA alarm cancellation signals BSTD_VACMA_CANC_OFF=1 BSTD_VACMA_CANC_ON=1 or BSTD_VACMA_CANC_OFF=0 BSTD_VACMA_CANC_ON=0 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.270
Description	Emergency brake shall be applied if the driver does not confirm the presence after the acoustic warning.The lamp on driver desk is lit.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.250	The DVD applies the emergency brake commanding by de energization of the emergency brake loop.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.251	The DVD commands the lamp VY-URG-VA lighting on when applies the emergency brake command.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.252	TCMS shall read the status of DVD EB intervention BSTD_Q1-VA_RELAY	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.253	The DVD shall provide to the JRU the DVD EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.254	The driver shall rearm VACMA system after EB intervention.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.255	TCMS shall read the status of VACMA rearm push button BSTD-VACMA_FEED_N BSTD-VACMA_FEED_R	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1018	If the signals are in the below state BSTD-VACMA_FEED_N=0 BSTD-VACMA_FEED_R=1 then VACMA rearm push button is not requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1019	If the signals are in the below state BSTD-VACMA_FEED_N=1 BSTD-VACMA_FEED_R=0 then VACMA rearm push button is requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1020	In case of incoherence between VACMA rearm push button signals BSTD-VACMA_FEED_N=1 BSTD-VACMA_FEED_R=1 or BSTD-VACMA_FEED_N=0 BSTD-VACMA_FEED_R=0 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.271
Description	In EB required by DVD, the train shall inform the cab radio (within 30s after train is stopped) in order to allow an alarm radio call.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.257	The DVD shall communicate to the cab radio the alarm EB intervention.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.254
Description	The DVD test shall be started manually by the driver in the active cab.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.259	The diagnostic shall foresee a switch to command the DVD test.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.260	TCMS shall read the selector Z(ES)VA position 'Test'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.261	The faulty DVD status shall be communicated trough digital input to TCMS (DEF.VA= VACMA FAULT) BSTD_Q-DJ_RELAY	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.697	TCMS shall inform the driver on the DMI if during the VACMA test a failure has been detected, If BSTD_Q-DJ_RELAY=1 no failure on VACMA has been detected. If BSTD_Q-DJ_RELAY=0 a failure on VACMA has been detected.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.262	The DVD test selector shall communicate to the train recorder JRU the status	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.696	TCMS shall inform the driver that the Z(ES)VA is yet on position 'Test' signals BSTD_VACMA_TEST_ON BSTD_VACMA_TEST_OFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1021	If the signals are in the below state BSTD_VACMA_TEST_ON=1 BSTD_VACMA_TEST_OFF=0 then the selector VACMA test is requested	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1022	If the signals are in the below state BSTD_VACMA_TEST_ON=0 BSTD_VACMA_TEST_OFF=1 then the selector VACMA test is not requested	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1023	In case of incoherence between VACMA test signals BSTD_VACMA_TEST_ON=0 BSTD_VACMA_TEST_OFF=0 or BSTD_VACMA_TEST_ON=1 BSTD_VACMA_TEST_OFF=1 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.111
Description	It shall be possible to Bypass the EB applied by the DVD in case of a faulty DVD system.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.264	A sealed selector Z-VA on the cab shall be foreseen to bypass the DVD brake application in the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.265	The TCMS shall acquire the status of the DVD isolation selector through a digital inputs signals BSTD_SWZVA_ON BSTD_SWZVA_OFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1024	If the signals are in the below state BSTD_SWZVA_ON=1 BSTD_SWZVA_OFF=0 then the selector DVD isolation selector in Normal position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1025	If the signals are in the below state BSTD_SWZVA_ON=0 BSTD_SWZVA_OFF=1 then the selector DVD isolation selector in Isolation position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1026	In case of incoherence between Z-VA selector signals BSTD_SWZVA_ON=1 BSTD_SWZVA_OFF=1 or BSTD_SWZVA_ON=0 BSTD_SWZVA_OFF=0 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.266	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.267	The DVD isolation selector shall communicate to the train recorder JRU the status	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.268	The DVD isolation selector shall inhibit DVD functions	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

4.1.2.3 RPS Emergency Brake Function

The RPS is implemented inside the control unit (ATESS 3G) within the card RS.

4.1.2.3.1 Function Design

The ATESS RPS receives binary information through brush (Brosse) (SO-open signal and SF-closed signal).
In case the driver does not acknowledges SF signal, the RPS shall require the emergency brake.

4.1.2.3.2 Normal condition

In case ATESS RPS receives binary information SO signal, the RPS shall require acoustic signalization.
In case ATESS RPS receives binary information SF signal, the RPS shall require acoustic and visual signalization LS-SF lamp on KVB panel.
The driver acknowledges SF signal through dedicated pushbutton or acquittement button on KVB panel. In case the driver does not acknowledges SF signal, the RPS shall require the emergency brake and light the lamp on the desk.
Both received binary information and EB brake intervention are stored in JRU.

4.1.2.3.3 Failure cases

ATESS RPS applies the brake when not needed => The train is stopped
ATESS RPS fails to apply the brake when commanded =>The train is not stopped when needed

4.1.2.3.4 Failure detection

Failure self-evident
Failure self-evident

4.1.2.3.5 Failure handling

The ATESS RPS shall be isolated and the system shall be repaired during maintenance
The ATESS RPS shall be isolated and the system shall be repaired during maintenance

4.1.2.3.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.272
Description	The ATESS RPS binary information SO and SF shall communicate acoustic and visual siganlisation to the driver
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.283	ATESS RPS shall communicate through sound generator SO/SF signal.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.284	ATESS RPS shall communicate through visual signalisation SF signal.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.285	ATESS RPS shall communicate to JRU the SO/SF signal.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.286	ATESS RPS shall command the light signalisation for SF signal.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.273
Description	Emergency brake shall be applied if the driver does not acknowledge or acquitt the SF signal after the acoustic warning.The lamp on driver desk is lit.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.288	ATESS RPS applies the emergency brake commanding by de energization of the emergency brake loop.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.289	ATESS RPS commands the lamp VY-URG-RPS lighting on when applies the emergency brake command.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.290	TCMS shall read the status of ATESS RPS EB intervention signal BSTD_QRPS_RELAY	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.291	ATESS RPS shall provide to the JRU the RPS EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.292	The driver shall release ATESS RPS system after EB intervention.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.293	TCMS shall read the status of ATESS RPS release push button digital signal input BSTD_KVB_REALIN_N BSTD_KVB_REALIN_R	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1027	If the signals are in the below state BSTD_KVB_REALIN_N=0 BSTD_KVB_REALIN_R=1 then ATESS RPS release is not requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1028	If the signals are in the below state BSTD_KVB_REALIN_N=1 BSTD_KVB_REALIN_R=0 then ATESS RPS release is requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1029	In case of incoherence between ATESS RPS release signals BSTD_KVB_REALIN_N=0 BSTD_KVB_REALIN_R=0 or BSTD_KVB_REALIN_N=1 BSTD_KVB_REALIN_R=1 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.294	ATESS RPS shall provide to the JRU the RPS release button	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.274
Description	It shall be possible to Bypass the EB applied by the ATESS RPS in case of a faulty ATESS RPS system.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.296	A sealed selector Z-RPS on the cab shall be foreseen to bypass the ATESS RPS brake application in the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.297	The TCMS shall acquire the status of the ATESS RPS isolation selector through a digital input signals: BSTD_RPS_SWOFF BSTD_RPS_SWON	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1030	If the signals are in the below state BSTD_RPS_SWOFF=0 BSTD_RPS_SWON=1 then the RPS isolation selector in Normal position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1031	If the signals are in the below state BSTD_RPS_SWOFF=1 BSTD_RPS_SWON=0 then the RPS isolation selector in Isolation position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1032	In case of incoherence between Z-RPS selector signals BSTD_RPS_SWOFF=1 BSTD_RPS_SWON=1 or BSTD_RPS_SWOFF=0 BSTD_RPS_SWON=0 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.298	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.299	The ATESS RPS isolation selector shall communicate to the train recorder JRU the status	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.300	The ATESS RPS isolation selector shall inhibit ATESS RPS functions	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

4.1.2.4 KVB Emergency Brake Function

The KVB signalling supervise train speed through the information received by ATESS tachymetry and KVB balise information received by the track. KVB also supervise signalling information as square signal crossing.

4.1.2.4.1 Function Design

In the case the driver does not respect speed information or square signalling crossing signal. the KVB shall inform by visual and acoustic information the driver.

4.1.2.4.2 Normal condition

In case the KVB balise information for square signalling or speed restriction is received the information is displayed in the KVB panel in the driver desk. The driver shall not override square signal crossing or don't respect speed limit KVB shall require the emergency brake and the train shall apply the emergency braking.

4.1.2.4.3 Failure cases

A. KVB applies the brake when not needed => The train is stopped
B. KVB fails to apply the brake when commanded=>The train is not stopped when needed

4.1.2.4.4 Failure detection

A. Failure self-evident
B. Failure self-evident

4.1.2.4.5 Failure handling

A. The KVB shall be isolated and the system shall be repaired during maintenance
B. The KVB shall be isolated and the system shall be repaired during maintenance

4.1.2.4.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.275
Description	The KVB information Square crossing signal and over speed shall communicate acoustic and visual signalisation to the driver
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.315	KVB shall communicate through acoustic (tut/bip) and visual signalisation	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.698	TCMS shall read the status of KVB Crossing/Override push button PB-FC by the digital signal input BSTD_QFC_DIA	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.276
Description	Emergency brake shall be applied if the driver does not override square crossing signal or overspeed after the acoustic and visual signalisation.The lamp on driver desk is lit.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.317	KVB applies the emergency brake commanding by de energization of the emergency brake loop.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.318	KVB commands the lamp VY-URG-KVB lighting on when applies the emergency brake command.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.319	TCMS shall read the status of KVB EB intervention signal BSTD_QKVB_RELAY.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.320	KVB shall provide to the JRU the KVB EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.321	The driver shall release KVB system after EB intervention.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.322	TCMS shall read the status of KVB release push button digital signal input BSTD_KVB_REALIN_N BSTD_KVB_REALIN_R	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1033	If the signals are in the below state BSTD_KVB_REALIN_N=0 BSTD_KVB_REALIN_R=1 then KVB release is not requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1034	If the signals are in the below state BSTD_KVB_REALIN_N=1 BSTD_KVB_REALIN_R=0 then KVB release is requested	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1035	In case of incoherence between KVB release signals BSTD_KVB_REALIN_N=0 BSTD_KVB_REALIN_R=0 or BSTD_KVB_REALIN_N=1 BSTD_KVB_REALIN_R=1 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.323	KVB shall provide to the JRU the KVB release button	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.277
Description	It shall be possible to Bypass the EB applied by the KVB in case of a faulty KVB system.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.325	A sealed selector Z-KVB on the cab shall be foreseen to bypass the KVB brake application in the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL0
2F_09.02.Zefiro-IF.CONCEPT.326	The TCMS shall acquire the status of the KVB isolation selector Z-KVB	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.327	TCMS shall read the position of Z-KVB isolation selector through a digital input signals BSTD_KVB_SWON BSTD_KVB_SWOFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1036	If the signals are in the below state BSTD_KVB_SWON=1 BSTD_KVB_SWOFF=0 then the selector KVB isolation in Normal position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1037	If the signals are in the below state BSTD_KVB_SWON=0 BSTD_KVB_SWOFF=1 then the selector RPS isolation in Isolation position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1038	In case of incoherence between Z-RPS selector signals BSTD_KVB_SWON=1 BSTD_KVB_SWOFF=1 or BSTD_KVB_SWON=0 BSTD_KVB_SWOFF=0 show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.328	The KVB isolation selector shall communicate to the train recorder JRU the status	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.329	The KVB isolation selector shall inhibit KVB functions	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

4.1.2.5 TVM Emergency Brake Function

The TVM monitors the train speed through the information received by the antennas.
In the cases foreseen (i.e.: the driver does not respect authorized speed information), the TVM shall require the emergency brake.

4.1.2.5.1.1 Function Design

TVM indicates to the driver by DMI the speed information and signalling signals constrains.

4.1.2.5.2 Normal condition

In case the TVM speed information indications and signalling signal are not executed by the driver as per TVM indication, the TVM shall require the emergency brake. TVM commands 2 independent EB channels

4.1.2.5.3 Failure cases

A. TVM applies the brake when not needed => The train is stopped
B. TVM fails to apply the brake when commanded=>The train is not stopped when needed

4.1.2.5.4 Failure detection

A. Failure self-evident
B. Failure can be detected during the periodic test initiated by the driver

4.1.2.5.5 Failure handling

A. The TVM shall be isolated and the system shall be repaired during maintenance
B. If a TVM failure is detected during the periodic test, the mission is cancelled

4.1.2.5.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.278
Description	The TVM signal and over speed shall communicate acoustic and visual signalisation to the driver
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.344	TVM shall communicate through DMI acoustic and visual signalisation.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.279
Description	Emergency brake shall be applied if the driver does not respect TVM inidcation, acoustic and visual signalisation.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.346	TVM applies the emergency brake commanding by de-energization of the emergency brake loop.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.347	TVM commands signalization on DMI when applies the emergency brake command.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1039	TCMS shall read the following digital inputs: BSTD_Q1_FEEDBACK and BSTD_Q2_FEEDBACK	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1040	If the selector: Z-ETCS/TVM is not set to A+B OFF and Z-KARM is not set to OFF Then the signals shall be: BSTD_Q1_FEEDBACK=0 BSTD_Q2_FEEDBACK=0 or BSTD_Q1_FEEDBACK=1 BSTD_Q2_FEEDBACK=1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1041	In case of incoherence between signals in above conditions: then show on TCMS DMI a diagnose message.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.348	TCMS shall read the digital inputs BSTD_EB_FBCOMN and BSTD_EB_FBCOMR	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1042	IF the selectors: Z(FU)ETCS is not in position 1-OFF or 2-OFF and Z-RPS is not in position ‘’Escluso’’ (Excluded) and Z-KVB is not in position ‘’Escluso’’ (Excluded) and Z-VA is not in position ‘’Escluso’’ (Excluded) Then the signals shall be BSTD_EB_FBCOMN=0 BSTD_EB_FBCOMR=0 or BSTD_EB_FBCOMN=1 BSTD_EB_FBCOMR=1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1043	With above conditions, if after 3s from the change status of the first signal, the incoherence between signals has been detected then show on TCMS DMI a diagnose message.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.349	TVM shall provide to the JRU the TVM EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.280
Description	It shall be possible to Bypass the EB applied by the TVM in case of a faulty TVM system.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.351	A sealed selector Z(FU)ETCS on the cab shall be foreseen to bypass channel 1 or channel 2 of brake application in the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.352	The TCMS shall acquire the status of channel 1 or channel 2 the Z(FU)ETCS selector through a digital input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.353	TCMS shall read the position of Z(FU)ETCS isolated channel 1	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.650	TCMS shall read the position of Z(FU)ETCS isolated channel 2	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.354	TVM shall communicate to the train recorder JRU the status of Z(FU)ETCS isolation channel 1 or channel 2	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.355	A sealed selector Z-ETCS on the cab shall be foreseen to bypass the bi-standard equipment chain A, B or A+B	Derived Requirement	4S_09.02.09.-Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.356	The TCMS shall acquire the status of the Z-ETCS isolation selector through a digital inputs BSTD_SWTVM_AOFF BSTD_SWTVM_ON BSTD_SWTVM_BOFF BSTD_SWTVM_ABOFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1044	If the signals are in the below state BSTD_SWTVM_AOFF=1 BSTD_SWTVM_ON=0 BSTD_SWTVM_BOFF=0 BSTD_SWTVM_ABOFF=0 then the selector the Z-ETCS isolation selector in 'Isolation of chain A'.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1045	If the signals are in the below state BSTD_SWTVM_AOFF=0 BSTD_SWTVM_ON=1 BSTD_SWTVM_BOFF=0 BSTD_SWTVM_ABOFF=0 then the selector the Z-ETCS isolation selector in 'Normal'.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1046	If the signals are in the below state BSTD_SWTVM_AOFF=0 BSTD_SWTVM_ON=0 BSTD_SWTVM_BOFF=1 BSTD_SWTVM_ABOFF=0 then the selector the Z-ETCS isolation selector in 'Isolation of chain B'.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1047	If the signals are in the below state BSTD_SWTVM_AOFF=0 BSTD_SWTVM_ON=0 BSTD_SWTVM_BOFF=0 BSTD_SWTVM_ABOFF=1 then the selector the Z-ETCS isolation selector in 'Isolation of chain A & B'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1048	For all different sequences other than the concepts 1044, 1045, 1046, 1047 (see above) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

2F_09.02.Zefiro-IF.CONCEPT.357	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.651	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.652	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.358	TVM shall communicate to the train recorder JRU the status of Z-ETCS isolation A, B or A+B.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0
2F_09.02.Zefiro-IF.CONCEPT.359	Configuration: Z-ETCS isolation selector shall inhibit ETCS EB functions	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL0

4.1.3 Train Input to ATP

4.1.3.1 Function design

The following Train input are received by the Bi-Standard.
The train shall transmit the following information signal with the required logic implementation as follows:

Cabin Selection/Sleeping

Cabin Selection	Sleeping Mode	Cabin Status
0	0	Stand-by
1	0	Error->Not possible
0	1	Sleeping
1	1	Leading

Table 5 :Cabine activation information to ATP

Direction Forward/Reverse

Direction Reverse	Direction Forward	Direction
0	1	Error → Neutral
1	1	Forward direction
0	0	Reverse direction
1	0	Neutral

Table 6: Direction Control information to ATP

Single Unit = ”1”/Multiple Unit = “0”
Pantograph lowering by the Driver = “1”
Main Switch opening by the Driver = “1”
French (25kV ,1.5kV ,25kV GV) Catenary voltage selection by the Driver = “1”
Z(BG)IS bogie isolation switch

4.1.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.282
Description	Bi-Standard in TVM supervision shall acquire the status of cab activation/sleeping, train direction forward/reverse, train composition Single Unit/Multiple Unit ,Pantograph lowering by the driver, Main Switch opening by the driver, French Line voltage selection by the driver.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.370	The train shall communicate to TVM the status of cabine activation.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL4
2F_09.02.Zefiro-IF.CONCEPT.371	The train shall communicate to TVM the status of cabine sleeping mode	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL4
2F_09.02.Zefiro-IF.CONCEPT.653	TCMS shall not consider the signal TTF output (§ 4.1.4) relevant in case the cab is not enabled (TVM is in sleeping mode)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.372	The train shall communicate to TVM the status of direction forward selected	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL4
2F_09.02.Zefiro-IF.CONCEPT.373	The train shall communicate to TVM the status of direction reverse selected	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL4
2F_09.02.Zefiro-IF.CONCEPT.374	The train shall communicate to TVM the status of train composition single unit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL4
2F_09.02.Zefiro-IF.CONCEPT.1079	The train shall communicate to TVM the status of train composition multiple unit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL4
2F_09.02.Zefiro-IF.CONCEPT.375	The train shall communicate to TVM the status of Pantograh lowering by the driver	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.376	The train shall communicate to TVM the status of Main Switch opening by the driver	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.699	TCMS compare the digital outputs N with the digital inputs R and verify misalignment between signals BSTD_PANTO_OPBYDR_N and BSTD_PANTO_OPBYDR_R	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.377	TCMS shall command through digital output the command of pantographs lowering by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1049	The command of pantograph lowering by TVM and by driver shall be sent by MVB to ATESS	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.700	TCMS compare the digital outputs N with the digital inputs R and verify misalignment between signals BSTD_DJ_OPBYDR_N and BSTD_DJ_OPBYDR_R	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.378	TCMS shall command through digital output the command the Main Circuit Breaker opening by the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1050	The command of Main Circuit Breaker by TVM and by driver shall be sent by MVB to ATESS	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.379	“The train shall communicate to TVM the status of French Catenary Voltage selection by the driver	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.380	“The train shall communicate to TVM the status of bogie isolated by the driver	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0

4.1.4 ATP output Train Technical Functions

The Bi-Standard shall give outputs in order to perform train technical functions.

4.1.4.1 TVM supervision

The Bi-Standard under TVM supervision shall give outputs in order to perform train technical functions.

4.1.4.1.1 Function design

The Bi-Standard under TVM supervision shall give outputs in order to perform train technical functions.
ATP shall command the main circuit breaker when needed (phase and voltage transition) see. 4.1.5. and 4.1.6.
TCMS shall forward the request to the traction system that will command the MCB opening. It is then a “Train” task to assure the execution of the command.
ATP shall command the pantograph lowering command when needed (voltage transition) see 4.1.6.
TCMS shall forward the request to the traction system that will command the pantograph lowering. It is then a “Train” task to assure the execution of the command.
ATP shall command the traction cut-off command when needed (voltage transition) see 4.1.6.
TCMS shall forward the request to the traction system that will command the pantograph lowering. It is then a “Train” task to assure the execution of the command.
ATP shall command the Line voltage selection when needed (voltage transition) see 4.1.6.
TCMS shall forward the request to the driver in order to set the correct position of Line Voltage Selector. It is then a “Driver” task to assure the execution of the command.
ATP shall command the Tunnel Entry/Exit section when needed.
TCMS shall acquire the information and command Train Tunnel mode.
There are Four Train Technical Functions that are managed by the Bi-Standard in TVM supervision:

Main Switch Off
Lower Pantograph
Catenary Indication
Air Tightness

These functions are managed by the following outputs of TVM:

Output(s)	TTF function	Input N TCMS	Input R TCMS
S_CODJT	Manual Main Switch Off (TVM 1/p)	BSTD_CODJTN_COM	BSTD_CODJTR_COM
S_CODJR	Automatic Main Switch Off (TVM n/p)	BSTD_CODJRN_COM	BSTD_CODJRR_COM
S_CFDJ	Automatic Main Switch Off (TVM n/p)	BSTD_CFDJN_COM	BSTD_CFDJR_COM
S_SECTAUTO	Automatic Main Switch Off (TVM n/p)	BSTD_SECTAUTON_SEG	BSTD_SECTAUTOR_SEG
S_CRET	Automatic Main Switch Off (TVM n/p)	BSTD_CRETN_COM	BSTD_CRETR_COM
S_COBP	Manual Lower Pantograph	BSTD_COBPN_COM	BSTD_COBPR_COM
S_ELGV	Catenary Indication	BSTD_ELGVN_SEG	BSTD_ELGVR_SEG
S_ELC25KV	Catenary Indication	BSTD_ELC25KVN_SEG	BSTD_ELC25KVR_SEG
S_ELC1KV5	Catenary Indication	BSTD_ELC1KV5N_SEG	BSTD_ELC1KV5R_SEG
S_ETU	Air Tightness Function		

TVM internal rack status faults:

Output(s)	Function	Input N TCMS	Input N TCMS
S_DJMIN	Status of internal TVM minor Fault	BSTD_DJMINN_MEM	BSTD_DJMINR_MEM
S_DJMAJ	Status of internal TVM major Fault	BSTD_DJMAXN_MEM	BSTD_DJMAXR_MEM

4.1.4.1.2 Parameters

Distance/Time	Meaning	[s]	[m]
T2	90000507.G00.EN §2.2.3.1 This variable is intended to set the time needed to execute the traction reduction command. Used only by the train technical functions to determine when traction cut off command must be issued so that traction cut off will be really cut when entering a neutral section.	5	-
T1	90000507.G00.EN §2.2.3.2 This variable is intended to set the time needed to execute the lowering of the pantograph. Used only by the train technical functions to determine when pantograph command must be issued so that pantograph will be really low when required.	8	-
L_DEGAGEMENT_US	90000507.G00.EN §2.2.3.3 This parameter is intended to set the distance at which is placed the last pantograph installed on the roof of the train from the train front end when running in Simple Unit configuration. When receiving from trackside the distance at which is placed a powerless section (pantograph management), the Bi-Standard will add the distance assigned to this parameter, in order to delay the command to permit to raise again a pantograph (really it is the opposite: the command to lower the pantograph is stopped). Note: ERTMS rule, about trackside powerless sections areas management, do not considers the train length; this parameter changes it.	-	180
L_DEGAGEMENT_UM	90000507.G00.EN §2.2.3.4 This parameter is intended to set the distance at which is placed the last pantograph installed on the roof of the train from the train front end when running in Multiple Unit configuration. When receiving from trackside, the distance of powerless sections (pantograph management), the Bi-Standard will add the distance fixed to this parameter, in order to delay the command. Note: trackside powerless sections areas do not consider the train length.	-	380
L_PT_AR	90000507.G00.EN §2.2.3.5 This variable is like L_Degagement_US (see 2.2.3.3), but L_PT_AR is only used by TVM system.	-	180

Table 7: Parameters Phase and Catenary Changing (TVM parameters)

4.1.4.1.3 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.284
Description	TVM shall inform its internal status faults to the train
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.391	TCMS shall acquire through digital intput the status of TVM signal BSTD_DJMINN_MEM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1061	If the signals TVM Minor Fault (PSTD-A) is in the below state BSTD_DJMINN_MEM=1 TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1062	TCMS shall acquire through digital intput the status of TVM signal BSTD_DJMINR_MEM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1063	If the signals TVM Minor Fault (PSTD-B) is in the below state BSTD_DJMINR_MEM=1 TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.392	TCMS shall acquire through digital intput the status of TVM signals BSTD_DJMAXN_MEM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1064	If the signals TVM Major Fault (PSTD-A) is in the below state BSTD_DJMAXN_MEM=1 TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1065	TCMS shall acquire through digital intput the status of TVM signals BSTD_DJMAXR_MEM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1066	If the signals TVM Major Fault (PSTD-B) is in the below state BSTD_DJMAXR_MEM=1 TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.285
Description	ATP shall command the Tunnel Entry/Exit section (predisposition)
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.394	TCMS shall acquire through digital input the status of TVM signal BSTD_ETUN/R_COM	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.395	TCMS shall acquire through digital intput the status of TVM signal BSTD_ETUN_COM BSTD_ETUR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1067	TCMS shall generate an event if the signals BSTD_ETUN_COM=1 or BSTD_ETUR_COM=1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1068	In case of incoherence between the signals ETU from PSTD-A BSTD_ETUN_COM=1 BSTD_ETUR_COM=0 or BSTD_ETUN_COM=0 BSTD_ETUR_COM=1 TCMS shall show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.701	TCMS by IP network, shall inform Saloon HVAC unit 1 and 2, the request to close the HVAC flaps. The HVAC cab unit, if the conditions are verified, provides the command by train line to all HVAC of the trainset.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.396	DELETED				

ID	2F_09.02.Zefiro-Europe.TRS.286
Description	ATP shall command Main Switch opening /closing
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.398	TCMS shall acquire through digital intput the status of TVM signal S_CODJT	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.399	TCMS shall acquire through digital intput the status of TVM signal S_CODJT TCMS signal: BSTD_CODJTN_COM BSTD_CODJTR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.400	TCMS shall command to open the main circuit breaker when it receives the TVM signal S_CODJT which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.401	TCMS shall acquire through digital intput the status of TVM signal S_CODJR	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.402	TCMS shall acquire through digital intput the status of TVM signal S_CODJR BSTD_CODJRN_COM BSTD_CODJRR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.403	TCMS shall open main circuit breaker when it receives the TVM signal S_CODJR which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.404	TCMS shall acquire through digital intput the status of TVM signal S_CFDJ	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.405	TCMS shall acquire through digital intput the status of TVM signal S_CFDJ BSTD_CFDJN_COM BSTD_CFDJR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.406	TCMS shall command main circuit breaker closing when it receives the TVM signal S_CFDJ which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.287
Description	ATP shall command the Traction reduction
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.408	TCMS shall acquire through digital intput the status of TVM signal S_CRET	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.409	TCMS shall acquire through digital intput the status of TVM signal S_CRET	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.410	TCMS shall command traction reduction of traction effort to zero with a ramp with duration 5s when it receives the TVM signal S_CRET, which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.288
Description	ATP shall command the Pantograph lowering
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.412	TCMS shall acquire through digital intput the status of TVM signal S_COBP.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	$\leq 10^{-5}$
2F_09.02.Zefiro-IF.CONCEPT.413	TCMS shall acquire through digital intput the status of TVM signal S_COBP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.414	When S_COBP is received and as long as it set TCMS shall: - command pantograph lowering - inhibit command pantograph raising	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.289
Description	ATP shall inform the line voltage catenary ELxx
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.416	TCMS shall acquire through digital intput the status of TVM output signals S_ELGV or S_ELC25KV or S_ELC1KV5	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.417	TCMS shall acquire through digital intput the status of TVM output signals S_ELGV or S_ELC25KV or S_ELC1KV5	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.702	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELGVN_SEG BSTD_ELGVR_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.703	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELC25KVN_SEG BSTD_ELC25KVR_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.704	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELC1KV5N_SEG BSTD_ELC1KV5R_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.418	DELETED				

ID	2F_09.02.Zefiro-Europe.TRS.310
Description	A check is performed at TCMS level to define misalignment between the digital inputs redundant
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.706	TCMS shall read the status of the below signals BSTD_DEFFSBN_INI BSTD_DEFFSBR_IRI BSTD_COBPN_COM BSTD_CODJRN_COM BSTD_CODJTN_COM BSTD_CRETN_COM BSTD_CFDJN_COM BSTD_COBPR_COM BSTD_CODJRR_COM BSTD_CODJTR_COM BSTD_CRETR_COM BSTD_CFDJR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.707	If the signals: BSTD_DEFFSBN_INI=0 and BSTD_DEFFSBR_IRI=0 Then compare the other digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant. If the respective signals are not alligned then show a diagnose message on DMI.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1069	If the signals: BSTD_DEFFSBN_INI=1 and BSTD_DEFFSBR_IRI=0 Then the TTF functions of PSTD-A are inhibited. Show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1070	If the signals: BSTD_DEFFSBN_INI=0 and BSTD_DEFFSBR_IRI=1 Then the TTF functions of PSTD-B are inhibited. Show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

4.1.4.2 KVB supervision

The Bi-Standard Under KVB supervision shall give outputs in order to perform train technical functions and KVB internal status faults.

4.1.4.2.1 Function design

The Bi-Standard Under KVB supervision shall give outputs in order to perform train technical functions.
ATP shall command the main circuit breaker when needed (phase and voltage transition) see. 4.1.5. and 4.1.6.
TCMS shall forward the request to the traction system that will command the MCB opening. It is then a “Train” task to assure the execution of the command.
ATP shall command the pantograph lowering command when needed (voltage transition) see 4.1.6.
TCMS shall forward the request to the traction system that will command the pantograph lowering. It is then a “Train” task to assure the execution of the command.
ATP shall command the Line voltage selection when needed (voltage transition) see 4.1.6.
TCMS shall forward the request to the driver in order to set the correct position of Line Voltage Selector. It is then a “Driver” task to assure the execution of the command.
ATP shall command the Tunnel Entry/Exit section when needed.
TCMS shall acquire the information and command Train Tunnel mode.
There are Four Train Technical Functions that are managed by the Bi-Standard in KVB supervision:

- Main Switch Off
- Lower Pantograph
- Catenary Indication
- Air Tightness

These functions are managed by the following outputs of KVB CSSP Card:

Output(s)	**TTF function**	**Input TCMS**
AODJ	Annonce ouverture DJ	BSTD_KVB_AODJ
EODJ	Exécution ouverture DJ	BSTD_KVB_EODJ
FODJ	Fin ouverture DJ	BSTD_KVB_FODJ
ABP	Annonce baisser panto	BSTD_KVB_ABP
FBP	Fin baisser panto	BSTD_KVB_FBP
ELGV	Exécution baisser panto pour entrer sur une Ligne Grande Vitesse	BSTD_KVB_ELGV
ELC25	Exécution baisser panto pour entrer en zône à 25 kV	BSTD_KVB_ELC25
ELC1,5	Exécution baisser panto pour entrer en zône à 1,5 kV	BSTD_KVB_ELC1_5
ELC3	Exécution baisser panto pour entrer en zône à 3 kV	BSTD_KVB_ELC3
ETU/STU	Announcement of tunnel / End of tunnel	BSTD_KVB_ETUST

KVB internal status faults:

Output(s)	Function	**Input TCMS**
ETATUE	Status of internal KVB recorder dedicated for maintenance	BSTD_KVB_ETAT
DV	Inconsistency between the two speeds information provided by the tachygraph ATESS 3G	BSTD_KVB_DVT

4.1.4.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.292
Description	KVB shall inform its internal status to the train
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.427	TCMS shall acquire through digital intput the status of KVB signal ETATUE	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.428	TCMS shall read the status of KVB signal ETATUE	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.429	TCMS shall acquire through digital intput the status of KVB signal DV	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.430	TCMS shall read the status of KVB signal DV	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.285
Description	ATP shall command the Tunnel Entry/Exit section.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.432	TCMS shall acquire through digital intput the status of KVB signal ETU/STU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.433	TCMS shall acquire through digital intput BSTD_KVB_ETUST the status of KVB signal ETU/STU	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.708	TCMS by IP network, shall inform Saloon HVAC unit 1 and 2, the request to close the HVAC flaps.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.434	DELETED				

ID	2F_09.02.Zefiro-Europe.TRS.311
Description	It is verified a KVB inconsistency between the two speeds information
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.710	The TCMS shall acquire the signal BSTD_KVB_DVT of the KVB inconsistency between the two speeds information provided by the tachy ATESS 3G, and show the diagnose message on TCMS DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.312
Description	KVB inform TCMS about the status of internal recorder
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.712	The TCMS shall acquire the signal BSTD_KVB_ETAT of the KVB internal recorder status, and show the diagnose message on TCMS DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.286
Description	ATP shall command Main Switch opening /closing
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.436	TCMS shall acquire through digital intput the status of KVB signal AODJ	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.437	TCMS shall acquire through digital intput BSTD_KVB_AODJ the status of KVB signal AODJ	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.438	TCMS shall prepare to open the main circuit breaker when it receives the KVB signal AODJ which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.439	TCMS shall acquire through digital intput the status of KVB signal EODJ	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.440	TCMS shall acquire through digital intput BSTD_KVB_EODJ the status of KVB signal EODJ	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.441	TCMS shall command main circuit breaker opening when it receives the KVB signal AODJ which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.442	TCMS shall acquire through digital intput the status of KVB signal FODJ	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.443	TCMS shall acquire through digital intput BSTD_KVB_FODJ the status of KVB signal FODJ	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.444	TCMS shall allow main circuit breaker closing when it receives the KVB signal FODJ is activated for 300ms	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.302
Description	ATP shall inform the line voltage catenary
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.446	TCMS shall acquire through digital intput the status of KVB signals ELGV or ELC25KV or ELC1KV5	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.447	TCMS shall acquire through digital intput the status of KVB signals ELGV or ELC25KV or ELC1KV5 BSTD_KVB_ELGV BSTD_KVB_ELC25 BSTD_KVB_ELC1_5	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.448	TCMS shall inform the driver about line voltage catenary when it receives the KVB signals ELGV or ELC25KV or ELC1KV5 which are 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.293
Description	ATP shall command the Pantograph lowering / rising
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.450	TCMS shall acquire through digital intput the status of KVB signal ABP	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.451	TCMS shall acquire through digital intput BSTD_KVB_ABP the status of KVB signal ABP	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.452	TCMS shall prepare to the pantograph lowering when it receives the KVB signal ABP which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.453	TCMS shall acquire through digital intput the status of KVB signal ELxx	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.454	TCMS shall acquire through digital intput the status of KVB signal ELxx	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.455	TCMS shall command pantograph lowering when it receives the KVB signal ELxx which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.456	TCMS shall acquire through digital intput the status of KVB signal FBP	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.457	TCMS shall acquire through digital input BSTD_KVB_FBP the status of KVB signal FBP	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.458	TCMS shall allow pantograph rising when it receives the KVB signal FBP which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.1.5 Phase Transition

4.1.5.1 Phase Transition with KVB Sequence and signalling

4.1.5.1.1 Function design

After passing the announcement balise, the KVB produces the high-active signal **S5** (AODJ = Annonce ouverture DJ) and thereby communicates to the TCMS system that there is a line section ahead on which the main switch must be open.
The TCMS system then activates the light unit VYSECT on the Driver Desk.
The pictogram will be disappearing again automatically by the TCMS system after the defined route distance of d1 = 1100 m.
Regarding operation, it is the responsibility of the driver to initiate voltage stability in the traction circuits and after successful initiation to open the main switch at the proper time.
If the driver does not manually shut off the main switch in due time after the visual request to shut off the main switch lights up, the main switch shall be shut off automatically by the TCMS system.
The command to shut off the main switch is generated via the control system after the train has covered the route distance of d2 = 900m past the announcement balise, i.e. after receiving signal **S5** (AODJ). The traction circuit undergoes a "hard" shut down here, i.e. immediately and without a torque ramp.
Normally, announcement balises are installed with successive execution balises in such a way that after a route distance of L1 = 1000m after the train crosses the announcement balise it crosses the corresponding execution balise and in doing so sends the high-active command **S0** (EODJ = Exécution ouverture DJ) to the KVB.
The command to shut off the main switch is immediately generated by the TCMS system upon crossing the execution balise (where the KVB sends the command **S0** = EODJ) (parachute function).
With receipt of the signal **S0** (EODJ), the TCMS system activates the pictogram VY_SECT on the driver desk for an additional route distance of d3 = L2 + d4 (d3 = 350m + 180m = 530m) regardless of which portion of route distance d1 has been covered after the announcement balise d4=180m =>ETR1000 IF Single unit length.
With receipt of the signal **S0** (EODJ), the TCMS system activates the pictogram VY_SECT on the driver desk for an additional route distance of d3 = L2 + d4 (d3 = 350m + 380m = 730m) regardless of which portion of route distance d1 has been covered after the announcement balise. d4=380m =>ETR1000 IF Multiple unit length.
The TCMS system ends the activation of pictogram VY_SECT on the driver desk if the end of the neutral section has been shown via the high-active signal **S6** (FODJ = Fin ouverture DJ), also if the route distance d3 has not yet been fully covered.
Important:
Announcement balises with the execution balises that follow are not necessarily installed in the track.
The signals **S5** (AODJ) and **S0** (EODJ) are therefore independent of each other and can be activated separately via the KVB, i.e. it is not necessary that both signals be activated one after another.
No main switch lock takes place due to the KVB.
Switching on of the main switch by the TCMS system within the neutral section is disabled because no line voltage is identified within the neutral zone

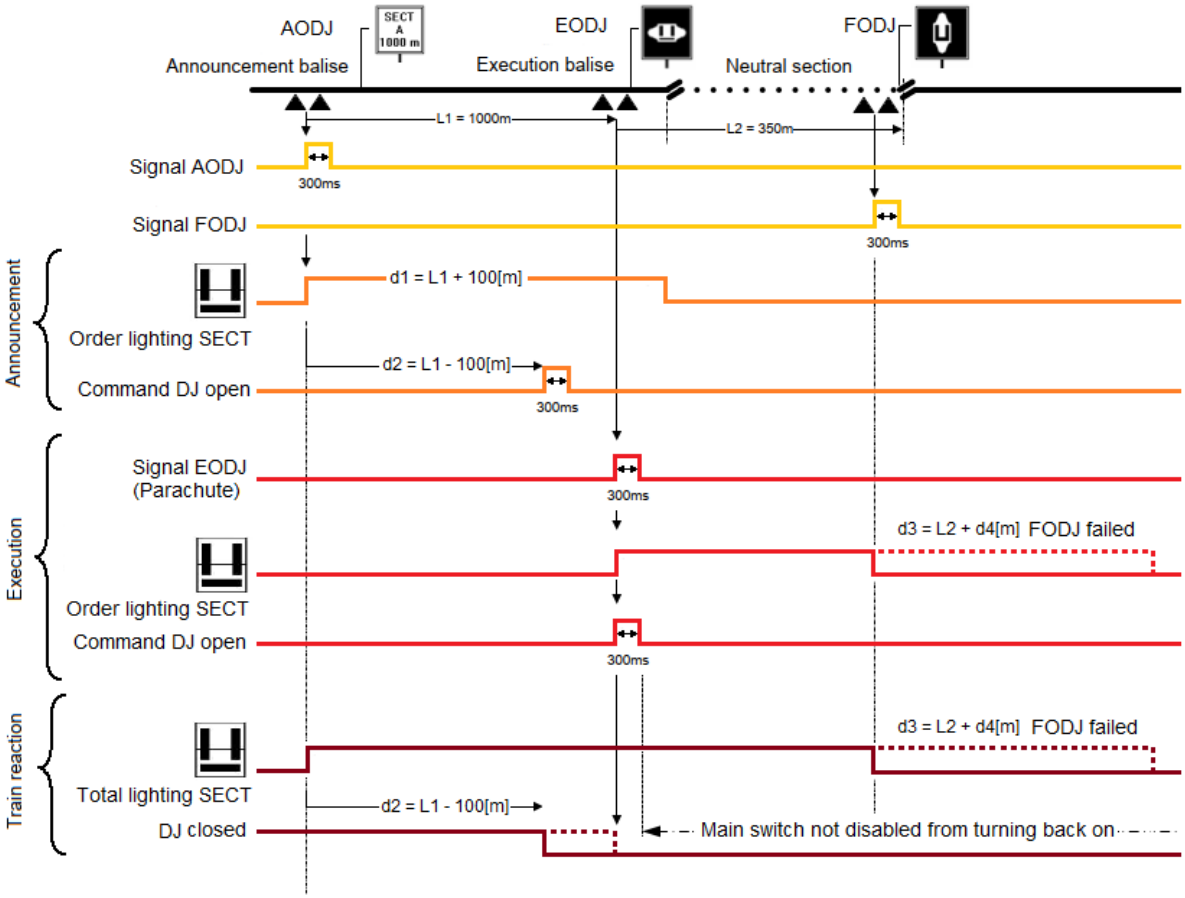


Figure 29:Phase Transition section signal sequence for switching off the main switch via the KVB.

4.1.5.1.2 Requirements

The TCMS will compute the distance where to open the MCB and will command it as described in 4.1.5.1.1
The computation of the location to command and revoke the SW traction block and MCB opening command is since the TCMS know which the active pantograph is (front or back pantograph)

ID	2F_09.02.Zefiro-Europe.TRS.294
Description	The phase separation sections and separation section sequences shall be activated by external sensors e.g. ATP Signal AODJ,EODJ,FODJ
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.656	In case of single configuration unit, the TCMS shall calculate the following distance for Phase Transition with KVB: Distance L1=1000m Distance L2=350m Distance d1=L1+100m=1000m+100m=1100m Distance d2=L1-100m=1000m-100m=900m Distance d3=L2+d4=350m+180m=530m Distance d4=180m	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1080	In case of multiple configuration unit, the TCMS shall calculate the following distance for Phase Transition with KVB: Distance L1=1000m Distance L2=350m Distance d1=L1+100m=1000m+100m=1100m Distance d2=L1-100m=1000m-100m=900m Distance d3=L2+d4=350m+380m=730m Distance d4=380m	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.468	The TCMS shall command through digital output the lighting on VY-SECT for distance d1 after receiving AODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.469	The TCMS shall command LCB opening at distance d2 after receiving the AODJ announce	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.470	If no other signals are receveid by ATP (EODJ,FODJ). The TCMS shall allow LCB closing command. After distance L1+d3 after receiving AODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.472	The TCMS shall command immediate LCB opening after receiving EODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.473	The TCMS shall command through digital output the lighting on VY-SECT after receiving EODJ command until receiving FODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.474	IF FODJ is missed The TCMS shall command through digital output the lighting on VY-SECT for distance d3 after receiving EODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.475	The TCMS shall allow LCB closing command After distance d3 after receiving EODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.477	The TCMS shall command through digital output the lighting off VY-SECT after receiving FODJ command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.478	TCMS shall allow command main circuit breaker closing when it receives the KVB signal FODJ is activated for 300ms	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.943	The sequence starts when: - VY-SECT is lighted on	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.944	The sequence is correctly terminated when: - FODJ is set or - d1 is passed from AODJ or d3 is passed from EODJ	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.313
Description	Cancellation phase separation sections and separation section sequences shall be activated by external sensors e.g. ATP Signal AODJ,EODJ,FODJ. **End of sequence:** this procedure will be performed if the driver carries out one of the following actions while the pictogram VY SECT is activated on the lamp panel on the driver desk (time for sections d1 and/or d3 after AODJ and/or EODJ): The sequence is cancelled when one of the following conditions occurs: 1. Train speed is less than 30km/h and pantographs switches are at "0" (passage for "0") or 2. The direction selector on the driver desk is set to reverse. The active function shall be cancelled in the TCMS. **Note:** If the direction switch is moved to "R", the pictogram too will be switched off. This way, the TCMS is again ready to receive further commands from the KVB. If cancellation occurs via v<30km/h and Lower Pantograph, the pictogram VY SECT remains if the train is located within route distance d1 and/or d3 after AODJ and/or EODJ. This way, the TCMS remains ready to receive further commands from the KVB within the activation phase of the pictogram VY SECT. **Note:** The pictogram VY SECT remains lit as long as it is activated and the isolation switch Z(CO)DJE (JD-S013) has not been set to position 'A+B OFF' and the isolation switch Z-KVB (JD-S017) is in the 'ON' position. All functions are interrupted and disabled by the isolation switch Z(CO)DJE (JD-S013) in position 'A+B OFF'
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.714	TCMS acquire the train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.715	TCMS control the command from pantograph switches	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.716	TCMS control the signal from direction push buttons	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.717	TCMS control the signal from Z(CO)DJE (JD-S013) rotating switch	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.718	The phase separation section sequences is cancelled when one of below conditions are verified: - train speed is less than 30km/h and pantograph switches are at "zero" (passage to "0" position) or - the direction is set to reverse at standstill or - the rotating switch Z(CO)DJE (JD-S013) has been set to position "A+B Off"	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.945	If the sequence is cancelled due to: - the direction is set to reverse then - the sequence is aborted - the pictogram VY-SECT is switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.946	If the sequence is cancelled due to: - train speed is less than 30km/h and pantograph switches are at "zero" (passage to "0" position) then - the sequence is aborted - the pictogram VY-SECT remains on until the distance d1 and/or d3 elapses	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.947	If Z-KVB (JD-S017) is not in the 'ON' position the pictogram VY-SECT is switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.948	If the Z(CO)DJE has been set to position "A+B Off" all the functions are interrupted and disabled and the pictogram VY-SECT shall be switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.949	When the sequence is cancelled the previus commands and inhibitions received and set by TCMS are reset. Signals received after the cancellation shall be considered by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.1.5.2 Phase Transition with TVM Manual Sequence and Signalling

4.1.5.2.1 Function design

The Phase transition begins with crossing the announcement balise SECT and the activation of the pictogram VY_SECT on the signalling DMI and ends after completing the section:

- D1 = L_ANNONCE + L_EXECUTION + L_DEGAGEMENT_US (in case of single configuration unit),
- D1 = L_ANNONCE + L_EXECUTION + L_DEGAGEMENT_UM (in case of multiple configuration unit).

The TVM300 system sends the commands for opening the main switch depending on the route distances covered (announcement and execution point). The first command to open (2 s impulse each time) is sent out after completing the route section D2 = L_ANNONCE – ΔlODJ and the second command after the route section L_ANNONCE.
Calculation of D1 (in case of single configuration unit):
D1 = L_ANNONCE + L_EXECUTION + L_DEGAGEMENT_US
D1 = 1124m + 205m + 180m = 1509m
Calculation of D1 (in case of multiple configuration unit):
D1 = L_ANNONCE + L_EXECUTION + L_DEGAGEMENT_UM
D1 = 1124m + 205m + 380m = 1709m
Calculation of D2:
D2 = L_ANNONCE – ΔlODJ
D2 = 1124m – 100m = 1024m

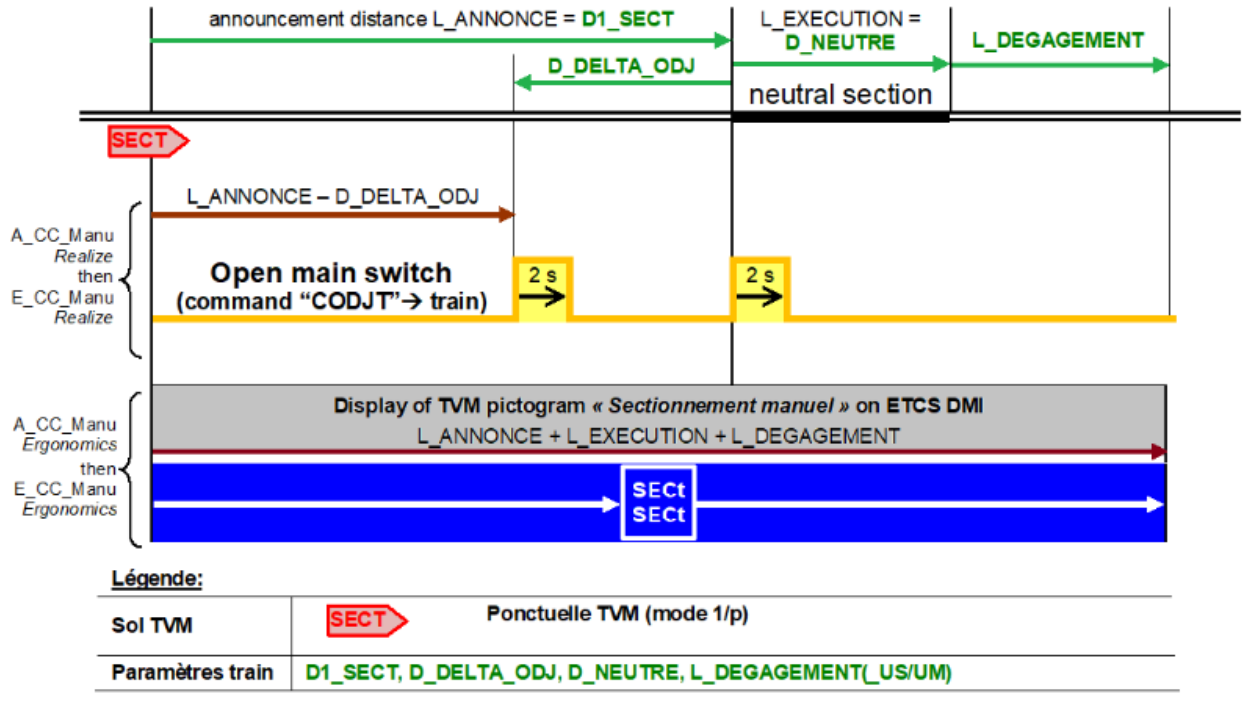


Figure 30:Phase Transition TVM Manual main switch off (TVM 1/p)

4.1.5.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.295
Description	The phase separation sections and separation section sequences in manual mode shall be activated by external sensors e.g. ATP Signal CODJT (See 4.1.4.1.3 Requirements)
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.719	TCMS shall acquire through digital intput the status of TVM signal S_CODJT	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.720	TCMS shall acquire through digital intput the status of TVM signal S_CODJT TCMS signal: BSTD_CODJTN_COM BSTD_CODJTR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.721	TCMS shall command to open the main circuit breaker when it receives the TVM signal S_CODJT which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.950	TCMS considers the sequence correctly terminated when the LCB is closed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.951	The sequence statrs when S_CODJT is received and COBP is not set	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

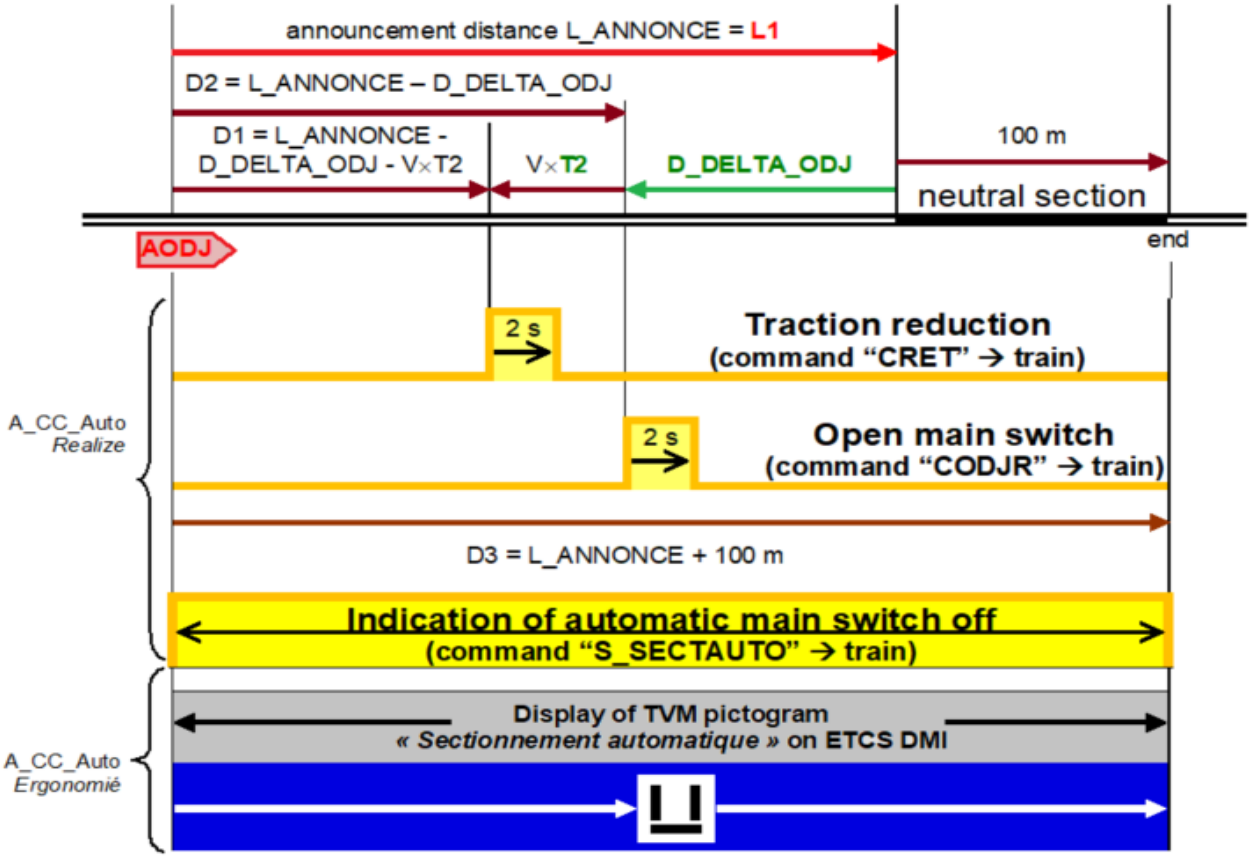
ID	2F_09.02.Zefiro-Europe.TRS.314
Description	Cancellation phase separation sections and separation section sequences in manual mode
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.723	TCMS acquire the train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.724	TCMS control the command from pantograph switches	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.725	TCMS control the signal from direction push buttons	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.726	TCMS control the signal from Z(CO)DJE (JD-S013) rotating switch	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.727	The phase separation sections and separation section sequences in manual mode <u>is cancelled</u> when one of below conditions are verified: - train speed is less than 30km/h and pantograph switches are at “zero” (passage to “0” position) or - the direction is set to reverse at standstill or - the rotating switch Z(CO)DJE (JD-S013) has been manipulated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.952	When the sequence is cancelled the previus commands and inhibitions received and set by TCMS are reset. Signals received after the cancellation shall be considered by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.1.5.3 Phase Transition with TVM Automatic Sequence and signalling

4.1.5.3.1 Function design

The Phase transition begins with crossing the announcement balise AODJ and the activation of the pictogram VY_SECTAU on the signalling DMI and ends after completing the section D3.
The TVM system sends the commands for traction reduction and for opening the main switch depending on the route distances covered.



Légende		
Sol TVM	AODJ	Ponctuelle TVM (mode n/p)
	L1	Paramètre transmis par la ponctuelle
Paramètres train	D_DELTA_ODJ, T2	
Paramètre fixe	100m	

Figure 31:Phase Transition TVM Automatic main switch off, Announcement TVM (n/p)

The EODJ function begins with crossing the execution balise EODJ and the sending of the command to open the main switch and ends with the disappearing of the pictogram VY_SECTAU on the signalling DMI after completing the section D4.

In this case of single configuration unit, D4 = L_EXECUTION + L_DEGAGEMENT_US (with L_DEGAGEMENT_US = 180m).
In case of multiple configuration unit, D4 = L_EXECUTION + L_DEGAGEMENT_UM (with L_DEGAGEMENT_UM = 380m).
L_EXECUTION is a value transmitted through the execution balise EODJ.
After a distance D5=L_EXECUTION+L_PT_AR, the Bi-Standard sends the command CFDJ to ask the train to close its main switch (with L_PT_AR=180m both in single configuration unit and multiple configuration unit).

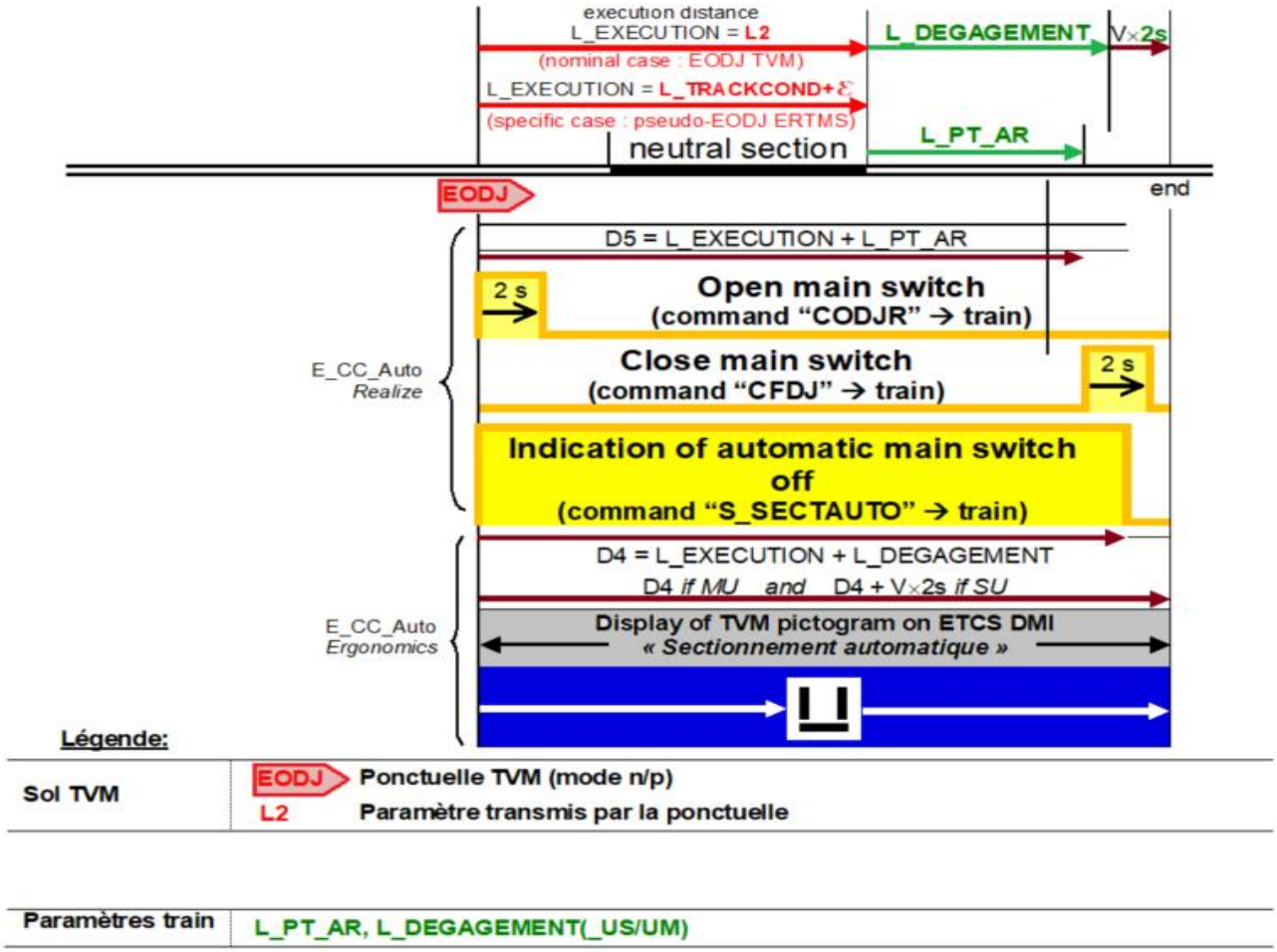


Figure 32:Phase Transition TVM Automatic main switch off, Execution TVM (n/p)

4.1.5.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.296
Description	The phase separation sections and separation section sequences in automatic mode shall be activated by external sensors e.g. ATP Signal CRET,CODJR,CFDJ (See 4.1.4.1.3 Requirements)
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.728	TCMS shall acquire through digital intput the status of TVM signal S_CRET	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.729	TCMS shall acquire through digital intput the status of TVM signal S_CRET	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.730	TCMS shall command traction reduction of traction effort to zero with a ramp with duration 5s when it receives the TVM signal S_CRET, which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.731	TCMS shall start the distance calculation by 2500m when S_CRET is received	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.732	TCMS shall acquire through digital intput the status of TVM signal S_CODJR	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.733	TCMS shall acquire through digital intput the status of TVM signal S_CODJR BSTD_CODJRN_COM BSTD_CODJRR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.734	TCMS shall open main circuit breaker when it receives the TVM signal S_CODJR which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.735	TCMS shall maintain the inhibition to the closing of the main switch for 2500m when the S_CRET is received. In case the signals of the following points did not arrive; after 2500m the authorization to close the main switch depends on the master controller in "coasting" position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.736	The sequence starts when: - CRET is received or - CODJR is received	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.737	The sequence is correctly terminated when: - CFDJ is received or - 2500m or 2100m elapses	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
Note: the actions foreseen in this phase will also be performed if the signals of the announcement phase have not arrived.					
2F_09.02.Zefiro-IF.CONCEPT.739	TCMS shall command instant DJ opening	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.740	TCMS shall leave the constraints linked to the distance of 2500m	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.741	TCMS shall start calculating distance 2100m when: - the S_CODJR is received and S_CRET is missing or - the S_CODJR is twice received after S_CRET	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

2F_09.02.Zefiro-IF.CONCEPT.742	TCMS shall keep the inhibition to the closing of the DJ for 2100m, if the following points were not received; once reaced 2100m the authorization to close the main switch depends on the master controller in "coasting" position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.743	TCMS shall acquire through digital intput the status of TVM signal S_CFDJ	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.744	TCMS shall acquire through digital intput the status of TVM signal S_CFDJ TCMS signal BSTD_CFDJN_COM BSTD_CFDJR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.745	TCMS shall command main circuit breaker closing when it receives the TVM signal S_CFDJ which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.953	TCMS shall leave of constraints linked to the distance of 2100m	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.954	TCMS shall command the immediate closure of DJ (Traction resumption with effort defined by the current lever)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.955	The CFDJ signal (closing of the DJ from TVM430) shall be masked by the logic and not memorized if it arrives outside the sequence "Automatic phase change from TVM430".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.956	The CFDJ signal shall be used and stored only within the described sequence and shall not have influence on other TCMS functions.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.315
Description	Cancellation phase separation sections and separation section sequences in automatic mode. **End of sequence in degraded mode**: this procedure will be performed if not all the expected signals arrive. 1. Missing arrival of the "Execution" and "End of sequence" signals (only "Announcement"): The authorization to close the DJ depends to the "zero" position of the traction manipulator, after 2500m from the arrival of the CRET signal. 2. Missing arrival of the "End Sequence" signal (only Announcement + Execution): The authorization to close the DJ depends to the "zero" position of the current manipulator, after 2100m from the arrival of the CODJR signal of the execution phase.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.747	TCMS acquire the train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.748	TCMS control the command from pantograph switches	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.749	TCMS control the signal from direction push buttons	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.750	TCMS control the signal from Z(CO)DJE (JD-S013) rotating switch	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.751	The phase separation sections and separation section sequences in automatic mode <u>is cancelled</u> when one of below condintions are verified: - train speed is less than 30km/h and pantograph switches are at “zero” (passage to “0” position) or - the direction is set to reverse at standstill or - the rotating switch Z(CO)DJE (JD-S013) has been manipulated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.957	When the sequence is cancelled the previus commands and inhibitions received and set by TCMS are reset. Signals received after the cancellation shall be considered by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.1.6 Voltage Transition

4.1.6.1 Voltage Transition with KVB Sequence and signalling

4.1.6.1.1 Function design

After crossing the announcement balise, the KVB generates the high-active signal **S7** (ABP = Annonce baisser panto) in order to communicate to the TCMS system that there is a line section ahead on which the pantograph must be lowered.
The TCMS system then activates the pictogram VY_BPT on the driver desk.
The light element will be switched off again automatically by the TCMS system after the defined route distance of d1 = 1500m.
Regarding operation, it is the responsibility of the driver to lower the pantograph in due time (lowering the pantograph always implies having shut off the main switch beforehand).
If the driver does not manually lower the pantograph in due time after the visual request to lower the pantograph lights up, the main switch shall be shut off automatically by the TCMS system and the pantograph automatically lowered.
The command to lower the pantograph (and thus to shut off the main switch) is generated by the control system after the train has covered the route distance d2 = L1 - 100m – 8s x v (where v = speed in m/s) after the announcement balise, i.e. after receiving signal **S7** ABP.
The traction undergoes a "hard" shut down here, i.e. immediately and without a torque ramp. The TCMS system activates the lowering of the pantograph up to the end of route distance d1.
Normally, announcement balises are installed with successive execution balises in such a way that after a route distance of L1 = 1400m after the train crosses the announcement balise it crosses the corresponding execution balise and in doing so sends the high-active command:
S1 (*"EBP"* > ELC25 = Exécution baisser panto pour entrer en zône à 25 kV) or
S2 (*"EBP"* > ELC1.5 = Exécution baisser panto pour entrer en zône à 1,5 kV) or
S3 (*"EBP"* > ELGV = Exécution baisser panto pour entrer sur une Ligne Grande Vitesse) or
The command to lower the pantograph is generated immediately in the TCMS system upon crossing the execution balise (where the KVB sends the command **S1** (*"EBP"* > ELC25 = Exécution baisser panto pour entrer en zône à 25 kV) or **S2** (*"EBP"* > ELC1.5 = Exécution baisser panto pour entrer en zône à 1,5 kV) or **S3** (*"EBP"* > ELGV = Exécution baisser panto pour entrer sur une Ligne Grande Vitesse).
With receipt of this information, the TCMS system activates the pictogram VY_BPT on the driver desk for an additional route distance of d3 = L2 + d4 (d3 = 450m + 180m = 630m) regardless of which portion of route distance d1 has been covered after the announcement balise. The d4 distance is Single unit composition of ETR1000 IF.
With receipt of this information, the TCMS system activates the pictogram VY_BPT on the driver desk for an additional route distance of d3 = L2 + d4 (d3 = 450m + 380m = 830m) regardless of which portion of route distance d1 has been covered after the announcement balise. The d4 distance is Multiple unit composition of ETR1000 IF.
The pictogram VY_BPT on the driver desk is likewise switched off by the TCMS system if the end of the section is shown by the KVB via the high-active signal **S8** (FBP = Fin baisser panto), at which the pantograph must be lowered.
Regardless of whether the pantograph has been lowered by the driver in due time or automatically, the TCMS system saves which voltage system was requested by the KVB, that is, which of the three signals **S1, S2. S3** was sent.
This information aids in checking the voltage system selection of the driver on the Line voltage selector. If the manual selection of the voltage system does not match the specification of the KVB, then the raising of the pantograph will be disabled by the TCMS system beyond the indicator light cycle until the driver selection has been corrected.
Disabling of pantograph raising can be withdrawn with the cancellation conditions below. The voltage system selection is set to OK here automatically.
If the driver attempts to raise the wrong pantograph, the diagnostics EVENT "Train voltage system not same as specification of train protection system" is generated, which is shown to the driver on the DMI TCMS.
Important:
No pantograph disabling takes place due to the KVB.
Switching on of the main switch by the TCMS system within the neutral section is disabled because no line voltage is identified within the neutral zone.

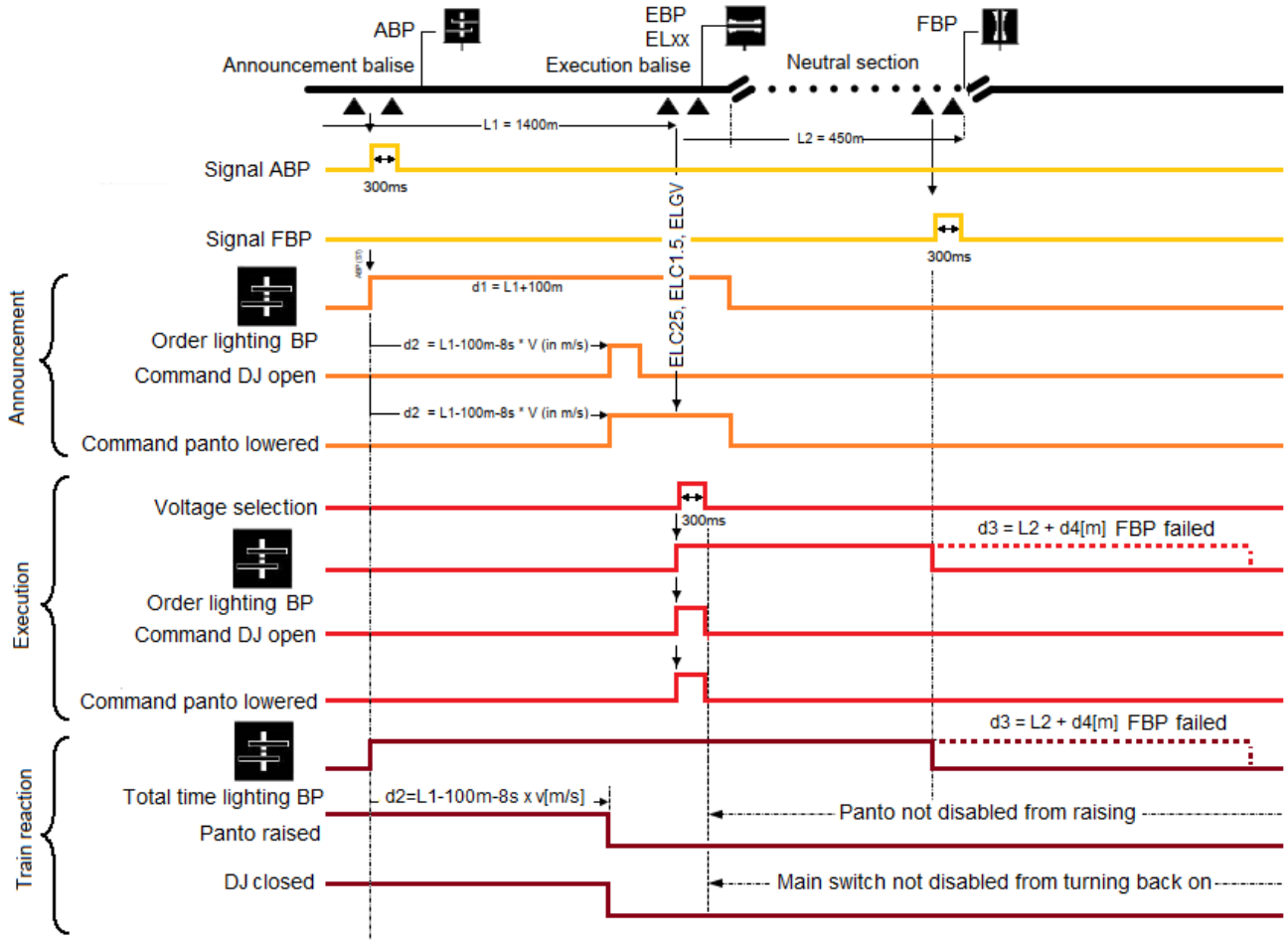


Figure 33:Voltage Transition section signal sequence for lowering of the pantograph via the KVB.

NOTE: the signal *"EBP"* does not foreseen as CSSP function output of the KVB, the TCMS shall calculate the distance d1, the TCMS will receive only one of the signals ELxx by balise EBP

4.1.6.1.2 Requirements

The TCMS will compute the distance where to lower the pantograph and open the MCB and will command it as described in 4.1.6.1.1.
The computation of the location to command and revoke the SW traction block, pantograph lowering and MCB opening command is since the TCMS know which the pantograph is active (front or back pantograph, and which system Line voltage is selected).

ID	2F_09.02.Zefiro-Europe.TRS.297
Description	The voltage separation sections and separation section sequences shall be activated by external sensors e.g. ATP (Signal ABP,FBP,ELC25,ELC1,5,ELGV)
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.657	In case of single configuration unit, the TCMS shall calculate the following distance for Voltage Transition with KVB: Train speed =v (m/s) Distance L1=1400m Distance L2=450m Distance d1=L1+100m=1400m+100m=1500m Distance d2=L1-100m-[8s x v(m/s)] Distance d3=L2+d4=450m+180m=630m Distance d4=180m	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1081	In case of multiple configuration unit, the TCMS shall calculate the following distance for Voltage Transition with KVB: Train speed =v (m/s) Distance L1=1400m Distance L2=450m Distance d1=L1+100m=1400m+100m=1500m Distance d2=L1-100m-[8s x v(m/s)] Distance d3=L2+d4=450m+380m=830m Distance d4=380m	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.507	The TCMS shall command through digital output the lighting on VY-BPT for distance d1 after receiving ABP command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.508	The TCMS shall prepare LCB opening at distance d2 after receiving ABP command.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.509	The TCMS shall command pantograph lowering at distance d2 after receiving ABP command and keep it commanded until covering distance d1 from ABP command.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.510	If no other signals are receveid by ATP (ELxx, FBP). The TCMS shall allow pantograph rising command and LCB closing command. After distance =L1+d3 after receiving ABP command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.513	The TCMS shall command immediate LCB opening command after receiving ELxx command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.514	The TCMS shall command immediate Pantograph lowering command after receiving ELxx command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.515	The TCMS shall command through digital output the lighting on VY-BPT after receiving ELxx command until receiving FBP command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.516	The TCMS shall do a congruency check between the line voltage ELxx command trasmitted by ATP,and Line voltage selector position commanded by the driver after receiving ELxx command	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.517	After receiving ELxx command the TCMS shall inhibit pantograph raising until Line voltage selector position commanded by the driver is congurent with atp information ELxx.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0 -5
2F_09.02.Zefiro-IF.CONCEPT.519	IF FBP is missed The TCMS shall command through digital output the lighting on VY-BPT after receiving ELxx command for distance d3	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.520	The TCMS shall allow pantograph rising command and LCB closing command after receiving ELxx command After distance d3	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.523	The TCMS shall command through digital output the lighting off VY-BPT after receiving FBP command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.524	The TCMS shall allow pantograph rising command and LCB closing command after receiving FBP command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/ h)
2F_09.02.Zefiro-IF.CONCEPT.752	TCMS shall acquire through digital intput the status of KVB signal ABP	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.753	TCMS shall acquire through digital intput BSTD_KVB_ABP the status of KVB signal ABP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.754	TCMS shall prepare to the pantograph lowering when it receives the KVB signal ABP which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.755	TCMS shall start calculation distance L1 (1400m)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.756	Activate the lighting VY-BPT by TCMS outputs BSTD_KVB_BPTR BSTD_KVB_BPTN and keep it active for L1 (1400m)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.757	After a distance of d2, TCMS shall command immediate opening of the DJ	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.758	After a distance of d2, TCMS shall command lowering of the pantographs and inhibition of manual control of the pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.759	TCMS shall acquire through digital intput the status of KVB signal ELxx	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.760	TCMS shall acquire through digital intput the status of KVB signal ELxx	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.761	TCMS shall command pantograph lowering when it receives the KVB signal ELxx which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
Note: the actions planned in this phase will also be performed if the signals of the announcement phase have not arrived					
2F_09.02.Zefiro-IF.CONCEPT.763	TCMS shall leave the constraints linked to the calculation distance of L1 (1400m)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.764	TCMS shall start the distance L2 (450m) calculation	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.765	Activate the lighting VY-BPT by TCMS outputs BSTD_KVB_BPTR BSTD_KVB_BPTN and keep it active for d3=L2+d4	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.766	TCMS shall command instant DJ opening	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.767	TCMS shall command immediate lowering of the pantographs and inhibition of the command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.768	TCMS shall storage of the type of catenary that will be engaged	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.769	TCMS shall acquire through digital intput the status of KVB signal FBP	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.770	TCMS shall acquire through digital input BSTD_KVB_FBP the status of KVB signal FBP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.771	TCMS shall allow pantograph rising when it receives the KVB signal FBP which is 300ms long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
Note: If FBP is missing the sequence must be considered completed after d3=L2+ d4[m] from the arrival of the execution signal ELxx					
2F_09.02.Zefiro-IF.CONCEPT.773	When the pantograph lift is required TCMS shall verify the congruence between the position of the catenary selector when the train reaching the distance of d3=L2+d4 [m] from ELxx	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.774	TCMS shall authorize for the manual control of raised of the pantograph in the presence of congruency and when the train reaching the distance of d3=L2+d4 [m] from ELxx	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.775	Deactivate the lighting VY-BPT by TCMS outputs BSTD_KVB_BPTR BSTD_KVB_BPTN	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.316
Description	Voltage separation sections and separation section sequences shall be activated by external sensors e.g. ATP (Signal ABP,FBP,ELC25,ELC1,5,ELGV) **End of sequence in degraded mode:** this procedure will be performed if not all the expected signals arrive. 1. Missing arrival of “Execution” signals: deactivation of “Pantograph Lowering” signal after L1+d3m from the arrival of the ABP signal and authorization to the manual command to raise the pantographs (concept 510). **Cancellation sequence**: this procedure will be performed if the driver carries out one of the following actions while the pictogram VY BP is activated the lamp panel on the driver desk (time for sections d1 and/or d3 after ABP and/or ELxx): 1. the train runs more slowly than 30km/h and the driver press the button Lower Pantograph or 2. the direction switch is moved to "Reverse". The active function shall be cancelled in the TCMS.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.777	TCMS acquire the train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.778	TCMS control the command from pantograph switches	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.779	TCMS control the signal from direction push buttons	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.780	The voltage separation section sequences is cancelled when one of below conditions are verified: - train speed is less than 30km/h and pantograph switches are at “zero” (passage to “0” position) or - the direction is set to reverse at standstill	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.959	If the sequence is cancelled due to: - the direction is set to reverse then - the sequence is aborted - the pictogram VY-BPT is switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.960	If the sequence is cancelled due to: - train speed is less than 30km/h and pantograph switches are at “zero” (passage to “0” position) then - the sequence is aborted - the pictogram VY-BPT remains on until the distance d1 and/or d3 elapses	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.961	If Z-KVB (JD-S017) is not in the 'ON' position the pictogram VY-BPT is switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.962	If the Z(CO)DJE has been set to position “A+B Off” all the functions are interrupted and disabled and the pictogram VY-BPT shall be switched off	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.963	When the sequence is cancelled the previus commands and inhibitions received and set by TCMS are reset. Signals received after the cancellation shall be considered by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.1.6.2 Voltage Transition with TVM Manual Sequence and signalling

4.1.6.2.1 Function design

The BPT function begins with crossing the announcement balise BPT and the activation of the pictogram VY_BPT on the DMI signalling and ends after completing the section:

- L_ANNONCE + L_EXECUTION + L_DEGAGEMENT_US (in case of single configuration unit),
- L_ANNONCE + L_EXECUTION + L_DEGAGEMENT_UM (in case of multiple configuration unit).

The TVM system sends the commands for opening the main switch, for lowering the pantograph and ELxx (information on voltage system) depending on the route distances covered.
In case of manual sequence TVM activate all the ELXX signals.

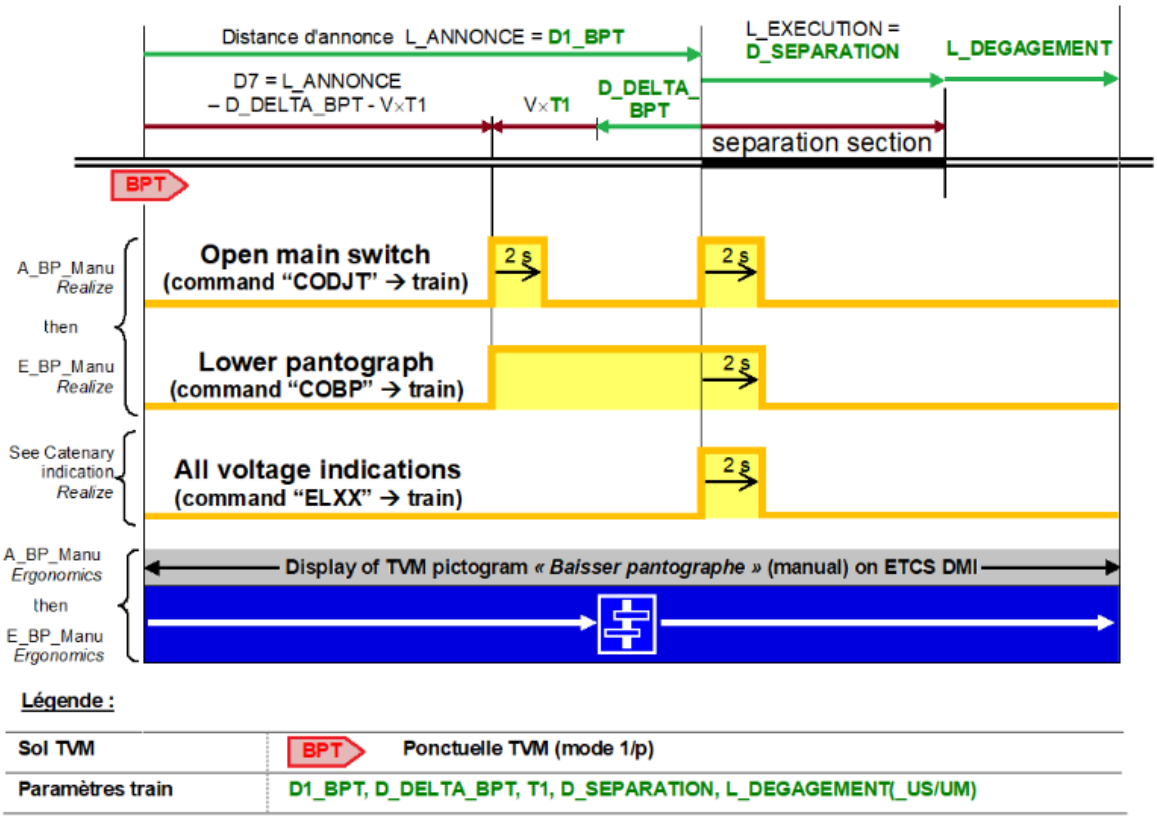


Figure 34:Voltage Transition TVM "Manual" with TVM300

4.1.6.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.298
Description	The voltage separation sections and separation section sequences manual mode shall be activated by external sensors e.g. ATP Signal CODJT,COBP,ELC25,ELC1,5,ELGV (See 4.1.4.1.3 Requirements)
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.781	TCMS shall acquire through digital intput the status of TVM signal S_CODJT	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.782	TCMS shall acquire through digital intput the status of TVM signal S_CODJT TCMS signal: BSTD_CODJTN_COM BSTD_CODJTR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.783	TCMS shall command to open the main circuit breaker when it receives the TVM signal S_CODJT which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.784	TCMS shall acquire through digital intput the status of TVM signal S_COBP.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.785	TCMS shall acquire through digital intput the status of TVM signal S_COBP S_COBP signal is activated at high level.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.786	When S_COBP is received and as long as it set TCMS shall: - command pantograph lowering - inhibit command pantograph raising	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.787	TCMS shall acquire through digital intput the status of TVM output signals S_ELGV S_ELC25KV S_ELC1KV5	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.788	TCMS shall acquire through digital intput the status of TVM output signals S_ELGV S_ELC25KV S_ELC1KV5	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.789	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELGV_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.790	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELC25KV_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.791	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELC1KV5_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

2F_09.02.Zefiro-IF.CONCEPT.792	TCMS shall inform the driver line voltage catenary when it receives the TVM output signals S_ELGV S_ELC25KV S_ELC1KV5 which are 2s long	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.793	If TCMS receives the following signals S_ELGV S_ELC25KV S_ELC1KV5 TCMS shall command pantograph lowering	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.964	The sequest strarts when: - CODJT and COBP is received	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.965	TCMS considers the sequence correctly terminated when the LCB is closed	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

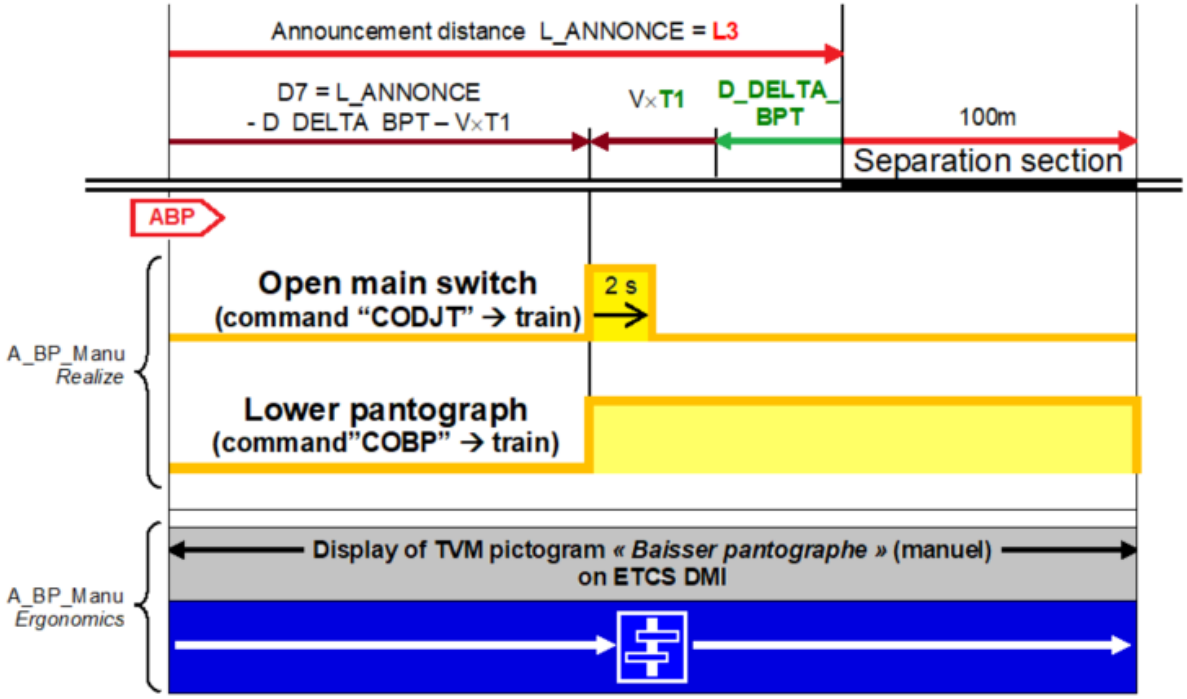
ID	2F_09.02.Zefiro-Europe.TRS.317
Description	Cancellation voltage separation sections and separation section sequences in manual mode **Note:** Once the pantograph has been lifted and before the consensus to the DJ closure, the TCMS will do a consistency check between the detected voltage value and the position of the catenary voltage selector. In the event of discrepancy, the pantograph will be automatically lowered, and a monitor message will inform the driver of the consistency error.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.795	TCMS acquire the train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.796	TCMS control the command from pantograph switches	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.797	TCMS control the signal from direction push buttons	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.798	TCMS control the signal from Z(CO)DJE (JD-S013) rotating switch	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.799	The voltage separation sections and separation section sequences in manual mode is cancelled when one of below conditions are verified: - train speed is less than 30km/h and pantograph switches are at “zero” (passage to “0” position) or - the direction is set to reverse at standstill or - the rotating switch Z(CO)DJE (JD-S013) has been manipulated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.966	When the sequence is cancelled the previus commands and inhibitions received and set by TCMS are reset. Signals received after the cancellation shall be considered by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.1.6.3 Voltage Transition with TVM Automatic Sequence and signalling

4.1.6.3.1 Function design

The ABP function begins with crossing the announcement balise ABP and the activation of the pictogram VY_BPT (Voyant Baisser Pantograph) on the DMI signalling and ends after completing the section L3 + 100m (L3 = L_ANNONCE). The TVM system sends the commands fopening the main switch and for lowering the pantograph depending on the route distances covered.
In case of automatic sequence (TVM430) TVM activate only the ELXX signal related to the next line voltage.



Légende :

Sol TVM	ABP **Ponctuelle TVM (mode n/p)** L3 **Paramètre transmis par la ponctuelle**
Paramètres train	D_DELTA_BPT, T1
Paramètre fixe	100m

Figure 35:Voltage Transition TVM Announcement phase Automatic sequence

The “EBP” function begins with crossing the execution balise EBP and the sending of the commands ELxx to open the main switch, to lower the pantograph the signal ELxx give the information on voltage system and ends with the completion of the section:

- L_EXECUTION + L_DEGAGEMENT_US (where L_DEGAGEMENT_US = 180m; single unit),
- L_EXECUTION + L_DEGAGEMENT_US (where L_DEGAGEMENT_UM = 380m; multiple unit).

L_EXECUTION is a value calculated through the execution balise EBP signal ELxx.

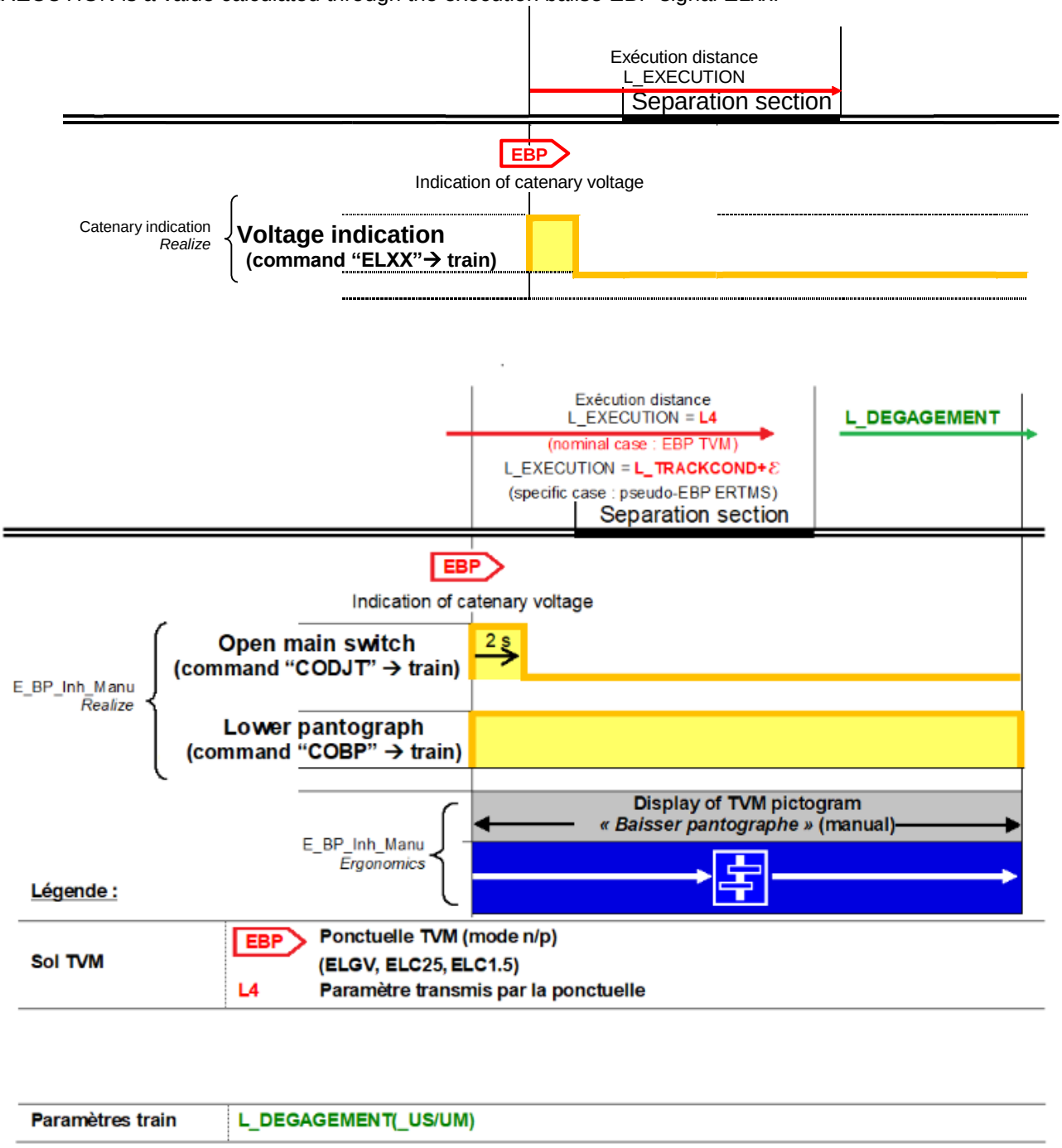


Figure 36:Voltage Transition TVM Execution phase Automatic sequence

4.1.6.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.321
Description	The voltage separation sections and separation section sequences automatic mode shall be activated by external sensors e.g. ATP Signal CODJT,COBP,ELC25,ELC1,5,ELGV (See 4.1.4.1.3 Requirements)
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.800	TCMS shall acquire through digital intput the status of TVM signal S_CODJT	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.801	TCMS shall acquire through digital intput the status of TVM signal S_CODJT TCMS signal: BSTD_CODJTN_COM BSTD_CODJTR_COM	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.802	TCMS shall command to open the main circuit breaker when it receives the TVM signal S_CODJT which is 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.803	TCMS shall acquire through digital intput the status of TVM signal S_COBP.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.804	TCMS shall acquire through digital intput the status of TVM signal S_COBP S_COBP signal is activated at high level.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.805	When S_COBP is received and as long as it set TCMS shall: - command pantograph lowering - inhibit command pantograph raising	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.806	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.807	TCMS shall acquire through digital intput the status of TVM output signals S_ELGV or S_ELC25KV or S_ELC1KV5	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW	SIL0
2F_09.02.Zefiro-IF.CONCEPT.808	TCMS shall acquire and through digital intput the status of TVM output signals S_ELGV or S_ELC25KV or S_ELC1KV5	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.809	TCMS memorize the signal and start counting three minutes	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.810	After 3 minutes from receiving one of the signals ELC1.5, ELC25, ELGV and the COBP doesn't go to zero: - the TCMS will authorize the manual command to lift the pantographs and the closing of the Main switch regardless of the status of the other signals - the TCMS reset the data of the stored catenary and allow to rise any type of pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.811	After 3 minutes if COBP is set to high level the TCMS consider it only when a new CODJT is received	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

2F_09.02.Zefiro-IF.CONCEPT.812	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELGV_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.813	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELC25KV_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.814	TCMS shall acquire through digital intput the status of TVM signal BSTD_ELC1KV5_SEG compare the digital inputs N with the digital inputs R and verify misalignment between DI Normal and DI Redundant	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.815	TCMS shall inform the driver line voltage catenary when it receives the TVM output signals S_ELGV or S_ELC25KV or S_ELC1KV5 which are 2s long.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.967	When the signal S_COBP goes to zero TCMS verify the congruency between the catenary signal and line voltage selector position commanded by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.968	If the congruency between the catenary signal and line voltage selector position commanded by the driver is verified then TCMS remove the inhibition to raise the pantograph and the driver can command the pantograph up	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.969	The sequest strarts when: - CODJT and COBP is received	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.970	TCMS considers the sequence correctly terminated when: - COBP is set to 0 for 2 sec or - 3 minutes counter lapses	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.318
Description	Cancellation voltage separation sections and separation section sequences in automatic mode End of sequence in degraded mode: this procedure will be performed if not all the expected signals arrive. 1. Missing arrival of the "Announcement" signal: Execution and End sequence phase is applied 2. Missing arrival of "Execution" signals: authorization to manually control the pantograph
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.817	TCMS acquire the train speed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.818	TCMS control the command from pantograph switches	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.819	TCMS control the signal from direction push buttons	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.820	TCMS control the signal from Z(CO)DJE (JD-S013) rotating switch	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.821	The voltage separation sections and separation section sequences in automatic mode is cancelled when one of below conditions are verified: - train speed is less than 30km/h and pantograph switches are at “zero” (passage to “0” position) or - the direction is set to reverse at standstill or - the rotating switch Z(CO)DJE (JD-S013) has been manipulated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.971	When the sequence is cancelled the previus commands and inhibitions received and set by TCMS are reset. Signals received after the cancellation shall be considered by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1071	When the EBP signal is missing and S_COBP goes to zero: - TCMS will authorize the manual command to lift the pantograph and close the main circuit breaker.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

4.2 Juridical Recorder Unit (ATESS)

4.2.1 Function design

The train recorder (ATESS JRU) shall record signal as per SAM S 704, coming from the train through MVB (TCMS) or through digital input (Hardware).

4.2.1.1 Normal condition

ATESS JRU records train signal as per SAM S 704.

4.2.1.2 Failure Cases

A. Fault recorder
B. Fault speed recorder.
C. Removable Cartridge presence fault

4.2.1.3 Failure Detection

A. ATESS JRU itself auto detected
B. ATESS JRU itself auto detected
C. ATESS JRU itself auto detected

4.2.1.4 Failure Handling

A. Isolation selector Z-ENR
B. Isolation selector Z-ENR
C. Isolation selector Z-ENR

4.2.1.5 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.319
Description	The train recorder (ATESS JRU) restart after 40th hours of continuous service
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.823	TCMS shall have an hours counter for service in France	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.824	TCMS shall increase the counter when: the Bistandard signalling is working in France mode (JD-S012 Selector I-F), see input signals: BSTD_SEL_I-F_1 BSTD_SEL_I-F_2 BSTD_SEL_I-F_3 and the power supply contactors (60 min) output signals are in one of the following configurations: OTC_BAT_BUS1.60=1 OTC_BAT_BUS2.60=1 or OTC_BAT_BUS1.60=0 OTC_BAT_BUS2.60=1 or OTC_BAT_BUS1.60=1 OTC_BAT_BUS2.60=0	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.825	TCMS after 18 hours shall show on DMI a message to the driver that within 3 hours a reset of Bistandard.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1072	TCMS shall reset the counter when: the selector JD-S012 is set to Italy. or the output signals are in the following configuration: OTC_BAT_BUS1.60=0 OTC_BAT_BUS2.60=0	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.826	Everytime the Bistandard signalling is restarted, the counter shall be set back to 0 and restarted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

ID	2F_09.02.Zefiro-Europe.TRS.299
Description	The train recorder (ATESS JRU) shall give itself diagnostic outputs faults, in case of fault shall be isolated.
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.560	TCMS shall read through digital input the status of Z-ENR selector postions signals BSTD_ATEREC_SWON BSTD_ATEREC_SWOFF and verify the coherence, show the diagnose message on DMI in case of incoherence	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.1073	If the signals are in the below state BSTD_ATEREC_SWON=1 BSTD_ATEREC_SWOFF=0 Then the selector is set in 'On'	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1074	If the signals are in the below state BSTD_ATEREC_SWON=0 BSTD_ATEREC_SWOFF=1 Then the selector is set in 'Off''	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1075	In case of incoherence between Z-ENR signal position (both signals at zero or at one) show the diagnose message on DMI	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.561	TCMS shall read through digital input the status of Recorder Fault case A BSTD_ATESS_R3ENR	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.658	TCMS shall read through digital input the status of Recorder Fault case B BSTD_ATESS_VENR	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.659	TCMS shall read through digital input the status of Recorder Fault case C BSTD_ATESS_R2ENR	Derived Requirement	4S_09.03.05.-.TCMS Softwa re		SIL0
2F_09.02.Zefiro-IF.CONCEPT.562	TCMS shall read the status of Z-ENR selector postion is OFF	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.563	TCMS shall read the status of ATESS RECORDER Fault Case A BSTD_ATESS_R3ENR	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.660	TCMS shall read the status of ATESS RECORDER Fault Case B BSTD_ATESS_VENR	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.661	TCMS shall read the status of ATESS RECORDER Fault Case C BSTD_ATESS_R2ENR	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.827	TCMS shall read the status of ATESS tachometry V1 Fault BSTD_ATESS_V1	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.828	TCMS shall read the status of ATESS tachometry V2 Fault BSTD_ATESS_V2	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.829	TCMS shall read the status of ATESS tachometry V3 Fault BSTD_ATESS_V3	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0
2F_09.02.Zefiro-IF.CONCEPT.830	TCMS shall read the status of ATESS 3km/h activation threshold tachometry V3 Fault BSTD_ATESS_SVMV	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL0

ID	2F_09.02.Zefiro-Europe.TRS.320
Description	The train recorder (ATESS JRU) shall record analog signals as per SAM S704, coming from the train through MVB (TCMS)
Safety level	SIL0

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL / TFFR (events/h)</th></tr>
<tr><td>2F_09.02.Zefiro-IF.CONCEPT.832</td><td>TCMS shall trasmitt via MVB to ATESS(JRU) the status of presence of tension in AC and DC lines
Voltage is present=1
Voltage is absent =0</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL0</td></tr>
<tr><td>2F_09.02.Zefiro-IF.CONCEPT.833</td><td>If the AC or DC line voltage is less to 200V for 10s, TCMS shall trasmitt via MVB to ATESS(JRU) the digital status: absence of line voltage (absence/not absence)</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL0</td></tr>
<tr><td>2F_09.02.Zefiro-IF.CONCEPT.834</td><td>TCMS shall trasmitt via MVB to ATESS(JRU) the level of the maximun line current</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL0</td></tr>
<tr><td>2F_09.02.Zefiro-IF.CONCEPT.835</td><td>In the AC lines the maximum current selectable is 1120A, the current step minimum is 80A.
At every 80A positive step, TCMS shall trasmit via MVB to ATESS(JRU) the coded values.
<table>
<tr><th colspan="5">Max current selectable under 25kVAC lines</th></tr>
<tr><th>Current range</th><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th></tr>
<tr><td>Sélection ≤80A</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >80A & ≤160A</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>Sélection >160A & ≤240A</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>Sélection >240A & ≤320A</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>Sélection >320A & ≤400A</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >400A & ≤480A</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>Sélection >480A & ≤560A</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>Sélection >560A & ≤640A</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Sélection >640A & ≤720A</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >720A & ≤800A</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>Sélection >800A & ≤880A</td><td>1</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>Sélection >880A & ≤960A</td><td>1</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>Sélection >960A & ≤1040A</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >1040A & ≤1120A</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>Reserve</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL0</td></tr>
<tr><td>2F_09.02.Zefiro-IF.CONCEPT.836</td><td>In the DC line the maximum current selectable is 6500A, the current step minimum is 100A.
At every 500A positive step, TCMS shall trasmit via MVB to ATESS(JRU) the coded values
<table>
<tr><th colspan="5">Max current selectable under 1,5kVDC line</th></tr>
<tr><th>Current range</th><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th></tr>
<tr><td>Sélection ≤100A</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >100 & ≤500A</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>Sélection >500A & ≤1000A</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>Sélection >1000A & ≤1500A</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>Sélection >1500A & ≤2000A</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >2000A & ≤2500A</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>Sélection >2500A & ≤3000A</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>Sélection >3000A & ≤3500A</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Sélection >3500A & ≤4000A</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >4000A & ≤4500A</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>Sélection >4500A & ≤5000A</td><td>1</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>Sélection >5000A & ≤5500A</td><td>1</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>Sélection >5500A & ≤6000A</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>Sélection >6000A & ≤6500A</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>Reserve</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL0</td></tr>
</table>

ID	2F_09.02.Zefiro-Europe.TRS.300

Description	The train recorder (ATESS JRU) shall record signal as per SAM S704, coming from the train through MVB (TCMS) or through digital input
Safety level	SIL0

Nr	Description	Type	Derived to	User Interface	SIL / TFFR (events/h)
2F_09.02.Zefiro-IF.CONCEPT.565	TCMS shall acquire the status of Horn activation(Activated /Not Activated) signal: OTC_HORN_HIGH or OTC_HORN_LOW	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.566	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Horn activation(Activated /Not Activated)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.567	TCMS shall trasmitt via MVB to ATESS(JRU) the status of the test Passenger Emergency Brake (OK/Failure)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.568	TCMS shall trasmitt via MVB to ATESS(JRU) the status Passenger Emergency Brake activation (Activated /Not Activated)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.569	TCMS shall trasmitt via MVB to ATESS(JRU) the status Passenger Emergency Brake override (Activated /Not Activated)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.570	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Main Switch (LCB)(Open/Close)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.571	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Door closed and locked Signalisation(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.572	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Traction Safe(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF CONCEPT.573	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Master Controller(Traction/Braking/Coasting)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.574	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Master controller postion.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.575	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.576	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Cab radio VACMA alarm	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.577	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Emergency Brake Loop (Activated = 0/Not Activated=1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.578	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Electircal Brake (Activated/Not Activated)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.579	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Pantographs (Raised/Lowered)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.580	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Traction converters exclusion	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.581	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Driving Mode (Manual/Automatic)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.582	TCMS shall trasmitt via MVB to ATESS(JRU) the status of WSP fault lamp(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.583	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Spring Brake fault lamp(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.584	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Axle Blocked fault lamp(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.585	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Instability Monitor fault lamp(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.586	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Gear Box oil pressure fault(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.587	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Hot box axle detection alarm(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.588	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Hot box axle detection pre alarm(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.589	TCMS shall read through digital input the status of Line voltage switch postions	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

2F_09.02.Zefiro-IF.CONCEPT.590	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Line voltage switch postion	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.591	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Brake Continues valve	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.592	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Brake test result	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.593	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Brake pipe pressure	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.594	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Fire fighting system intervention	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.595	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Fire fighting system status(Ok/Fail)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.596	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Fire fighting system acoustic alarm status(Warning/No warning)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.597	TCMS shall trasmitt via MVB to ATESS(JRU) the status of emergency brake pushbutton	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.598	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.599	TCMS shall trasmitt via MVB to ATESS(JRU) the status of ITA-FR selector positon(ITA/FR)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.600	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.601	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Z(CO)DJ selector positon(A-OFF,B-OFF,A+B-OFF,Normal)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.602	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.603	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Z-VIT selector positon(On/Off)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.604	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.605	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Z(FU)ETCS isolation selector channel 1 or channel 2	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.606	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.607	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Z(EXPL) position(TVM/TVM-ERTMS)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.608	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.609	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Z(ES)ETCS position(Normal,TVM,ODOMETRIE)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.610	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.611	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.612	TCMS shall trasmitt via MVB to ATESS(JRU) the status of Z(BG)IS position (0,1,2,3,4,5,6,7)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.613	TCMS shall trasmitt via MVB to ATESS(JRU) the brake status(direct brake,indirect EP,indirect IP)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.614	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.615	DELETED				
2F_09.02.Zefiro-IF.CONCEPT.616	TCMS shall trasmitt via MVB to ATESS(JRU) the Right door opening command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.617	TCMS shall trasmitt via MVB to ATESS(JRU) the Left door opening command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.618	TCMS shall trasmitt via MVB to ATESS(JRU) the door closing command	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_09.02.Zefiro-IF.CONCEPT.1077	TCMS shall trasmitt via MVB to ATESS(JRU) the status of the selector Z(SEL)DMI.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

5 Signal Interface

5.1 I/O Interface

TCMS SW
See 6 RD4
Inputs Type:
C=common
N=normal
R=redundant

Nr	Mnemonic name	Description	Logic Level "1"	Logic Level "0"
1	BSTD_KVB_EODJ	KVB execution command to open the main switch (**HIGH LEVEL 300ms long**)	1 = Command present	0=No Action
2	BSTD_KVB_ELC25	KVB execution command to lower the pantograph and advanced notice of a 25kV Conventional Catenary section (**HIGH LEVEL 300ms long**)	1 = Command present	0=No Action
3	BSTD_KVB_ELC1_5	KVB execution command to lower the pantograph and advanced notice of a 1.5kV Conventional Catenary section (**HIGH LEVEL 300ms long**)	1 = Command present	0=No Action
4	BSTD_KVB_ELC3	KVB execution command to lower the pantograph and advanced notice of a 3kV Conventional Catenary section (**HIGH LEVEL 300ms long**)	1 = Command present	0=No Action
5	BSTD_KVB_ELGV	KVB execution command to lower the pantograph and advanced notice of a 25kV High Speed Catenary section (**HIGH LEVEL 300ms long**)	1 = Command present	0=No Action
6	BSTD_KVB_AODJ	KVB announcement to open the main switch.(**HIGH LEVEL 300ms long**)	1 = Opening announced	0=No Action
7	BSTD_KVB_FODJ	KVB end of neutral section; the main switch can be closed.(**HIGH LEVEL 300ms long**)	1 = Close	0 = Open
8	BSTD_KVB_ABP	KVB announcement to lower the pantograph.(**HIGH LEVEL 300ms long**)	1 = Lowering announced	0 =Lowering not announced
9	BSTD_KVB_FBP	KVB end of neutral section; the pantograph can be raised.(**HIGH LEVEL 300ms long**)	1 = Raise	0 = Lower
10	BSTD_KVB_ETUST	KVB announcement of tunnel / end of tunnel	1 = Inside tunnel	0 = Outside tunnel
11	BSTD_KVB_ETAT	KVB internal recorder status.	1 = Fault	0=No Action
12	BSTD_KVB_REALIN_N	Release intervention KVB/RPS	1=Pressed	0=No Action
13	BSTD_KVB_REALIN_R	Release intervention KVB/RPS	1=No Action	0=Pressed
14	BSTD_KVB_DVT	KVB Inconsistency between the two speeds information provided by the tachy ATESS 3G	1 = Fault	0=No Action
15	BSTD_QKVB_RELAY	KVB Emergency Brake Intervention	1 = KVB EB intervention	0=No Action
16	BSTD_CVRACK1_FEED	KVB Converter 1 dc/dc 110/72 status	1=No Action	0 = Fault
17	BSTD_CVRACK2_FEED	KVB Converter 2 dc/dc 110/72 status	1=No Action	0 = Fault
18	BSTD_KVB_SWON	Z-KVB selector on the position 'Normal'	1=Pos. 'Normal'	0=No Action
19	BSTD_KVB_SWOFF	Z-KVB selector on the position 'Isolation'	1=Pos. 'Isolation'	0=No Action
20	BSTD_SWZVA_ON	Z-VA selector on the position 'Normal'	1=Pos. 'Normal'	0=No Action
21	BSTD_SWZVA_OFF	Z-VA selector on the position 'Isolation'	1=Pos. 'Isolation'	0=No Action
22	BSTD_SEL_I-F_1	ITA-FR selector on the position "France"	1=Pos.'France'	0=Pos.'Italy'
23	BSTD_SEL_I-F_2	ITA-FR selector on the position "France"	1=Pos.'France'	0=Pos.'Italy'
24	BSTD_SEL_I-F_3	ITA-FR selector on the position "France"	1=Pos.'France'	0=Pos.'Italy'
25	BSTD_ETCSTVMTEST_TVM	Z(ES)ETCS selector on the position 'TVM'	1=Pos. 'TVM'	0=No Action
26	BSTD_ETCSTVMTEST_ODM	Z(ES)ETCS selector on the position 'ODO'	1=Pos. 'Odometric'	0=No Action
27	BSTD_ETCSTVMTEST_N	Z(ES)ETCS selector on the position 'Normal'	1=Pos. 'Normal'	0=No Action
28	BSTD_ETCSTVMTEST_ET	Note:It is a predisposition. NOT USED	NOT USED	NOT USED
29	BSTD_TVM_VOIE1	Push button TVM manual arming on track 1	1 = Pressed	0=No Action
30	BSTD_TVM_VOIE2	Push button TVM manual arming on track 2	1 = Pressed	0=No Action

31	BSTD_TVM_VOIEDM	Push button TVM manual disarming	1 = Pressed	0=No Action
32	BSTD_SELDMI_1	Z(SEL)DMI selector on the position 'Left'	1 = Pos.'Left'	0=No Action
33	BSTD_SELDMI_2	Z(SEL)DMI selector on the position 'Right'	1 = Pos.'Right'	0=No Action
34	BSTD_OTC_HORN	Horns H/L tones	1=Horn Activation	0=No Action
35	BSTD_SECTAUTON_SEG	Note:TBD answer from EPSF if we use or Predisposition	1 = Command present	0=No Action
36	BSTD_SECTAUTOR_SEG	Note:TBD answer from EPSF if we use or Predisposition	1 = Command present	0=No Action
37	BSTD_DATAENTRYN_1	Z(BG)IS safety discret input1	See §4.1.1	
38	BSTD_DATAENTRYR_1	Z(BG)IS safety discret input1		
39	BSTD_DATAENTRYN_2	Z(BG)IS safety discret input2		
40	BSTD_DATAENTRYR_2	Z(BG)IS safety discret input2		
41	BSTD_DATAENTRYN_3	Z(BG)IS safety discret input3		
42	BSTD_DATAENTRYR_3	Z(BG)IS safety discret input3		
43	BSTD_ELC25KVN_SEG	25kV Conventional Line Catenary section (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
44	BSTD_ELC25KVR_SEG	25kV Conventional Line Catenary section (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
45	BSTD_ELC1KV5N_SEG	1,5kV Conventional Line Catenary section (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
46	BSTD_ELC1KV5R_SEG	1,5kV Conventional Line Catenary section (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
47	BSTD_ELGVN_SEG	25kV High speed Line Catenary section (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
48	BSTD_ELGVR_SEG	25kV High speed Line Catenary section (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
49	BSTD_CODJTN_ COM	Open main switch Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
50	BSTD_CODJTR_ COM	Open main switch Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
51	BSTD_CODJRN_COM	Open main switch Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
52	BSTD_CODJRR_COM	Open main switch Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
53	BSTD_COBPN_COM	Lower pantograph Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
54	BSTD_COBPR_COM	Lower pantograph Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
55	BSTD_CRETN_COM	Traction reduction Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
56	BSTD_CRETR_COM	Traction reduction Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
57	BSTD_CFDJN_COM	Close main switch Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
58	BSTD_CFDJR_COM	Close main switch Command (**HIGH LEVEL 2s long**)	1 = Command present	0=No Action
59	BSTD_MDJARMN_MEM	Note:It is a predisposition. NOT USED	NOT USED	NOT USED
60	BSTD_MDJARMR_MEM	Note:It is a predisposition. NOT USED	NOT USED	NOT USED
61	BSTD_ETUN_COM	Tunnel mode Command	1 = Tunnel mode ON	0=No Action
62	BSTD_ETUR_COM	Tunnel mode Command	1 = Tunnel mode ON	0=No Action
63	BSTD_DJMINN_MEM	Minor Fault	1 = Fault	0=No Action
64	BSTD_DJMINR_MEM	Minor Fault	1 = Fault	0=No Action
65	BSTD_DJMAXN_MEM	Major Fault	1 = Fault	0=No Action
66	BSTD_DJMAXR_MEM	Major Fault	1 = Fault	0=No Action
67	BSTD_DEFFSBN_INI	Inhibition of TTF TVM commands	1 = Inhibition	0=No Action
68	BSTD_DEFFSBR_INI	Inhibition of TTF TVM commands	1 = Inhibition	0=No Action
69	BSTD_Q1_FEEDBACK	Q1-ETCS EB Intervention	1 = No Action	0=BSTD EB intervention
70	BSTD_Q2_FEEDBACK	Q2-ETCS EB Intervention	1 = No Action	0=BSTD EB intervention
71	BSTD_CODJE_SWABOFF	Z(CO)DJE selector on the position 'A+B Off'	1 = Pos.'A+B Off'	0=No Action
72	BSTD_CODJE_SWAOFF	Z(CO)DJE selector on the position 'A Off'	1 = Pos.'A Off'	0=No Action
73	BSTD_CODJE_SWBOFF	Z(CO)DJE selector on the position 'B Off'	1 = Pos.'B Off'	0=No Action
74	BSTD_CODJE_SWN	Z(CO)DJE selector on the position 'Normal'	1 = Pos.'Normal'	0=No Action
75	BSTD_SWTVM_ABOFF	Z-ETCS selector on the position 'Isolation of chain A + B'	1 = Pos.'A+B Off'	0=No Action

76	BSTD_SWTVM_AOFF	Z-ETCS selector on the position 'Isolation of chain A'	1 = Pos.'A Off'	0=No Action
77	BSTD_SWTVM_BOFF	Z-ETCS selector on the position 'Isolation of chain B'	1 = Pos.'B Off'	0=No Action
78	BSTD_SWTVM_ON	Z-ETCS selector on the position 'Normal'	1 = Pos.'Normal'	0=No Action
79	BSTD_ETCSMO_SWON	Z(EXPL) selector position "TVM" Usage of position « TVM » by default	1 = Pos.'ON'	0=No Action
80	BSTD_ETCSMO_SWOFF	Z(EXPL) selector position "TVM/ERTMS" not implemented ERTMS	1 = Pos.'Off'	0=No Action
81	BSTD_ETCSGEN_SWON	Z-BISTANDARD selector on the position 'ON'	1 = Pos.'ON'	0=No Action
82	BSTD_ETCSGEN_SWOFF	Z-BISTANDARD selector on the position'OFF'	1 = Pos.'Off'	0=No Action
83	BSTD_EBL_SW1OFF	Z(FU)ETCS selector on the position 'Isolation of EB channel 1'	1 = Pos.'1 Off'	0=No Action
84	BSTD_EBL_SW2OFF	Z(FU)ETCS selector on the position 'Isolation of EB channel 2'	1 = Pos.'2 Off'	0=No Action
85	BSTD_EBL_SWON	Z(FU)ETCS selector on the position 'Normal'	1 = Pos.'Normal'	0=No Action
86	BSTD_EB_FBCOMN	One of ATP FR EB intervention	1 = No Action	0=One of ATP system EB intervention
87	BSTD_EB_FBCOMR	One of ATP FR EB intervention	1 = No Action	0=One of ATP system EB intervention
88	BSTD_QFC_DIA	BP-FC push button pressed	1=Pressed	0=No Action
89	BSTD_QLN-2_RELAY	TVM arming	1=TVM armed	0=No Action
90	BSTD_RPS_SWON	Z-RS selector on the position 'ON'	1 = Pos.'ON'	0=No Action
91	BSTD_RPS_SWOFF	Z-RS selector on the position 'Off'	1 = Pos.'Off'	0=No Action
92	BSTD_QRPS_RELAY	RPS Emergency Brake intervention	1=RPS EB intervention	0=No Action
93	BSTD_FAN601	FAN cubicle 601 diagnostic	1 = No Action	0=Fault
94	BSTD_FAN606	FAN cubicle 606 diagnostic	1 = No Action	0=Fault
95	BSTD_ATESS_V1	ATESS Tachometry V1 Fault	1 = Fault	0= No Action
96	BSTD_ATESS_V2	ATESS Tachometry V2 Fault	1 = Fault	0= No Action
97	BSTD_ATESS_V3	ATESS Tachometry V3 Fault	1 = Fault	0= No Action
98	BSTD_ATESS_VENR	ATESS Recorder fault speed recording	1 = No Action	0=Fault
99	BSTD_ATESS_R2ENR	ATESS Recorder fault removable cartridge	1 = Fault	0= No Action
100	BSTD_ATESS_R3ENR	ATESS Recorder fault	1 = No Action	0=Fault
101	BSTD_ATESS_SVMV	ATESS 3km/h activation threshold	1 = V<3km/h	0=V>3km/h
102	BSTD_ATESS_ACK_N	Acknowledge BRS push button	1 = Pressed	0= No Action
103	BSTD_ATESS_ACK_R	Acknowledge BRS push button	1 = No Action	0=Pressed
104	BSTD_VACMA_FEED_N	Vacma arming push button	1 = Pressed	0= No Action
105	BSTD_VACMA_FEED_R	Vacma arming push button	1 = No Action	0=Pressed
106	BSTD_ATESS_SWON	Z-VIT selector on the position 'Normal'	1 = Pos.'ON'	0=No Action
107	BSTD_ATESS_SWOFF	Z-VIT selector on the position 'Isolation'	1 = Pos.'Off'	0=No Action
108	BSTD_ATEREC_SWON	Z-ENR selector on the position 'Normal'	1 = Pos.'ON'	0=No Action
109	BSTD_ATEREC_SWOFF	Z-ENR selector on the position 'Isolation'	1 = Pos.'Off'	0=No Action
110	BSTD_VACMA_CANC_ON	Vacma alarm cancellation push button	1 = Pressed	0= No Action
111	BSTD_VACMA_CANC_OFF	Vacma alarm cancellation push button	1 = No Action	0=Pressed
112	BSTD_VACMA_TEST_ON	Z(ES)VA selector on the position 'Test' (Stable position)	1 = Pos.'Test'	0=No Action
113	BSTD_VACMA_TEST_OFF	Z(ES)VA selector on the position 'Normal'	1 = Pos.'ON'	0=No Action
114	BSTD_ETCS_ACK_N	ETCS_Acknowledge	1 = Pressed	0= No Action
115	BSTD_ETCS_ACK_R	ETCS_Acknowledge	1 = Pressed	0= No Action
116	BSTD_Q1-VA_RELAY	VACMA Emergency Brake intervention	1=VACMA EB intervention	0=No Action
117	BSTD_Q-DJ_RELAY	VACMA Failure	1=VACMA system failure	0=no VACMA system failure
118	BSTD_COM1_FCBAUX	CB Comand rack 1 Diagnostic	1 = No Action	0=Cb Tripped
119	BSTD_COM2_FCBAUX	CB Comand rack 2 Diagnostic	1 = No Action	0=Cb Tripped
120	BSTD_COM3_FCBAUX	CB Comand rack 3 Diagnostic	1 = No Action	0=Cb Tripped
121	BSTD_609A_FCBAUX	CB cubicle 609A Diagnostic	1 = No Action	0=Cb Tripped
122	BSTD_KARM_SWON	Z-KARM selector on the position 'ON'	1 = Pos.'ON'	0=No Action
123	BSTD_KARM_SWOFF	Z-KARM selector on the position 'OFF'	1 = Pos.'Off'	0=No Action
124	BSTD_QDEFKARM_DIA	TVM Arming control Default	1 =TVM armed	0= TVM non armed

Outputs Type:

C=common
N=normal
R=redundant

Nr	Mnemonic name	Description	Logic Level "1"	Logic Level "0"
1	BSTD_KVB_BPT_N	Pantograph Down KVB lamp	1=Lamp Is lit	0=No Action
2	BSTD_KVB_BPT_R	Pantograph Down KVB lamp	1=Lamp Is lit	0=No Action
3	BSTD_KVB_SECT_N	Main Switch Open KVB lamp	1=Lamp Is lit	0=No Action
4	BSTD_KVB_SECT_R	Main Switch Open KVB lamp	1=Lamp Is lit	0=No Action
5	BSTD_PANTO_OPBYDR_N	Pantograph Down by the Driver input or ATP	1= No Action	0= Panto down commanded by the driver
6	BSTD_PANTO_OPBYDR_R	Pantograph Down by the Driver input or ATP	1= No Action	0= Panto down commanded by the driver
7	BSTD_DJ_OPBYDR_N	Main Switch Open by the Driver input or ATP	1= No Action	0= Main Switch open commanded by the driver
8	BSTD_DJ_OPBYDR_R	Main Switch Open by the Driver input or ATP	1= No Action	0= Main Switch open commanded by the driver
9	BSTD_TUNNEL_MODE	Tunnel Mode On commanded by ATP	1=Tunnel mode On	0=No Action

5.2 Bus Interface

5.2.1 TCMS SW to ATESS JRU (MVB)

See: 6 RD5

Nr	Functional name	Nr	Functional name
1	tcmstromba	26	tcmsprscndg
2	tcmseffiAllarmePass	27	tcmsintantinc
3	tcmsstatoir	28	tcmsstcirantinc
4	tcmsalrpass	29	tcmssegacantinc
5	tcmsntalrpass	30	tcmsfungo
6	tcmssegnprtchs	31	tcmsItaFra
7	tcmsTractionSafe	32	tcmsCoDje
8	tcmsstatotraz	33	tcmsVit
9	tcmslevainvert	34	tcmsEnr
10	tcmsradio	35	tcmsEtcsIsolQ1
11	tcmasEmergLoop	36	tcmsSelDmi
12	tcmsElBrActive	37	tcmsEtcsMode
13	tcmstrolley	38	tcmsEsEtcs
14	tcmsstazion	39	tcmsEtcsIsolQ2
15	tcmslevacomtraz	40	tcmsVa-ES
16	tcmssegnavantp	41	tcmsBgIs
17	tcmsintfrmolla	42	tcmsDoorsOpnDxCmd
18	tcmssegnasbloc	43	tcmsDoorsOpnSxCmd
19	tcmssegnserpcar	44	tcmsDoorsClsCmd
20	tcmssegninspreoil	45	tcmsV
21	tcmsBocCalda	46	tcmsAmax
22	tcmsPrebocCalda	47	tcmsStatoFreno
23	tcmsimpcatenaria	48	tcmsLevaTrazFren
24	tcmspszrbfrcaba		
25	tcmsprvfrmon		

6 References

Nr	Code	Title	Source	Revision
RD1	90000446.S07.EN	ETR1000 System Architecture Breakdown	HSTS	Rev. 01.00
RD2	90000056.D04.EN	SCHEMATIC DIAGRAM for FRENCH ATP SPECIFIC APPLICATION on the ETR1000/V300ZEFIRO I-F	HSTS	Rev. 04.00
RD3	3ECP307052-0001	V300ZEFIRO-DM8_IF	BT	Rev. _D
RD4	3ECP413534-0001	ICD ATESS ETR1000-V300ZEFIRO for Italy - France corridor	BT	Rev. _B
RD5	ZA580366.300	ETR-1000_ATESS_System_Decription	FT	Rev. A1
RD6		EventReport_VZIF_17112021	BT	Rev. __